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AIRPORT COOPERATIVE RESEARCH PROGRAM

ACRP RESEARCH REPORT 189

Design Considerations for Airport EOCs

Heidi Ann Benaman
FAITH GROUP, LLC
St. Louis, MO

Subscriber Categories
Aviation • Design • Security and Emergencies

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TRB
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2018

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AIRPORT COOPERATIVE RESEARCH PROGRAM

Airports are vital national resources. They serve a key role in transportation of people and goods and in regional, national, and international commerce. They are where the nation's aviation system connects with other modes of transportation and where federal responsibility for managing and regulating air traffic operations intersects with the role of state and local governments that own and operate most airports. Research is necessary to solve common operating problems, to adapt appropriate new technologies from other industries, and to introduce innovations into the airport industry. The Airport Cooperative Research Program (ACRP) serves as one of the principal means by which the airport industry can develop innovative near-term solutions to meet demands placed on it.

The need for ACRP was identified in *TRB Special Report 272: Airport Research Needs: Cooperative Solutions* in 2003, based on a study sponsored by the Federal Aviation Administration (FAA). ACRP carries out applied research on problems that are shared by airport operating agencies and not being adequately addressed by existing federal research programs. ACRP is modeled after the successful National Cooperative Highway Research Program (NCHRP) and Transit Cooperative Research Program (TCRP). ACRP undertakes research and other technical activities in various airport subject areas, including design, construction, legal, maintenance, operations, safety, policy, planning, human resources, and administration. ACRP provides a forum where airport operators can cooperatively address common operational problems.

ACRP was authorized in December 2003 as part of the Vision 100—Century of Aviation Reauthorization Act. The primary participants in the ACRP are (1) an independent governing board, the ACRP Oversight Committee (AOC), appointed by the Secretary of the U.S. Department of Transportation with representation from airport operating agencies, other stakeholders, and relevant industry organizations such as the Airports Council International-North America (ACI-NA), the American Association of Airport Executives (AAAE), the National Association of State Aviation Officials (NASAO), Airlines for America (A4A), and the Airport Consultants Council (ACC) as vital links to the airport community; (2) TRB as program manager and secretariat for the governing board; and (3) the FAA as program sponsor. In October 2005, the FAA executed a contract with the National Academy of Sciences formally initiating the program.

ACRP benefits from the cooperation and participation of airport professionals, air carriers, shippers, state and local government officials, equipment and service suppliers, other airport users, and research organizations. Each of these participants has different interests and responsibilities, and each is an integral part of this cooperative research effort.

Research problem statements for ACRP are solicited periodically but may be submitted to TRB by anyone at any time. It is the responsibility of the AOC to formulate the research program by identifying the highest priority projects and defining funding levels and expected products.

Once selected, each ACRP project is assigned to an expert panel appointed by TRB. Panels include experienced practitioners and research specialists; heavy emphasis is placed on including airport professionals, the intended users of the research products. The panels prepare project statements (requests for proposals), select contractors, and provide technical guidance and counsel throughout the life of the project. The process for developing research problem statements and selecting research agencies has been used by TRB in managing cooperative research programs since 1962. As in other TRB activities, ACRP project panels serve voluntarily without compensation.

Primary emphasis is placed on disseminating ACRP results to the intended users of the research: airport operating agencies, service providers, and academic institutions. ACRP produces a series of research reports for use by airport operators, local agencies, the FAA, and other interested parties; industry associations may arrange for workshops, training aids, field visits, webinars, and other activities to ensure that results are implemented by airport industry practitioners.

ACRP RESEARCH REPORT 189

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FOREWORD

By Marci A. Greenberger
Staff Officer
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ACRP Research Report 189: Design Considerations for Airport EOCs will guide airport staff in the planning and design of an airport emergency operations center (EOC). Whether updating an existing center or establishing a new EOC in an existing facility or in an as yet-to-be-built facility (greenfield), this guidebook will help airport planners and emergency management staff consider all of the factors necessary in planning and designing an EOC that best fits their environment.

Early incarnations of airport EOCs typically included only few personnel such as operations, police, and fire managing emergencies and/or unplanned events. Over time, airports have begun to realize the benefits of having all relevant stakeholders in one place, working together to mitigate the effects of an emergency event, as well as planned events that impact operations. Airports are thinking about ways in which their EOCs can be redesigned to accommodate these stakeholders, either by enhancing existing space or shifting to a new environment.

Faith Group, LLC was selected to conduct research identifying EOC design considerations in the areas of physical space, technology, function, operations, and governance. The guidance also includes information on (1) roles and responsibilities; (2) concept planning; (3) site selection; (4) ergonomic needs, technology, and human factors; and (5) architectural considerations that airports of all sizes can use as they begin planning and designing an EOC. The guidance includes lessons learned, a Concept and Development Planning Decision Tree flow chart, and several useful checklists to help airport staff work through all of the considerations involved in establishing or enhancing an EOC.

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SECTION 1

Introduction

This Guidebook will help airports with Emergency Operations Center (EOC) planning and design considerations, such as (1) establishing an EOC in an existing facility; (2) upgrading a current EOC facility; (3) designing and building a new facility within the terminal (or other existing campus building); and (4) designing and building a greenfield project, which means the airport is not bound by any constraints with existing buildings or infrastructure such as the terminal or other existing campus structures. The use of "facility" within this Guidebook is referring to the EOC space and not necessarily a building site per se. The Guidebook is relevant to all size airports, including general aviation (GA), small-hub and non-hub, medium-hub, and large-hub airports and provides information on (1) roles and responsibilities; (2) concept planning; (3) site selection; (4) ergonomic needs, technology, and human factors; and (5) architectural considerations. It is critical to plan a facility that accommodates the individuals who will be directly involved in managing airport incidents and events of all sizes and complexities, such as aircraft disasters, severe weather, terrorist or active shooter incidents, and for coordinating large-scale events, such as presidential, very important person (VIP) visits and airshows.

The Guidebook offers several tools for the reader that include an Airport EOC "Best Practices and Lessons Learned" matrix, EOC Concept and Development Planning Decision Tree (Decision Tree) flow chart for guiding the reader through the EOC consideration processes, and several action checklists correlated to the type of EOC improvements or manner of expansion that an airport decides to pursue.

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SECTION 2

Background

History

Most airports are familiar with the National Incident Management System (NIMS) and the Incident Command System (ICS). It is commonly accepted that most airports try to incorporate the use of these tools, plans, and practices for all types of response management under the NIMS/ICS umbrella. It is valuable for airports and their regional disaster response partners to share a coordinated preparedness platform and have the appropriate facility in which to accommodate proper flow of operations and personnel. Stakeholder organizations such as airlines, Transportation Security Administration (TSA), Federal Aviation Administration (FAA), Customs and Border Patrol (CBP), Federal Bureau of Investigation (FBI) and other specialists typically are brought in to provide subject-matter expertise and assist the airport staff during an incident. Thus, the airport EOC must be designed to handle the anticipated number of personnel based on the size and impact of calculated risks and the likelihood of their occurrence.

In the mid-1990s, the ICS structure started to make its way into the airport environment beyond just Aircraft Rescue and Fire Fighting (ARFF) departments, with the FAA encouraging the use of the system and updating Advisory Circular (AC) 150/5200-31A to include recommendations for airports. Code of Federal Regulations (CFR) Part 139 directs that certificated airports are no longer able to opt out from utilizing an NIMS/ICS approach. Rather, the FAA implies that airports reflect NIMS in their Airport Emergency Plans (AEPs) and pursue ICS and NIMS training for their employees in accordance with the most current AC 150/5200-31C. As airports become more sophisticated with their inclusion of NIMS/ICS practices, so too must the EOC environment contribute to a successful, coordinated, and well-communicated response.

The need for a viable EOC was not purely driven by the FAA. Most airport managers and directors realize that their place during an event is not in the field, but in an enclosed environment that can accommodate functional needs such as supporting the Field Command (FC), updating the board of directors and other authorities, dealing with the media, and isolating the event in the hopes of maintaining some semblance of airport operations. Depending on the size of the airport and the incident, many participants will need specially designated space to coordinate efforts. The coordination time can range from a couple of hours to days, months, or longer.

The first EOCs airports utilized were commonly large conference or training rooms. Often, the EOC was only utilized during the required triennial drill or perhaps the annual tabletop. Over time, many airports have come to value bringing key subject-matter experts (SMEs) together to create a cohesive plan to mitigate the effects of an event on operations. Quicker mitigations may save lives, property, and accelerate the return to operations. There could be a cost savings as well, especially for air carriers wanting to avoid further delays and cancellations because of an event hindering operations. EOCs are evolving into facilities that must fit the needs and budgets of any size airport. For some small airports, making technology or equipment updates to an existing

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conference room may be the appropriate investment. For larger airports, the size, concept and planning may need to be more complex to meet their needs. Co-utilization of space shared with other day-to-day functions is still often appropriate and economical; but choosing the right location is important. Security, access, and proximity to other functions and facilities must be considered. There is also increasing awareness of the need to protect from potential threats to emergency operations, such as weather and terrorism. It is important to first realize the current and projected needs of the EOC and then establish the priorities to support those needs.

Guidebook Parameters

The Guidebook will assist airports through all phases of planning and designing an EOC. Many airports have attempted concept planning internally but did not realize all the questions that needed to be asked and answered. Questions the Guidebook will answer include, but are not limited to, the following:

- Who, at a maximum, needs to be accommodated in the EOC?
- How large should the EOC be to meet current and future needs?
- What are some of the best practices being used in other EOC applications and geographies?
- How does an airport select the best site?
- What threats and vulnerabilities are associated with the site?
- What security and protective design considerations are appropriate for the selected site?
- How will the EOC be secured from access by the public, media, or others not requested to respond?
- Is it best to position the EOC in conjunction with an airport communications/dispatch center?
- Does the airport EOC need to replicate a local city or county EOC?
- What are possible operational and capital advantages/disadvantages of EOC consolidation with other functional areas (communication center, operations, ARFF, police, etc.)?
- Are there other location requirements regarding infrastructure (power, water, etc.)?
- Should there be a backup EOC in case the primary is rendered unusable?
- What are floor layout considerations?
- What equipment and infrastructure need to be incorporated in the EOC?
- What information technology (IT) support systems should be considered?
- What cyber requirements should be considered?
- What ergonomic considerations should be included?

This Guidebook was constructed with input from several airports throughout many of the tasks to create guidance that is not only relevant, but relatable.

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SECTION 3

Roles and Responsibilities

Purpose

This section defines the various roles and responsibilities of staff in both the planning process and as players within the EOC. This section encourages airport managers to use a sound organization business model such as the NIMS and ICS or similar organizational system such as Standardized Emergency Management System (SEMS).

ICS will serve as the common nomenclature for this Guidebook. Using an ICS-type structure for managing an event influences the size and flow of the EOC facility. That, in turn, influences the planning process described in Section 4 of the Guidebook.

Roles and Responsibilities

Airport Staff Roles and Responsibilities

There are various players within airport staff who should be involved in the planning and design of the EOC. They include, but are not limited to, representatives from airfield and landside operations, maintenance and facilities, security, airport law enforcement officers (LEOs), ARFF, media relations, planning and engineering, IT, finance, and executive management. At a minimum, these are the people who would respond to the EOC and therefore should be part of the planning.

Airport Stakeholder Roles and Responsibilities

There are various stakeholders who have a role and responsibility to assist airport management and staff in the planning and design of the airport EOC to ensure their needs are met when they assist in the EOC. These stakeholders include, but are not limited to, airlines, tenants, FAA Air Traffic Control, FBI, TSA, Customs and Border Control (CBP), mutual aid responders, the American Red Cross (ARC), and the National Transportation Safety Board (NTSB). The process guideline for involving these entities is mapped out in Section 4 of the Guidebook.

The Use of ICS Within the EOC

As mentioned earlier, most airports are familiar with NIMS and ICS. It is generally agreed that because airports rely upon outside resources when an incident or event outstrips their ability to manage the incident or event on their own, it is valuable for airports and their regional disaster response partners to share a similar preparedness platform. Assets such as hazardous materials (HAZMATs) response teams, bomb squads, hostage negotiation units,

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decontamination units, volunteer organizations, and other specialists typically are brought in from outside the airport when needed during an incident. Those assets and others exist within government structures such as counties, states, or federal agencies. Typically, these agencies follow NIMS, so it is important that all responders utilize a common structure and terminology. The structure of choice should be communicated and training provided for all responders.

The Federal Emergency Management Agency (FEMA) has oversight of NIMS and ICS as far as structure, training, and the supplemental event forms that are used for reimbursements. A federally declared event, such as a tornado, hurricane, and flood usually encompasses a large geographic area. During these types of events, an airport operator is usually supporting the local or state emergency management agency as another resource and is not the lead agency. The ICS structure for large-scale disasters can be a bit overwhelming to an airport operator. Airports should construct an ICS response system and supporting EOC in a way that reflects the needs of typical airport emergencies. Airports usually rely on immediate resources and mutual aid from the nearby community. Airport emergencies are relatively short term, from less than a day to maybe a few days as in the case of an aircraft accident.

The ICS originally utilized the Command Post (CP) as the main location where all major decisions concerning an event took place. Today, the CP for airports is regarded as the gathering place to coordinate tactical operations. Usually, when an airport experiences an event the Incident Commander (IC) manages tactical operations at the CP, but the airport is still open and operating or working toward recovery to resume operations. The IC is focused on life, safety, and protection of property and the environment. Some airport managers use the ICS nomenclature of "IC" for both the field commander and the EOC manager. There are many titles employed for the lead manager of the EOC, and they include, but are not limited to, IC, EOC manager, EOC Commander, and EOC Director, to name a few. It is not important what the name is, but it is important that communication and training take place, so all responders understand who is in what role and what their responsibilities are. The Guidebook gives several naming convention options for airports to consider based on research.

For most organizations like airports, the EOC provides a functional area away from the incident scene, but with the communications capabilities to monitor what is evolving on site. The airport EOC is usually seen as the part of the organization that is looking out airport-wide and maybe community-wide. They keep the big picture and support the needs of the IC at the CP. This inclusion of an EOC into the ICS structure has led to various approaches on how to mesh the two so that each provides the other with that they need to do their job. This Guidebook provides scenarios based on the survey outcomes of various ways to integrate an EOC and CP in the field. Communication is key.

Airports sometimes try to stay as close to the formal ICS organization chart as possible. This Guidebook incorporates suggestions to help airports make ICS fit into their structure rather than the airport fitting into ICS. Some of these issues include naming the top person in the EOC, as discussed above, and what positions are filled within the formal ICS structure. Some airports have added a Day-to-Day Section Chief to help manage resources that are dedicated to keeping airport operations flowing. Flexibility of the ICS is key to helping personnel embrace the concept. Many airports are not comfortable with the ICS or the use of their EOC because they feel ICS protocols are intransigent. This Guidebook provides some flexible options airports may adopt to tailor the ICS to their own needs.

Another variance may be what airports call their EOC. Some airports do not call their designated response facility an EOC, often because that is the term used for city, county, or state EOCs. Governments and their EOCs represent a larger geographic area than the airport. In a regional

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emergency, the city or county EOC is often the main point of communication and logistics management. So as not to confuse organizations, some states have mandated that only certain organizations may call their coordination center an EOC. California is an example of a state that restricts the use of the title "EOC." They call other organizations, such as airports, Department Operations Centers (DOCs). Other airports such as Phoenix Sky Harbor Airport (PHX) are referred to as the Airport Emergency Operations Center (AEOC) so as not to be confused with City of Phoenix EOC. It is not so important what an airport names its EOC if it is communicated to stakeholders and mutual aid responders. For the purposes of this Guidebook, the terms ICS and EOC are used as the generic names for organization structure and the designated facility to support and provide oversight of an event.

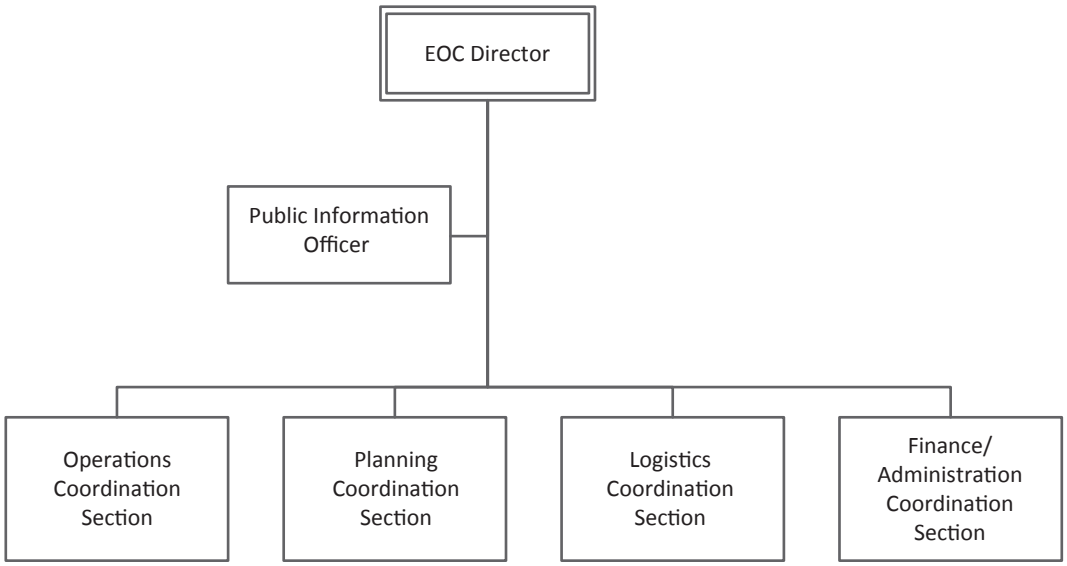
It is critical for airport management to understand how they plan to incorporate ICS into their emergency operations on a regular basis as this drives the amount of space an EOC requires. The flow of the floor plan and furniture arrangements are all conducive to the management of events of various sizes, impacts, and length of time the EOC is in operation. In October of 2017, FEMA published a revision of NIMS, which outlines three possible EOC structures:

- ICS or ICS-like (Figure 3-1),
- Incident Support Model (Figure 3-2), and
- Departmental Structure (Figure 3-3).

Each of these variations are described on the next pages.

This Guidebook acknowledges there are several incident management models. The important consideration is planning an EOC layout based on an airport's model of choice and communicating that model to all applicable stakeholders. This Guidebook references the ICS or ICS-like modeling because these are likely the most familiar for airports.

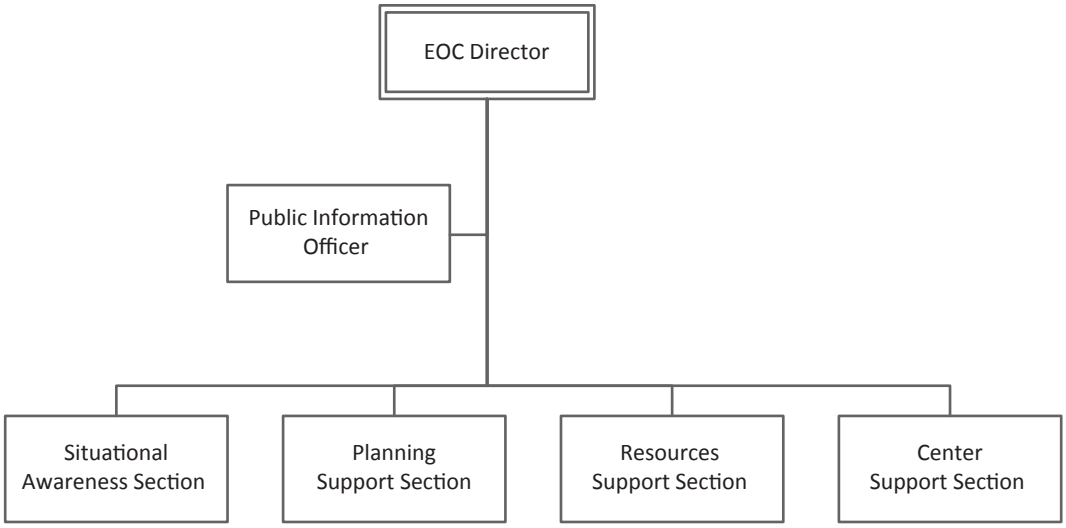
Figures 3-4 through 3-6 are ICS charts showing examples for the EOC staffing and interaction with tactical FC operations.



Source: Faith Group, LLC.

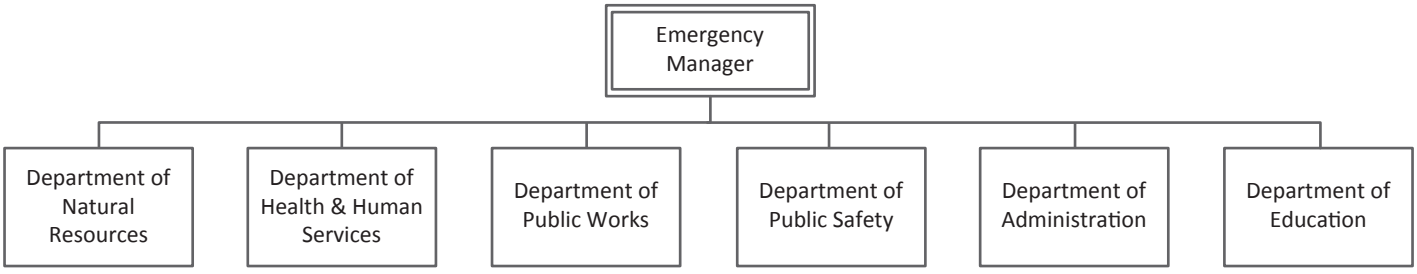
Figure 3-1. ICS or ICS-like. The ICS structure is probably the most familiar to organizations and aligns with the tactical response in the field.

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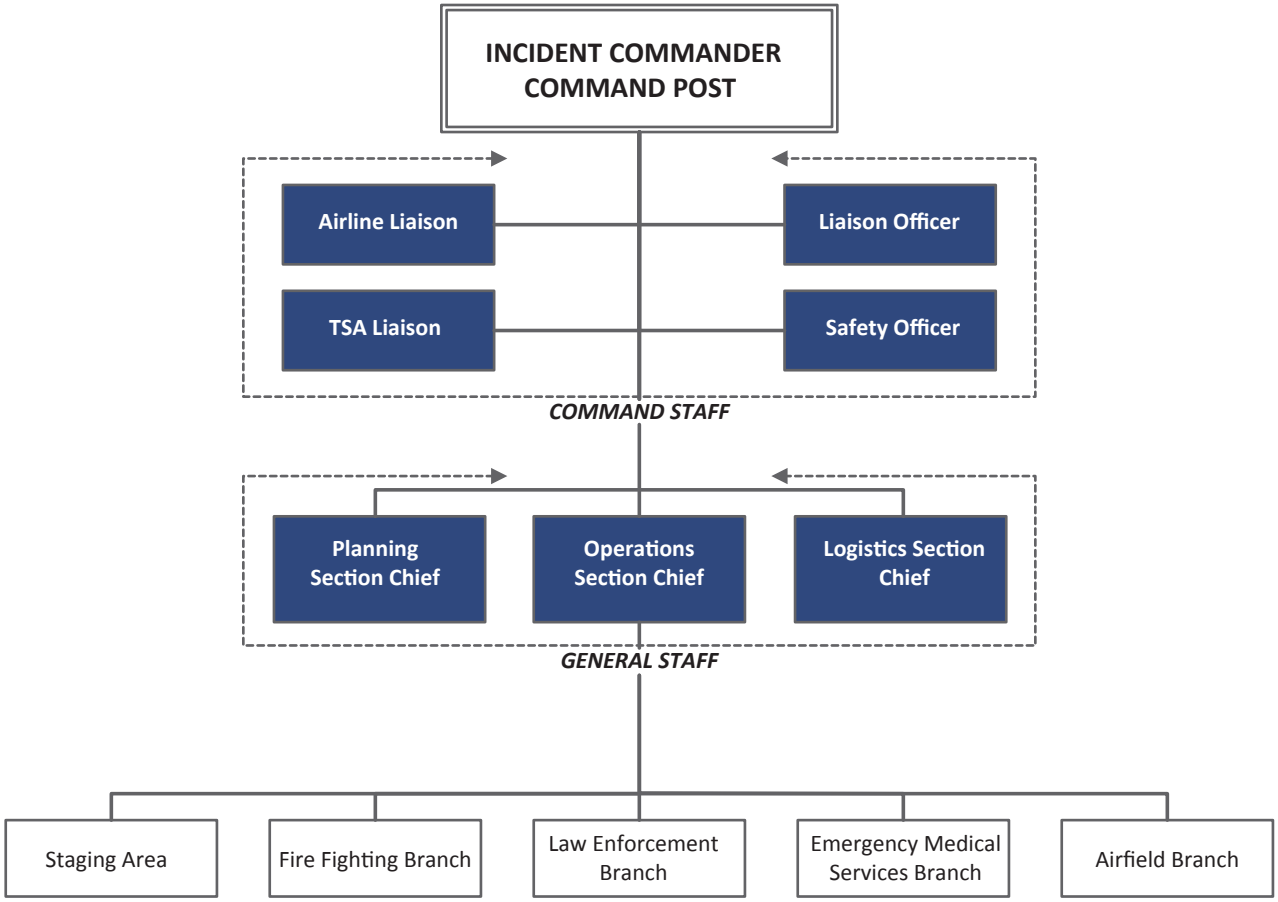
Figure 3-2. Incident support model (ISM). The ISM Model may be used by organizations that focus their EOC team's efforts on information, planning, and resource support, and that separate out the situational awareness function from planning and combine operations and logistics functions into an incident support structure.



Source: Faith Group, LLC.

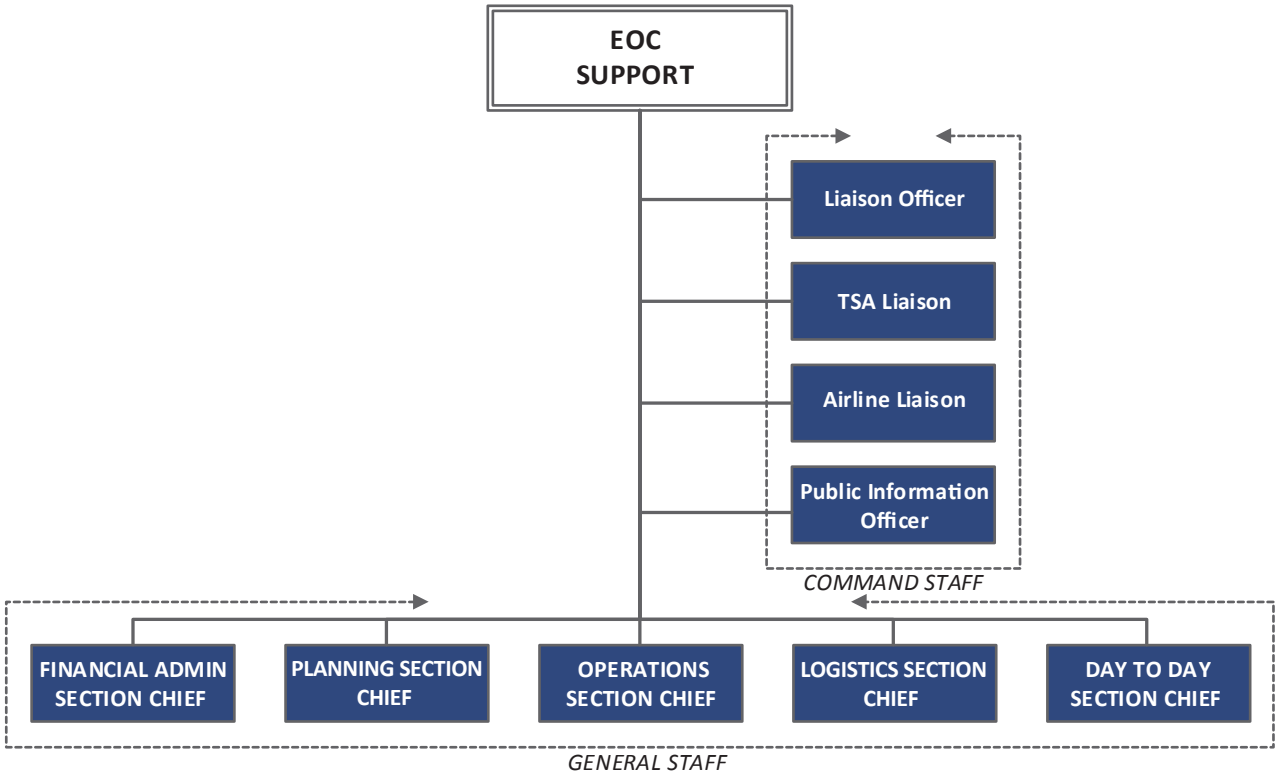
Figure 3-3. Departmental structure. The Department structure utilizes the day-to-day organization structure and titles in the EOC. Utilizing the day-to-day titles and organization, individuals can function with minimal preparation and start-up time.

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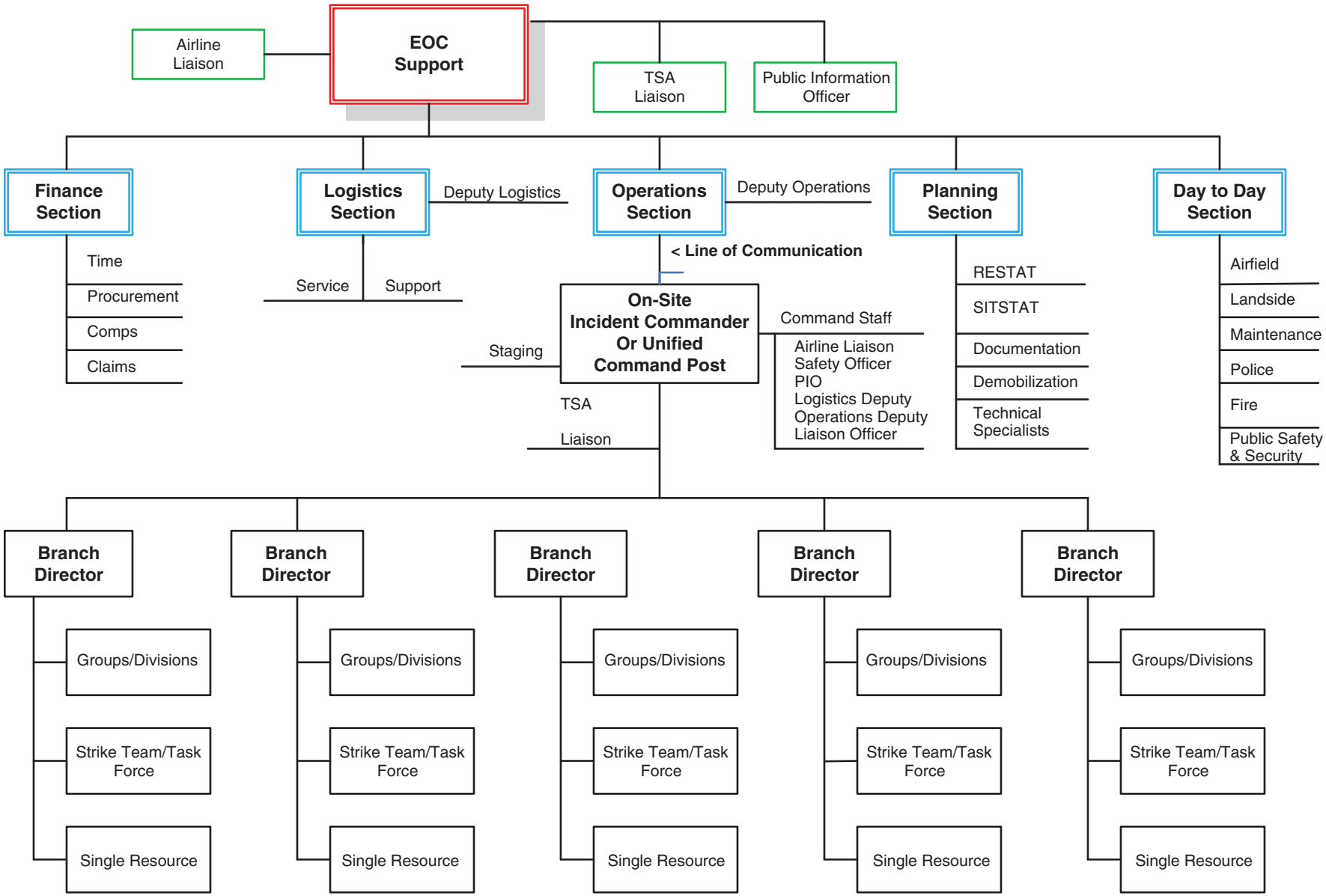
Figure 3-4. Basic ICS chart.



Source: Faith Group, LLC.

Figure 3-5. EOC ICS organization.

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Source: Faith Group, LLC.

Figure 3-6. Full ICS chart with EOC and FC best practices and lessons learned.

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ICS Structure

Figure 3-4 reflects the ICS structure in its purest form indicating the IC in charge of the field CP supported by the Command Staff and the general staff. The general staff represents what is commonly seen in airports. In the field, the IC usually works directly with the specific departments or SMEs and often they are considered "branches." The Section Chief positions are not normally staffed, per se, but their functionality is retained by the IC unless otherwise assigned. When the IC finds they are managing outside a reasonable span of control (five to seven personnel or resources) in the field or they require long-term support and resources, this is typically an indication that an EOC should be activated. When the EOC is activated, it may be up to the IC as to whether the Section Chiefs' roles remain in the field or are run through the EOC. Either the EOC or IC may utilize deputies for any of these positions.

EOC Structure

Figure 3-5 is representative of staffing for the EOC for airports. What an airport calls the person in charge of the EOC may vary as discussed previously in this section.

EOC and FC Structure

Figure 3-6 shows an example of an ICS inclusive of the EOC and FC. This is an example of how the EOC and FC could be organized. The hierarchy of the structure from EOC to FC is not so much lines of authority, but rather, lines of communication between the FC and the EOC.

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SECTION 4

Planning Processes

Purpose

This section will provide direction for airport management to make decisions concerning the complexities of their airport and what the needs may be for EOC facility requirements. Section 3 encouraged airport managers to consider the use of a sound business model such as the ICS during emergencies and events. Using an ICS structure for managing an event influences the size and flow of the EOC facility. Section 4 is based on best practices and lessons learned, how to budget for various concepts, and a Decision Tree to guide airport managers through the concept planning process. The conclusion of this section directs airport managers to consider this a continuous improvement process, which should be readdressed periodically.

Best Practices and Lessons Learned

EOC development needs to consider many things. Airport owners and operators are challenged to include all the issues that may or may not affect their EOC development, such as:

- Who will be accessing and utilizing the EOC when it is activated?
- Will it be just airport staff, or will it include outside agencies?
- Should the EOC be considered a Joint Operations Center (JOC)?
- Was the EOC envisioned as part of the master plan and shown on the Airport Layout Plan (ALP)?

These and many other questions are vital to understanding what will be needed as part of an airport's EOC. These questions and challenges are germane to the EOC development process, regardless of the airport size or complexity. If the process is followed thoroughly and corners are not cut, then the critical questions will be answered. Different components of the decision-making process will need to be conducted simultaneously. One component will feed the next and so on with valuable information. Certain components that are best practices regardless of the physical space, layout, and planning process include training and having standard operating procedures in place that people are familiar with or that can be accessed via a written document or checklist during an event.

Best Practices and Lessons Learned Matrices

Twenty-seven survey interviews were conducted during the research phase, inclusive of large-hub, medium-hub, small-hub/non-hub, and GA airports. Information gathered concerning best practices and lessons learned is documented in Tables 4-1 through 4-6.

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Table 4-1. Large-hub airport best practices and lessons learned.

Large-Hub Airport	Best Practices	Lessons Learned
George Bush Intercontinental Airport (IAH)	1. Easy access for non-badged responders. 2. Not used as a conference room. 3. Controlled access. 4. Standard operating procedures (SOPs) in place. 5. Virtual EOC as a stand-by mode for monitoring. 6. Formal mass notification system (Everbridge). 7. Minimize attendance, regardless of titles. 8. Redundancy for systems such as generators. 9. Paint walls with dry-erase white paint.	1. It is too small for the size of their airport. 2. Better proficiency with WebEOC. 3. Sidebar conversations do not get tracked. 4. Long and narrow facility is not good for flow. 5. Make sure to have a break room. 6. Have breakout rooms to cut down on sidebars. 7. They are vulnerable in the terminal. 8. Utilize compartmentalized workspace rather than table rows.
Dallas Ft. Worth International Airport (DFW)	1. Planning for new EOC based on experience with current space. Expected in 24–36 months. Would increase space to 60,000 square feet and transition into a JOC—an open concept with large and permanent pods for the terminal, IT help desk, TSA coordination center, CBP, American Airlines, North Texas Fusion Center, and other occupants. 2. Current facility has adequate computers and good security, including a camera at the door accessed only with secure badge.	1. Temperature in EOC was too cold. Personnel became ill; some refused to work there. Situation was corrected. 2. Moving away from ICS/NIMS based on experience, and back to Emergency Support Functions (ESFs). Airport EOC positions are identified by function now. 3. Poor flow and management of information and thus, limited situational awareness (SA). Used older information management procedures. Will be upgrading and adding WebEOC in new EOC.
Denver International Airport (DEN)	1. Have a backup EOC. 2. Controlled access. 3. The Airport Duty Manager works in the EOC, so it is always in stand-by mode. 4. Have good event tracking software. 5. SOPs. 6. Checklists. 7. Formal mass notification system (Everbridge). 8. Practice using often, and it runs well during larger events.	1. Need more breakout rooms. 2. Have Section Chiefs with their own work areas. 3. Not large enough. 4. Located between other operational areas makes for bad flow, e.g., people from maintenance traverse through EOC to get to and from work. 5. Do not have people in the EOC unless they have a specific role. 6. Noise levels are high. 7. Vulnerable position on 10th floor and one side all glass windows during a tornado.
Ft. Lauderdale Hollywood International (FLL)	1. Access plan in place for non-badged responders. 2. If EOC in terminal, have secure access with law enforcement posted at door when necessary (during activation). 3. Controlled access. 4. Bunk areas. 5. Locker rooms/shower facilities. 6. Make sure external people have a method of responding back to EOC so tasks can be received and carried out then reported on. 7. Plan for extended opening of EOC. 8. Work collectively when planning a new EOC. Speak to every division: TSA, Fire, law enforcement, airlines, etc. Make sure EOC will accomplish what it needs to accomplish without going overboard. Put logic and realism to it. 9. If situation is expanding, then people need to be asked to leave when necessary.	1. It is too small for the size of their airport. 2. Better proficiency with WebEOC. 3. Make sure to have a kitchen. 4. Have breakout rooms to cut down on sidebars. 5. They are vulnerable in the terminal if situation is active within terminal (active shooter, fire etc.). 6. Have a logistics plan (resources such as generators tend to be difficult to obtain during activation). 7. Do not rely solely on terminal tenants to sustain EOC (for food etc.). 8. Having EOC outside of terminal makes parking easier, especially during activation when more people (badged and non-badged) are present. 9. EOC outside of terminal building makes it less complicated to escort non-badged personnel. 10. Each event makes the EOC better. 11. Extended opening of EOC burned personnel out. 12. Each incident is unique so plan for as many different types of scenarios as possible.
Minneapolis St. Paul International Airport (MSP)	1. Backup EOC in terminal. 2. Controlled access. 3. Command staff and general staff are arranged accordingly to work within their groups. 4. Highly equipped for IT logistics, laptop-charging cart, video walls, plenty of phone and local area network (LAN) access. 5. SOPs. 6. Checklists. 7. Formal notification system (Everbridge). 8. Positions are placarded so people know where to go. 9. Keep a stock of various charging cords, protein bars, and bottled water to sustain people who respond in haste not knowing how long to plan for.	1. Because the EOC is not on site it does not get used much except for exercises. 2. They do not have their own event tracking SA software and must log into different software of working with the county or state. 3. It is not a dedicated facility and setup can take 30 minutes. 4. Exercises usually attract more people than a real event might. 5. People are in disparate areas of the airport and it is hard to remember to brief everyone.

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Table 4-1. (Continued).

Large-Hub Airport	Best Practices	Lessons Learned
Phoenix Sky Harbor Airport (PHX)	1. They use the AEOC on a regular basis and are very familiar. 2. They did a formal concept planning approach to meet the needs. 3. They realistically decided how many chairs and how much breakout space is needed. 4. They did a threat and vulnerability analysis, which helped eliminate some options. 5. They involved all the key stakeholders in the planning process.	1. The state uses Webbook for managing events and it does not really integrate with the event tracking system, so efforts are duplicated. 2. Managing accurate information within groups who are getting intel from various places. 3. Have someone from the communications center in the EOC to bridge information.
Portland International Airport (PDX) Current Day	1. Controlled access. 2. Concept planning for new facility. 3. Pre-identified responders and their roles. 4. Org chart. 5. SOPs. 6. Checklists. 7. Identifiable gear. 8. People are trained. 9. Bring in more SMEs like IT and GIS.	1. Too small. 2. Vulnerable location in main terminal. 3. If you do not use technology regularly like WEB EOC, it is hard to keep current. 4. Collocation to Com Center or Operations is not as important as having a liaison. 5. Layout is not good as one long room and table, would prefer smaller breakout set up for face-to-face communication. 6. Need better technology for good tracking and SA. 7. Finding a common area to access electronic manuals and checklists has been challenging, SharePoint, a file transfer platform (ftp), did not work well. 8. The setup of the EOC is as important as the process and procedures. 9. Would rather lose proximity to the terminal in lieu of secure facility. 10. Getting people to think about the EOC for smaller events, not just the bigger ones.
Washington Reagan (DCA)	1. Airport Police Chief obtained a grant to help fund emergency planning for the airport. 2. Two EOCs: large one (1,200 sq. ft.) located on the Security Identification Display Area (SIDA) side of the airport in a hangar. Alternate EOC is in the Ops Center. 3. Use WebEOC and are conducting training on it. 4. Have separate conference area and do video conferencing there. 5. Seats for all airport EOC personnel and all stakeholders (including hospital rep and Red Cross plus Feds, fire and police, etc.). 6. Put EOC on stand-by status for national security events (Inaugural and other major events). 7. Good continuous information displays.	1. Space in alternate EOC is insufficient. 2. For better logistics, wants to establish pods and more flexible space arrangements. Layout is extremely important. Should be able to compartmentalize and change. 3. Noise problems; wants more virtual EOC capabilities to solve problem.
Washington Dulles (IAD)	1. Emergency manager has purchased 30 new laptops along with a storage "cage" where they are kept. The computers are on a permanent connection to periodic updates and will be available for use whenever the EOC is activated. 2. Currently training all staff on WebEOC. 3. Good representation for positions at EOC: Air ops, police, fire, airlines, Federal partners (traffic control, FBI, CBP, FAA), Red Cross, hospital rep, state and local jurisdictions.	1. Space in EOC is insufficient. Planning to build new facility. 2. Must be able to compartmentalize space according to the type of incident and personnel in the EOC. 3. Emergency Management Director is a proponent of virtual EOC where possible and preferred, e.g., public information officer (PIO) functions.

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Table 4-2. Medium-hub airport best practices and lessons learned.

Medium-Hub Airport	Best Practices	Lessons Learned
Cincinnati/Northern Kentucky International Airport (CVG)	1. Dedicated EOC = minimal stand-up time. 2. Backup EOC location with cartload of automatically updated laptops. 3. Effective software: WebEOC, Everbridge. 4. Dry-erase marker maps.	1. Too small = noisy and crowded. 2. Remote from terminal area makes it less likely to be used. 3. Need adjacent breakout rooms. 4. Desktop monitors block face-to-face interaction – should be depressed in tables. 5. Too much brightness/glare without window blinds.
Indianapolis International Airport (IND)	1. Dedicated EOC—co-located with the Airport Operations Center (AOC) in a separate non-descript building away from the terminal. 2. The EOC/AOC facility was the result of the security master plan developed in 2002 (after 9/11). 3. Breakout rooms 2,000 sq. ft. total area for the EOC. 4. Single computer access into all systems is available for each representative. 5. The EOC has one console, which can be operated by a dispatcher and can access all systems in real time and not affect the AOC. 6. The area is Access Control System– (ACS-) and closed-circuit television– (CCTV-) controlled. 7. Can be arranged for training events as well (modular desks and computers). 8. One person responsible for EOC's operation and care. 9. Supports multiple agencies in the area. 10. The EOC is activated several times per year (weather, VIP arrivals, major events, etc.). 11. Everbridge is used for mass notification. 12. Purpose-built facility from a greenfield. 13. Updated computers to provide more workspace on a single point of entry into the systems for representatives. 14. No pending upgrades currently. (Everyone is satisfied with the facility.)	1. Should have designed the IT systems first and then designed the building around them; it was vice versa. (Resulted in several change orders during construction).
Mineta San Jose International Airport (SJC)	1. Controlled access. 2. Large main room expandable. 3. Many breakout rooms nearby. 4. Checklists. 5. Formal mass notification system (Everbridge).	1. Originally collocated with police; made them vulnerable. 2. Because it is not dedicated facility, they must set it up for use, which can take 45 minutes. 3. Because the EOC and ICS are not used often, staff and stakeholders are not comfortable with the processes. 4. Phone and LAN jacks limit table arrangements. 5. Exercises probably attract more people than a real event would. 6. Vulnerable position near roadway and all windows facing it. 7. The setup time serves as an excuse not to use the EOC or practice using it.
Portland International Airport (PDX) 1995-2006	1. Controlled access. 2. Pre-identified responders by shift. 3. Checklists.	1. Too small. 2. Vulnerable location in main terminal.
Sacramento International Airport (SMF)	1. Controlled access. 2. Utilize ICS structure. 3. Have binders for each position. 4. Do not use the EOC for anything other than emergencies or training. 5. Google Earth works great for drill down into any area of the airport.	1. The TVs are not all the same make and model and remote controls cannot differentiate among them. 2. Too small, but that may be a function of the furniture and configuration not being well laid out. 3. May be vulnerable in the terminal non-sterile side. 4. Direct TV may not be sustainable during events, especially those that are weather-related. 5. Make sure to routinely update the computers so they are not doing updates in the middle of an event.

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Table 4-3. Small-hub airport best practices and lessons learned.

Small-Hub Airport	Best Practices	Lessons Learned
Jackson Evers International Airport (JAN)	1. Controlled access. 2. Organization chart. 3. SOPs. 4. Formal notification system (Everbridge).	1. Provide checklists for EOC positions. 2. Better technology for tracking. 3. People do not always understand their role and start to freelance. 4. Vulnerable location in main terminal.
Des Moines International Airport (DSM)	1. Agreement with Polk County to share their EOC as a backup in case it is impacted by event. 2. Access controlled. 3. Multiple breakout rooms available as well as an all glass tower cab for visuals.	1. Older facility, no elevator, not Americans with Disabilities Act (ADA) compliant. 2. It takes a long time to set up, about 1-½ hours. Some of this is due to staff response times and logistics to bring them in. 3. Hard for some people to find. 4. Vulnerable location near terminal and roadway. 5. In a real event, facility will likely be too small.
Memphis International Airport (MEM)	1. Dedicated single-purpose EOC. 2. Assigned breakout spaces. 3. Backup EOCs available. 4. Weekly computer check, monthly tabletops, yearly exercises. 5. Adjacent to Airport Comm. Center.	1. Sterile area location makes access difficult for some. 2. Somewhat larger facility is desired.
Harrisburg International Airport (MDT)	1. Ensured the main EOC was close to airport operations, bathrooms, and communication center; is in a separate building with ample parking. 2. Replacing desk top computers with new laptops which will open more work space. 3. All EOC personnel have checklists and are familiar with their responsibilities. 4. For small events use the alternate EOC inside the terminal at the police station. 5. Went to the county's EOC for a tour and a meeting regarding best practices for the EOC. Obtained helpful information and strengthened working relationship/awareness between county and airport. 6. We are anticipating a whiteboard and computer-assisted displays. That will be our working log/action tracker and will help with SA.	1. Challenged by nearby hazards: rail line is next to airport and Three Mile Island nuclear plant is a few miles away. 2. Encountered problems with IT during triennial exercise. We realized improvements were needed in terms of access to Wi-Fi, ease of using laptops, and addition of power cords in conference table itself. Decided to upgrade main EOC. 3. Airports need better training in emergency management for airport personnel and their roles. They need to learn what their role is and how to carry it out with other members of the team. The American Association of Airport Executives (AAAE) should consider setting up EOC training classes. 4. Key factors to planning most functional EOC are who is going to be there, what resources needed, and how will the room function to resolve the incident.
Tulsa International (TUL)	1. Access from sterile and non-sterile side. 2. Primary and secondary electrical feeds as well as backup generators. 3. Separate area designated for media updates.	1. Slightly less space than required.
Colorado Springs Airport (COS)	1. EOC on third floor of terminal—co-located with communication center. 2. EOC used to be on different floor—relocation jointly planned with ops, internal IT, electronic specialists, and a local architect. 3. Two ergonomic workstations; each has six monitors. 4. Badged access to elevator and to room. 5. For big incidents, plan to put other stakeholders in a separate conference room with plug-in phones and Wi-Fi capability. This would become the adjunct EOC. It is 22 yards away from the main EOC. 6. Review SOPs once a quarter so personnel do not have to look up their duties when activated. 7. You must plan five years out if you want to have the budget to do what you want.	1. Airport Ops and EOC overwhelmed communication center and vice versa. Had to install a barn-style door to separate functions. 2. High windows created a glare. Put adjustable blinds on windows and a dimmer switch for the lights.

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Table 4-4. Small non-hub airport best practices and lessons learned.

Small Non-Hub Airport	Best Practices	Lessons Learned
Bangor International Airport (BGR)	1. Camera connections between EOC and breakout rooms, vehicles for field view. 2. WebEOC connection with county and state. 3. Controlled access. 4. Proximity of management to staff prevents them from going to the field. 5. Configurable space, flexible IT infrastructure. 6. Enlarged FEMA forms laminated to write on. 7. Provide identifiable gear to prevent others from wearing their organization vest and having numerous ICs. 8. Used often so people are familiar with it. 9. Keep it simple; what is used daily works best in crisis.	1. Vulnerable location in the terminal. 2. Not big enough for a large event.
Lincoln Airport (LNK)	1. They train together and with their mutual aid agencies a lot. Everyone knows everyone and works well together. 2. They utilize the ICS structure very well. 3. They have three adaptive spaces to be used depending on the size of the event. 4. They are taking advantage of existing space coming available to turn it into an Operations and Unified Command Control Center (UCC). 5. They have practiced for large events taking advantage of training during the large air show every five years. 6. Each facility is access controlled. 7. Train often.	1. Noise levels in all three areas can be problematic. 2. The heating and cooling of the fire station bay the largest room is difficult to control. 3. When they are in the large room they are a bit far from the CP and if an immediate briefing needed to happen, it could be difficult to do a face-to-face meeting. 4. They must set up each facility and bring in any equipment except for the smallest room, which has phones and computers ready. The large facility for a big event takes time to set up.
Midland, Bay City and Saginaw - MBS International Airport (MBS)	1. Use the county EOC for large incidents or any incidents involving an airline. 2. Access controlled. 3. Immediately staff the EOC with one member in advance of inclement weather or any concerns outside of normal operations. 4. Always set up for an emergency so if it is used outside EOC operations, those users adjust to the room, as opposed to the other way around.	1. Lighting is not adequate. 2. Most users are standing yet the workstations are seated only.
Tallahassee Regional Airport (TLH)	1. City-owned airport so they can use the city's WebEOC as well as the county's. 2. EOC members notified via Incident Resource Inventory System (IRIS). It is user friendly and affordable—good for small airports. 3. Cut down audible tone on phones to reduce noise—use a light on the phone to notify when a call is coming in. 4. Made sure that bathrooms were close and easily accessible within the secure area. 5. Has tertiary backup for phone and radio communications. 6. Standing procedure is to conduct twice-daily updates/situation reports in the EOC. 7. Use the city's PIO staff to help; they schedule press conferences and control access. 8. Visual displays include connection to Sheriff Department's helicopter visual feeds for bigger picture of the impacted area. 9. Chose EOC location without windows due to security concerns. Also wanted to be next to communication center.	1. We needed to do a more comprehensive review of our needs because the facility is too small. Has only 240 sq. ft. plus a couple breakout rooms. Would prefer a more formal EOC with designated workstations. Must use duct tape to secure wires and line. Traffic flow is impeded. 2. Be careful with the press. Media placed a highly sensitive microphone not far from the EOC leaders and the press picked up conversations. The hot mic topic can be a very sensitive issue and should be highly monitored by airport staff during media interviews.

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Table 4-5. General aviation airport best practices and lessons learned.

General Aviation Airport	Best Practices	Lessons Learned
Addison Airport (ADS)	1. The building is very sustainable made of all concrete block. 2. They have updated new technology equipment including TVs, monitors, and projectors. 3. Their EOC is in the fire station so it is well stocked with supplies, kitchen, bathrooms, etc. 4. They use it consistently for monitoring weather events. 5. It easily accessible.	1. They do not have any formal access control. 2. Parking is limited. 3. They do not utilize the ICS structure enough. Many town Directors would not know their role. 4. They need a designated person to formalize their emergency procedures and organize support materials.
Spirit of St. Louis (SUS)	1. Several options for space include admin offices, terminal for large events, and a mobile command vehicle. 2. Controlled access plans for all options. 3. Identifiable gear.	1. There are no monitors for the admin space. 2. All logistics such as furniture, computers, etc. would have to be brought to the terminal from the fixed-base operator (FBO).
Prosser Airport (RQE)	1. They are working together with multiple agencies to find a suitable EOC facility. 2. They train together well.	1. They have a big airshow coming in 2018 and realized they do not have a good coordinated place to manage the event. 2. An EOC does not have to be just for airplane accidents.

Source: Faith Group, LLC.

Table 4-6. City/county EOCs best practices and lessons learned.

City/County EOCs	Best Practices	Lessons Learned
Northeast, State	1. Good on-going training and evaluation. 2. Detailed plans. 3. Separate breakaway rooms for conferencing. 4. Good leadership. 5. Very secure access.	1. Need to move Emergency Medical Services (EMS) Com/Dispatch to a separate room from main EOC. 2. Need to plan for relief shifts for longer duration events.
Northeast City, Big	1. Good working relationship with local/regional/state authorities. 2. Excellent plans. 3. Good training. 4. Strong leadership. 5. Positions designated per ICS/NIMS.	1. Space too small. 2. Location in city not optimal. Limited parking and far from other agencies. 3. Should have been on stand-by status. 4. Too many senior leaders on vacation at critical time.
Western City, Medium Size	1. Some good training. 2. Basic plans. 3. Good leadership.	1. Space not well utilized. 2. EOC weak; not customarily used to coordinate response; handled by agency-specific ops centers instead.
Southeast University, Large		1. Must pre-determine rapid-decision line of authority. 2. Delayed public warning/alert; must have robust public information capability. 3. No designated EOC with established ESFs. 4. No plan for coordination with local/regional/state authorities.

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The surveys were also utilized to discern the four airports, representing each size to serve as a Use Case to vet the draft sections of the Guidebook. Full Use Case reports may be requested through ACRP.

- Addison (ADS), Texas—GA;
- Bangor International Airport (BGR), Maine—Small Non-Hub;
- San Jose International Airport (SJC), California—Medium-Hub; and
- Large Airport (Anonymous)—Large-Hub.

The Hartsfield-Jackson Atlanta International Airport (ATL) was not a survey participant for the best practices lessons learned but has contributed the following quote, concerning the importance of an EOC, from a real-world event, power outage, which took place during the development of the Guidebook.

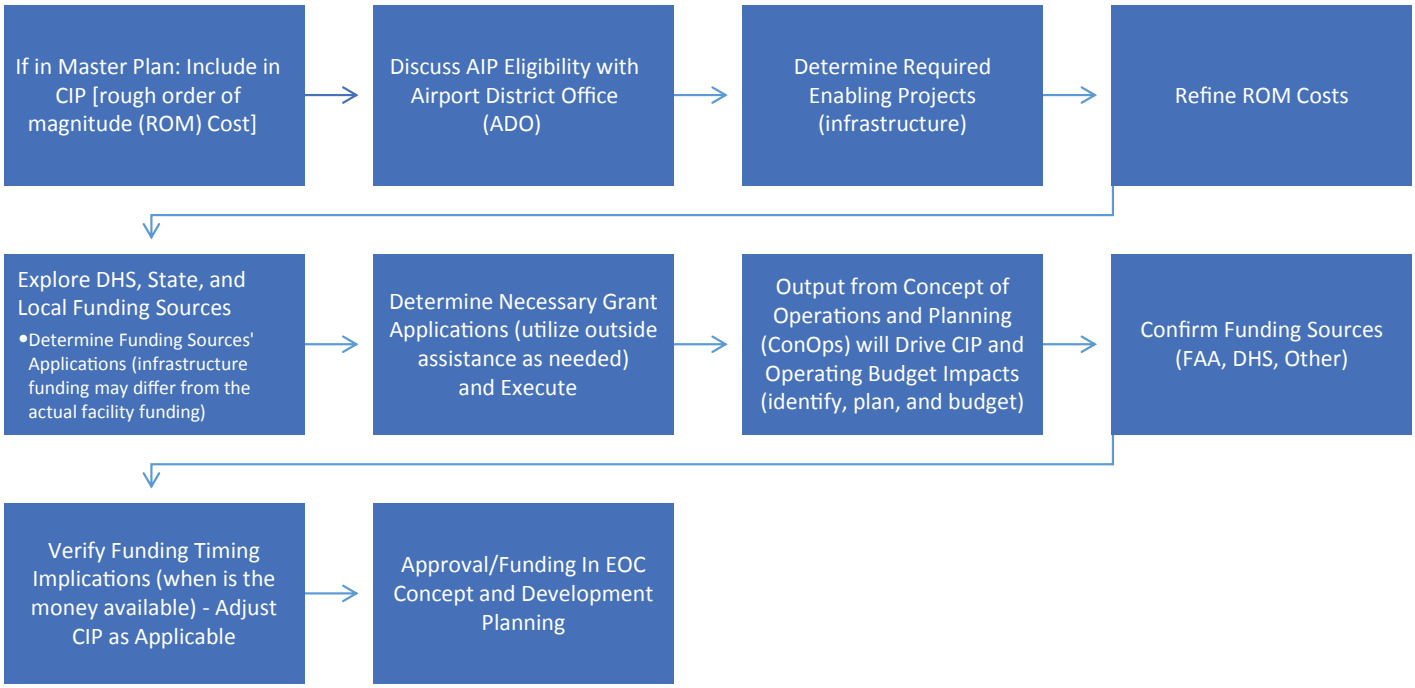
> On Dec. 17, 2017, Hartsfield-Jackson Atlanta International Airport (ATL) experienced a major power outage. While the Airport did not experience a total loss of power,* the event proved significant enough to warrant the activation of ATL's Emergency Operations Center (EOC) for coordination of response and recovery efforts.
>
> The EOC allowed Airport command staff, policy makers, elected officials, airlines and tenants to meet, share information and develop a coordinated response to the incident. EOC operational processes also facilitated quick recovery from the outage, as the Airport never closed and regained flight operations at 5 a.m. the following morning.
>
> The power outage offered many lessons learned for managing future incidents, including:
>
> (1) the need to communicate to Airport employees and passengers faster with increased frequency;
> (2) the need to invest in redundant power sources for communications systems;
> (3) the need to refine some emergency processes to ensure a more effective response; and
> (4) the need to mobilize all Airport stakeholders on site (airlines, concessions, facility operators, etc.) as a team of responders.
>
> As a result, the City of Atlanta Department of Aviation developed a comprehensive incident management training for Airport staff, airline and tenant employees. In addition, ATL will provide additional preparation through tabletop discussions, drills, and full functional exercises.
>
> The EOC proved to be a valuable asset for Hartsfield-Jackson, airlines and other Airport stakeholders as an information hub and center for operational and policy decisions.
>
> *During the Dec. 17 event at Hartsfield-Jackson, all airfield electrical systems remained on and life safety systems in terminal buildings and concourses were powered by backup generator power.

Budget

Airport managers first must determine their EOC needs before budgets can or should be set. Setting aside capital funds in an airport's Capital Improvement Program (CIP) is not unusual. Going through a deliberate and thoughtful process will provide airport operators with the information needed so that appropriate budget decisions can be made.

Budgeting for an EOC in most cases will be part of the CIP planning process. At most airports, the CIP contains the larger capital investment projects. Investing in the EOC, either through the development of a greenfield facility or the repurposing of existing space or something in between will most likely require a substantial enough investment to be included in the airport's CIP. Therefore, knowing how to budget for the EOC becomes an exercise in planning well in advance of the CIP being finalized. Within this section of the Guidebook, an EOC Concept and Development Planning workflow diagram is provided. In addition to that diagram, a funding decision and planning workflow diagram is also provided (see Figure 4-1). The funding decision workflow needs to be looked at as a concurrent process with the Concept and Development Planning. The intent of this information is to encourage airport operators to secure funding for the project as it is being planned.

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Source: Faith Group, LLC.

Figure 4-1. Funding decision workflow diagram.

Sources of Funding

AIP Funding

Sources of funding for an airport EOC include the FAA AIP, which compares and competes for discretionary dollars based on need and safety impacts, in addition to entitlement funds, which are based on enplanements. These funds can be used for site preparation and infrastructure, as well as building construction. According to the FAA Airports' AIP webpage: https://www.faa.gov/airports/aip/overview/, eligible projects include those improvements related to enhancing airport safety, capacity, security, and environmental concerns.

AIP funding must be coordinated through the airport's FAA Airport District Office (ADO) and be included in the 5-year AIP plan, which is a subset of an airport's 10-year CIP plan (typically).

State and Local Grants

Other sources of funding include state and local grants for emergency management. These sources are managed through state and local emergency management offices. Airport EOCs that are identified as an integral part of a state or region's emergency response system compete for grant funding. Further, there have been joint-use EOCs built using airport capital funds and county funds to develop an EOC located on airport property but utilized by county sheriffs' officers as a master station and communications center, for example, South Bend Airport (IN). Shared spaces often mean shared costs for airport operators.

FEMA

FEMA provides preparedness grants for emergency response and for EOCs that are associated with emergency response. The grant application process includes a checklist for vulnerabilities of an existing EOC to ensure that any grant funds are spent appropriately to improve existing conditions. Further, the Department of Homeland Security (DHS) puts out an annual

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memo that explains any nuances of the current year's grant program. Excerpts from the 2017 memo include (https://www.dhs.gov/news/2017/06/02/dhs-announces-funding-opportunity-fiscal-year-fy-2017-preparedness-grants):

> *WASHINGTON—Secretary of Homeland Security John Kelly today announced the release of Fiscal Year (FY) 2017 Notices of Funding Opportunity for 10 DHS preparedness grant programs totaling more than $1.6 billion. The grant programs provide funding to state, local, tribal, and territorial governments, as well as transportation authorities, nonprofit organizations, and the private sector, to improve the nation's readiness in preventing, protecting against, responding to, recovering from and mitigating terrorist attacks, major disasters and other emergencies . . .*
>
> *. . . Grant recipients are encouraged to use grant funding to maintain and sustain current critical core capabilities through investments in training and exercises, updates to current planning and procedures, and lifecycle replacement of equipment. New capabilities that are built using homeland security grant funding must be deployable if needed to support regional and national efforts. All capabilities being built or sustained must have a clear linkage to the core capabilities in the National Preparedness Goal.*

There are certain aspects of EOC design and construction that potentially would be eligible for grant support through FEMA funds that flow from the federal level through states and local jurisdictions. DHS/FEMA grants could be combined with FAA funding.

The Emergency Management Division of Michigan State Police produced EOC guidance for the state's local governments in the paper, *Design Recommendations and Criteria for Emergency Operations Centers*. In that document are helpful examples of allowable and non-allowable costs related to EOC construction, see Table 4-7. These give an idea of some of the costs that might be eligible for grant support, though airports would need to refer to the specific instructions of any grant source being considered for a final ruling.

Determine Costs

There are key decisions that must be made before budgeting for an EOC can take place. The first and most important decision is where the EOC will be located and to what extent the EOC needs to be upgraded. The following lists (starting with Figure 4-2 for Cost Considerations) are intended to be used as checklists for issues and items to be considered and or further investigated as needed. These lists are intended to build upon one another and be inclusive of each other. They are presented in this manner for ease of understanding and organization.

Table 4-7. Examples of allowable and non-allowable costs.

Allowable Costs	Non-allowable Costs
❑ Design fees	❑ Plumbing and electric
❑ Excavations for construction	❑ Landscaping
❑ Building shell construction and interior finishing	❑ Parking lots
❑ Modifications to existing building	❑ Construction of any space not related to emergency management operations
❑ Antenna and towers	❑ Space less than 50 square feet per person
❑ Heating, ventilation, and air conditioning (HVAC) equipment	❑ Equipment designed for daily non-emergency-related use
❑ Equipment designed for daily non-emergency use	❑ Maintenance
❑ Furniture for operations room	❑ Land purchase
❑ Radio/communications equipment	❑ Demolition
❑ Emergency generator	
❑ Kitchen/break room equipment	

Source: Emergency Management Division of Michigan State Police, Design Recommendations and Criteria for Emergency Operations Centers.

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Checklist: Cost Considerations

Site Selection—Modify Existing EOC Space

- ❑ Furniture
- ❑ Technology growth and upgrades
- ❑ Furnishings
- ❑ Modifications to the building, including access
- ❑ HVAC
- ❑ Power
- ❑ Other utilities

Site Selection—Greenfield Space

- ❑ Utilities in the area
- ❑ Site security
- ❑ New construction costs
- ❑ Threats and vulnerabilities (storm loads, as an example)
- ❑ Public access vulnerabilities

Site Selection—Move to a New but Existing Facility (See Existing EOC Costs)

- ❑ Who and how many organizations will be included (planned) in the EOC operations
- ❑ Growth needs and anticipated technology upgrades
- ❑ Inflation costs considered for current versus future spending

Operational functions

- ❑ What systems and functions are anticipated to take place when the EOC is activated
- ❑ What is the EOC concept for operations

Source: Faith Group, LLC.

Figure 4-2. Cost considerations.

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Design and Construction

The last important component of budgeting for the EOC is how the work will be designed and constructed. Several organization use a "design–build" model in place of the older "design-bid-build" model. These terms represent, in simplest terms, how the airport will hire the designers and contractors for the EOC construction and or redevelopment. A design–build approach enables the airport to hire a team that includes a contractor and a designer under one contract with the general contractor. The traditional design-bid-build model requires sequential steps so it can be more time consuming and produce more overhead costs.

Some airports have been utilizing a construction manager at risk (CMAR) model. The CMAR approach allows the airport to hire the general contractor who will work directly with the architects and engineers to ensure the facility is built and or reconditioned with a budget number as the not-to-exceed cost to the airport. Therefore, the general contractor is the one who is "at risk."

While a couple of these approaches to a capital project are relatively new to airports, they do provide an opportunity to streamline the process and allow the airport to have more direct control over the project scope, schedule, and budget. Of course, the airport's local procurement laws and regulations ultimately govern how capital projects can be procured.

Concept Planning

Concept planning needs to occur during the scoping and development process. Variations as to the size, style, location, etc. should all be considered as part of this process. Once a preferred concept is established, decisions can be made regarding findings. As with any project, EOCs need to be designed in phases or steps. Typical construction projects generally follow a 30%, 60%, 90%, and 100% document/design process. This translates very well to EOC concept planning and design. Each step in the decision process needs to be thoroughly thought through and executed to ensure enough information is carried over to the next step.

Most airports collaborate with other stakeholders during events. The needs of airlines, FAA, or FBI representatives in the EOC need to be considered. Some EOCs are joint-use facilities with local emergency management, police, or fire, for example. If the EOC will be used daily by more than just airport staff, a working group of interested agencies might need to be formed to review and vet design concepts. Joint-use facilities often benefit the airport because project funding can be shared. Organizing all the interested parties will help to facilitate buy-in and will help to ensure holistic and successful concept planning.

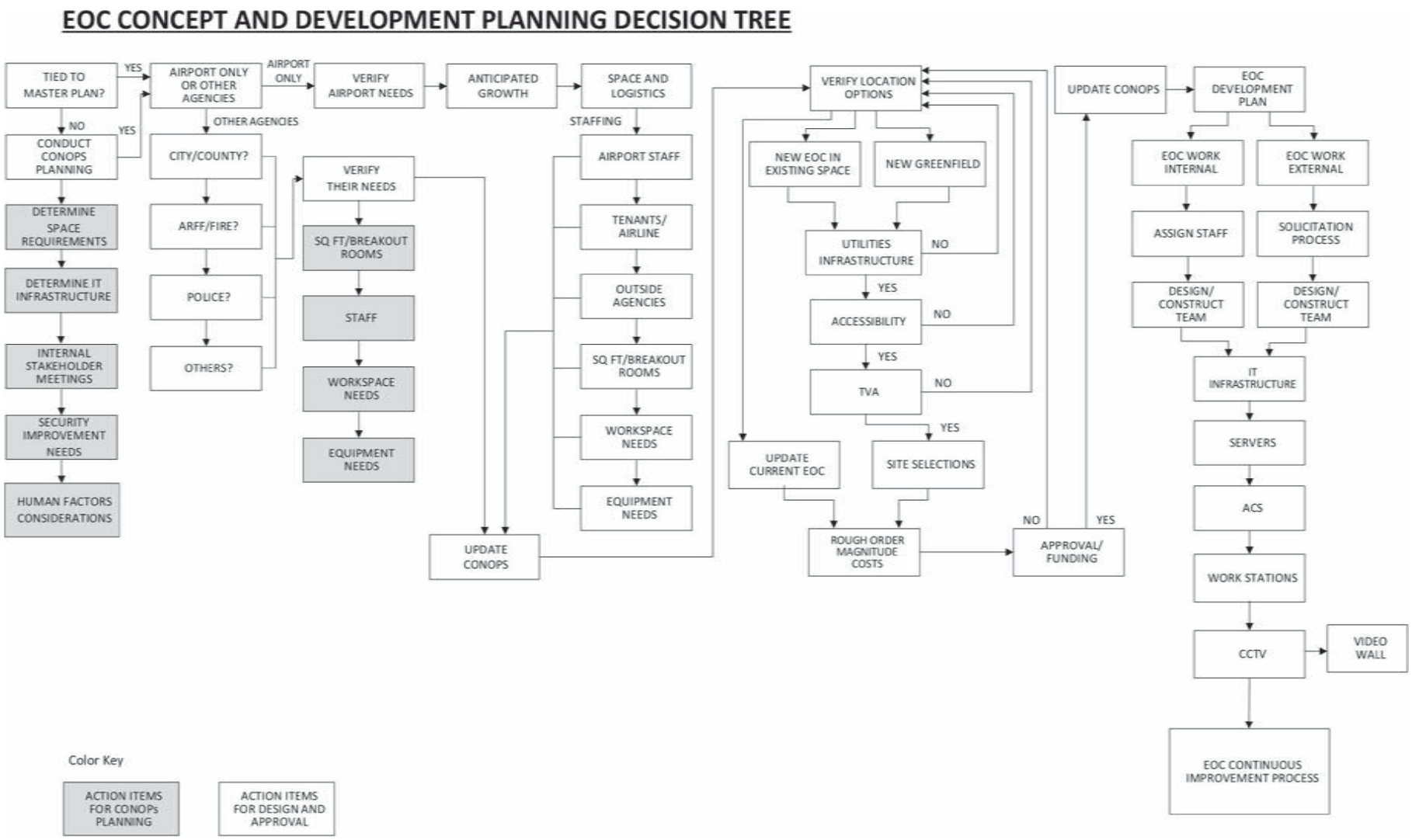
EOC Concept and Development Planning Decision Tree (Decision Tree)

This section contains two planning tools to help airports improve their EOC facility. One is a description and corresponding checklist of the tasks and decisions inherent in each of three EOC improvement options to help airports of any size and complexity navigate through the planning, budgeting, approval, and design process. The other is a Decision Tree that captures the processes in a visual illustration. The three potential EOC improvement options are:

- Update current EOC,
- Relocate to existing space, or
- Build a greenfield.

See Figure 4-3 for a Decision Tree depicting these steps.

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Source: Faith Group, LLC.

Figure 4-3. EOC concept and development planning Decision Tree.

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Update Current EOC. The airport has existing facility it is currently using as an EOC and they wish to make updates and improvements to the facility (see Figure 4-4 for associated checklist). Consider the following:

- If it is tied to the Master Plan, some high-level planning may have already been done and the airport can move forward to verify needs both internally and externally.
- If it is not tied to the Master Plan, the airport should consider conducting some Concept of Operations Planning (ConOps), which includes determining facility requirements, determining IT infrastructure needs, having discussions with internal stakeholders, and identifying needs for security improvements and human factors such as lighting, acoustics, and proper workstations.
- Once the Master Plan concepts or ConOps is completed, the airport can begin to consider the needs of outside agencies.
- Will the facility be a joint-use facility on a regular basis with other agencies such as fire or local emergency management, for example? If so, their needs must be verified for space, workstations, and equipment.
- If it is an airport-only EOC, the airport should determine the needs of all internal and external stakeholders who will operate at the center during emergencies such as the FAA, airlines, or FBI.
- Next, the airport would determine total number of staffing needs for internal and external stakeholders while considering anticipated growth, floor plan needs and size, workstations, equipment, and IT.
- It also is necessary to determine total number of staffing needs for internal and external stakeholders, floor plan needs and size, workstations, equipment and IT.
- Once personnel and space needs are determined and verified, the ConOps can be updated and the airport can continue toward verifying location options.
- If it is determined the airport will remain in the existing EOC, a rough order of magnitude (ROM) of costs needs to be documented.
- The next step is approval. If approval is achieved, the ConOps is updated again and the EOC development plan is put into action for design and implementation.
- During EOC plan development, either the selection of an external design–build contractor is executed, or internal staff are selected to work on the project.
- Elements of the project to be designed and implemented include IT infrastructure, server needs, ACS, workstations, and CCTV, which may or may not be tied to monitors or a video wall.

New EOC Utilizing Existing Space. The airport desires to move to a new but existing facility on or off the airport campus (see Figure 4-5 for associated checklist). Consider the following:

- If it is tied to the Master Plan, some high-level planning may have already been done and the airport can move forward to verify needs both internally and externally.
- If it is not tied to the Master Plan, the airport should consider conducting some Concept of Operations Planning (ConOps), which includes determining facility requirements, determining IT infrastructure needs, having discussions with internal stakeholders, and identifying needs for security improvements and human factors such as lighting, acoustics, and proper workstations.
- Once the Master Plan concepts or ConOps is complete, the airport can begin to consider the needs of outside agencies.
- Will the facility be a joint-use facility on a regular basis with other agencies such as, for example, fire or local emergency management? If so, their needs must be verified for space, workstations and equipment.

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Checklist: Update Current EOC

Following are steps that can be considered if the airport has a facility it is currently using as an EOC, and only wishes to make updates and improvements to that facility.

- ❑ ConOps or a Master Plan has been completed:
  - ❑ Space requirements have been determined
  - ❑ IT infrastructure needs have been accounted for (current and future)
  - ❑ Security features are included
  - ❑ Workplace ergonomics have been defined for best possible operations (lighting, acoustics, workstations, break room, restrooms, etc.)
  - ❑ Internal stakeholders have been consulted and involved in the planning
- ❑ Joint-use facility requirements have been verified by stakeholders for space, workstations and equipment, and appropriate accommodations
- ❑ Airport stakeholder (airlines, TSA, FAA, etc.) space requirements have been verified for work stations, equipment and appropriate accommodations
- ❑ Number of internal and external stakeholder staff expected
- ❑ Floorplan considerations
- ❑ Workstations, equipment, and IT
- ❑ Anticipated growth requirements
- ❑ ConOps Plan has been updated accordingly
- ❑ A rough order of magnitude (ROM) of costs to update current EOC has been calculated and documented
- ❑ The ROM is approved
- ❑ The ConOps has been updated again
- ❑ EOC design and implementation plans are initiated
- ❑ Either an external design–build contractor or internal staff are selected to carry out the project
- ❑ EOC update elements include these:
  - ❑ IT infrastructure
  - ❑ Server needs
  - ❑ ACS
  - ❑ Work stations
  - ❑ CCTV (including whether to tie this to monitors or a video wall)
  - ❑ Others

Source: Faith Group, LLC.

Figure 4-4. Update current EOC checklist.

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Checklist: New EOC Utilizing Existing Space

Following are steps that can be considered if the airport wants to move to a different facility on or off the airport campus.

- ❑ ConOps or a Master Plan has been completed:
  - ❑ Space requirements have been determined
  - ❑ IT infrastructure needs have been accounted for (current and future)
  - ❑ Security features are included
  - ❑ Workplace ergonomics have been defined for best possible operations (lighting, acoustics, workstations, break room, restrooms, etc.)
  - ❑ Internal stakeholders have been consulted and involved in the planning
- ❑ Joint-use facility requirements have been verified by stakeholders for space, workstations and equipment, and appropriate accommodations
- ❑ Airport stakeholder (airlines, TSA, FAA, etc.) space requirements have been verified for workstations, equipment and appropriate accommodations
- ❑ Number of internal and external stakeholder staff expected
- ❑ Floorplan considerations
- ❑ Workstations, equipment, and IT
- ❑ Anticipated growth requirements
- ❑ ConOps has been updated accordingly
- ❑ The new EOC facility has been evaluated and can accommodate the space, layout, and operational needs of the airport's EOC
- ❑ Other verifications have been completed regarding sufficiency of utilities and accessibility
- ❑ A threat and vulnerability assessment (TVA) has been conducted to rank survivability
- ❑ A rough order of magnitude (ROM) calculation has been completed for each of proposed sites
- ❑ The preferred site has been selected
- ❑ The ROM for the site and project are approved
- ❑ The ConOps is updated again
- ❑ EOC design and implementation plans are initiated
- ❑ Either an external design–build contractor or internal staff are selected to carry out the project
- ❑ EOC update elements include these:
  - ❑ IT infrastructure
  - ❑ Server needs
  - ❑ ACS
  - ❑ Workstations
  - ❑ CCTV (including whether to tie this to monitors or a video wall)
  - ❑ Others

Source: Faith Group, LLC.

Figure 4-5. New EOC utilizing existing space checklist.

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- If it is an airport-only EOC, the airport should determine the needs of all internal and external stakeholders who will operate at the center during emergencies such as the FAA, airlines, or FBI.
- Next, the airport would determine total number of staffing needs for internal and external stakeholders while considering anticipated growth, floor plan needs and size, workstations, equipment, and IT.
- Once personnel and space needs are determined and verified, the ConOps can be updated and the airport can continue toward verifying location options.
- If it is determined that the airport will want to use an existing facility (which will likely be modified to the tailored needs of an EOC), then other verifications need to be made for utility needs, accessibility, and finally, a Threat and Vulnerability Assessment (TVA) should be conducted for ranking survivability.
- Once these tasks are satisfactorily completed and the site selection(s) are finalized, a rough order of magnitude (ROM) of costs needs to be documented for each potential site (if more than one).
- The next step is approval. If approval is achieved, the ConOps is updated again and the EOC development plan is put into action for design and implementation.
- If the approval is not achieved, airport personnel return to verifying location options and proceed through the appropriate steps until approval for funding is achieved.
- Once approval is achieved, the ConOps is updated again and the EOC development plan is put into action for design and implementation.
- During EOC plan development, either the selection of an external design–build contractor is executed, or internal staff are selected to work on the project.
- Elements of the project to be designed and implemented include IT infrastructure, server needs, ACS, workstations, and CCTV, which may or may not be tied to monitors or a video wall.

Build a New Greenfield Space. The airport desires to construct a greenfield EOC (see Figure 4-6). Consider the following:

- If it is tied to the Master Plan, some high-level planning may have already been done and the airport can move forward to verify needs both internally and externally.
- If it is not tied to the Master Plan, the airport should consider conducting some Concept of Operations Planning (ConOps), which includes determining facility requirements, determining IT infrastructure needs, having discussions with internal stakeholders, and identifying needs for security improvements and human factors such as lighting, acoustics, and proper workstations.
- Once the Master Plan or ConOps is complete, the airport can begin to consider the needs of outside agencies.
- Will the facility be a joint-use facility on a regular basis with other agencies such as fire or local emergency management? If so, their needs must be verified for space, workstations, and equipment.
- If it is an airport-only EOC, the airport should determine the needs of all internal and external stakeholders who will operate at the center during emergencies such as FAA, airlines or FBI for example.
- Next, the airport should determine total number of staffing needs for internal and external stakeholders while considering anticipated growth, floor plan needs and size, workstations, equipment, and IT.
- Once personnel and space needs are determined and verified, the airport should update the ConOps and continue toward verifying location options.
- If it is determined the airport will continue with building a greenfield, then other verifications need to be made for utility needs, accessibility, and finally, a TVA should be conducted for ranking survivability.

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Checklist: Build a New Greenfield Space

Following are steps that can be considered if the airport wants to construct a greenfield EOC.

- ❑ ConOps or a Master Plan has been completed:
  - ❑ Space requirements have been determined
  - ❑ IT infrastructure needs have been accounted for (current and future)
  - ❑ Security features are included
  - ❑ Workplace ergonomics have been defined for best possible operations (lighting, acoustics, workstations, break room, restrooms, etc.)
  - ❑ Internal stakeholders have been consulted and involved in the planning
- ❑ Joint-use facility requirements have been verified by stakeholders for space, workstations and equipment, and appropriate accommodations
- ❑ Airport stakeholder (airlines, TSA, FAA, etc.) space requirements have been verified for workstations, equipment, and appropriate accommodations
- ❑ Number of internal and external stakeholder staff expected
- ❑ Floorplan considerations
- ❑ Workstations, equipment and IT
- ❑ Anticipated growth requirements
- ❑ ConOps has been updated accordingly
- ❑ Other verifications have been completed regarding sufficiency of utilities and accessibility
- ❑ A TVA has been conducted to rank survivability
- ❑ A rough order of magnitude (ROM) calculation has been completed for each potential site
- ❑ The preferred site has been selected
- ❑ The ROM for the site and project are approved
- ❑ The ConOps is updated again
- ❑ EOC design and implementation plans are initiated
- ❑ Either an external design/build contractor or internal staff are selected to carry out the project
- ❑ EOC design elements include these:
  - ❑ IT infrastructure
  - ❑ Server needs
  - ❑ ACS
  - ❑ Workstations
  - ❑ CCTV (whether to tie this to monitors or a video wall)
  - ❑ Others

Source: Faith Group, LLC.

Figure 4-6. Build a new greenfield space checklist.

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- Once these tasks are satisfactorily completed and the site selection(s) are finalized, a ROM of costs needs to be documented for each potential site (if more than one).
- The next step is approval. If approval is achieved, the ConOps is updated again and the EOC development plan is put into action for design and implementation.
- If the approval is not achieved, return to verifying location options and proceed through the appropriate steps until approval for funding is achieved.
- Once approval is achieved, the ConOps is updated again and the EOC development plan is put into action for design and implementation.
- During EOC plan development, select project delivery method, designer, and contractor to carry out the project.
- Elements of the project to be designed and implemented include IT infrastructure, server needs, ACS, workstations, and CCTV, which may or may not be tied to monitors or a video wall.

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SECTION 5

Site Selection

Introduction

This section of the Guidebook builds on the planning process outlined in Section 4, which described three options for EOC improvements: update a current EOC within existing space, relocate to an existing facility, or build a new greenfield site.

There are many factors to consider when evaluating existing or new sites, including topography and environmental constraints along with adjacent conditions involving any existing and planned support facilities such as operations offices, including fire departments and communications centers. Environmental protection, health, safety and zoning code requirements also come into play. Other considerations include noise protection from adjacent land use or flight approach and takeoff corridors, and the ability to provide a safe and sustainable environment during a crisis. In addition, an assessment of surrounding vulnerabilities, overlaid with the national terrorist profile will need to be taken into consideration. Alternate site locations that can provide minimum redundant infrastructure should be considered in the evaluation.

The following sections provide planning considerations to help airports of all sizes and complexities address the adequacy of their current EOC, as well as aid in the identification and evaluation of new sites. Planning considerations will focus on selection criteria for both existing and new sites including:

- Accessibility,
- Serviceability,
- Safety and security,
- Size,
- Environmental constraints,
- Versatility and adjacency, and
- Resiliency and TVA.

A site evaluation checklist (Figure 5-6) is provided at the end of this section, to help airport operators and their various stakeholders make informed decisions regarding site selection.

Site Evaluation

The primary goal of any EOC site, whether existing or new, is to provide a safe and operational effective area for the emergency management staff to carry out their duties and responsibilities. The key evaluation components for existing and new sites are described below. Section 4 provided a Decision Tree to help airports choose among the three potential site scenarios. For scenarios where a new site will be considered, conduct and document independent site visits, and rank them based on their ability to meet the site evaluation criteria outlined.

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Existing Sites and Buildings

If the Decision Tree identifies the EOC to be updated or relocated to an existing site, the guidelines to consider include the following.

Accessibility

The existing building or site should be readily accessible in the event of an emergent event for airport personnel and various stakeholders involved in the emergency response management and support.

- **Public Roadway Access—**For EOCs located outside of airport property-controlled access points, assess the site for quick and convenient access from the public roadway system, while also surveying potential congested areas that could hinder access. Survey the existing site for roadways that may become flooded during heavy periods of rain or buried by drifting snow. Road access that is elevated above low-lying areas helps avoid such incidents during inclement weather. Review the site against routes that may permit unimpeded access to the site, such as those that cross rail lines or those areas with low underpass clearances that may not permit large emergency vehicles to access the site. While limiting access to the EOC through a single controlled point of entry enhances security of the site, consider alternate points of access and egress should the primary become compromised. Additionally, a separate service access away from the main point of entry may be desirable.[1]
- **Airfield and Perimeter Road Access—**For sites being considered within the airport secured perimeter, assess the building's location from the entry point for long routes around the airfield system that could hinder access response time. Locate the building far enough inside the airport perimeter fence to enhance security effectiveness, supported by a large open space between the building and the perimeter access road or fence to mitigate the effect of potential blast effects.[2]
- **Parking—**Provide ample, well-lit parking in a secured environment with enough capacity to serve all stakeholder requirements, such as small vehicles, buses, mobile command units, and large emergency response vehicles such as those used by the fire department.
- **Existing Building Access—**If the EOC site is within the existing building such as a terminal, wayfinding and signage will be critical for personnel to quickly access the area. If the area is on the SIDA side of the terminal or within the existing Airport Operations Area (AOA) requiring badged or permit access to the site, appropriate measures to allow personnel secure access to these areas will need to be considered. Additionally, assess the location for ADA compliance. Section 6 provides additional information on building accessibility.
- **Counter-Surveillance—**Assess views of the building from the non-secure side of the site; e.g., does the building have unrestricted visual access from street corners, is its purpose obvious, can potential surveillance be detected via CCTV or security patrols?

Serviceability and Infrastructure

To ascertain the ability of the existing site or building to support the required infrastructure needed to access and operate an EOC, conduct a survey of existing site conditions while considering the following:

- The site must support required infrastructure such as water, power, gas, fiber cabling, and communications systems. While Ethernet provides the most reliable, stable, and fast source for LAN communication, Wi-Fi provides freedom of travel around the building and site. However, consider evaluating Wi-Fi for inside and outside sources that may interfere with network stability. Likewise, for voice communication, evaluate potential cellular interference from nearby structures or existing building systems. Ideally, the existing building or site will be located away from radio interference or other tall metal structures,[3] which may interfere with communications requirements.[4]

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- Assess the building for convenient and adequate access to fire hydrants.[5]
- Review the building or site for critical redundant utility systems that may be needed to support site security, life safety, and rescue functions in the EOC.
- Assess the building location for potential vulnerabilities to wind-blown debris.[4]
- Ideally, existing structures being considered for an EOC will be located on higher ground to avoid potential flooding.
- EOCs being considered on upper floors of an existing building must provide adequate vertical circulation such as stairs and elevators, with assessment of how they might be compromised during severe weather events or a fire in the building. Higher floors could also be more susceptible to windstorm damage depending on the location of the building.[4]
- If desired, assess the site or existing building for opportunities to implement sustainable design measures such as Leadership in Energy and Environmental Design (LEED). At a minimum, consider best practices associated with sustainable design.
- Consider conducting a noise analysis to ensure adjacent surroundings will not affect the operation of the EOC during an emergency crisis and, if so, what steps will be taken to mitigate the issue.
- Assess and review existing building systems (HVAC access and service, electrical redundancy, wall, ceiling, floor engineering, etc.). Systems, to the extent possible, should be compartmentalized or segregated from other building systems.

Safety and Security

Safety and security are critical to the site functioning and remaining secure during emergent situations. Security for the existing site or building is discussed in the TVA section. Consider the following principles[6] outlined below when evaluating the existing site's ability to protect the EOC from outside threats.

- **Wayfinding and signage—**Wayfinding is important to the security of the site. Measures need be taken to reduce the number of persons with knowledge concerning sensitive areas of the site such as communications or utility areas. A means to provide stakeholders with the required information to access the EOC will be established and distributed on a need-to-know basis only.
- **Controlled access—**Ideally, existing sites will have pull-off areas at the entrance for vehicle checks if parking is set within a controlled access area. Avoid existing buildings with perpendicular or straight-line of vehicle access. Buildings sited with parallel access mitigate the potential for vehicles to ram the building from the entrance. Study the site for existing service access and its location with respect to potentially affecting critical functions of the building, such as utilities, during a blast event.
- **Parking—**In review of existing vehicle access, remove all "under building" or inside drives from consideration, and assess the addition of structures that support keeping vehicles on the designated roadway: e.g., curves, curbs, trees, berms, speed bumps, etc., while allowing for emergency response access if needed. At a minimum, consider inspecting vehicles prior to entering the underground parking area. For any parking areas adjacent to the controlled EOC site that are utilized for general parking, review existing setbacks to minimize the effects of potential vehicle-borne improvised explosive devices (VBIEDs).
- **Lighting—**Evaluate existing lighting for its effectiveness to help prevent or minimize potential intruders from gaining undetected access to the controlled area of the site or building. Consider backup or emergency power as practical. Critical areas to evaluate include any essential utility pathways, including power, communications, HVAC, and water distribution.
- **Weather protection—**Assess the existing site against potential flooding and ponding due to severe weather if located in a depressed area within the airport surroundings. Analyze an existing building for proper structural protection against earthquakes and high wind events such as those from tornadoes and hurricanes.

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Size

The size of the site or existing building plays a key role in the ability of the EOC to provide all the necessary functions and supporting infrastructure, including safety and security, needed to operate the facility. Section 7 provides information on size and layout requirements for the building. Consider the following when evaluating the existing site or building for the location of the potential EOC:

- Identify all stakeholders to ascertain the site's ability to support the required building envelope and necessary parking spaces.
- The initial driver to design will be the intended ConOps of the facility—how it will work, who will be there, policies and procedures during an event response—then design to support that function.
- Assess the site or building for flexibility to expand the EOC, if required, beyond its current envelope, including the associated infrastructure requirements needed to operate a larger facility as well as security enhancements required for proper protection.
- Assess the site for adequate clear zones and setbacks from the perimeter, which provide necessary security for the building.
- The site must be clear from runway protection zone (RPZ) penetrations.

Versatility and Adjacency

Flexibility is key to successful EOC site planning including the ability to accommodate emerging technologies and future functional requirements. See Section 7 for additional information on building flexibility and adaptability. Consider the following factors to ascertain the existing site's ability to provide the following:

- If the EOC will be co-utilized with different non-emergency functions, conduct a study to determine if separate secured access and increased security such as wall hardening is required and any impacts to the other tenants.
- If the facility will be co-utilized or shared with other stakeholders such as Operations, Emergency Management, fire, police, etc., a survey of the existing building's area will be required to ensure proper space is available to meet the needs of all the stakeholders' programmatic requirements.
- Assess locations within the non-secure portions of the terminal for any potential security vulnerabilities, especially various attack vectors using public space.
- Make dedicating an EOC facility a goal so that there can be quick set up and start up for EOC functions during crisis management. See Section 7 for additional information on co-utilization.
- Study buildings under consideration for an EOC remote from the main stakeholder locations for quick and convenient access during emergency situations. See Section 7 for additional information on adjacencies.

New or Greenfield Sites

If the Decision Tree referenced in Section 4 identifies a new site, guidelines to consider include the following.

Accessibility

- **Public roadway access—**For EOCs located outside of Airport controlled access points, assess the site for quick and convenient access from the public roadway system while also avoiding congested areas that could hinder access. Avoid placing the EOC off roadways that may become flooded during heavy periods of rain or buried by drifting snow. Elevate new

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access roads above low-lying areas to avoid such incidents during inclement weather. Consider locating the EOC along routes that permit unimpeded access to the site. Avoid roads that cross rail lines or those with low underpass clearances as they may impede access or not permit large emergency vehicles to access the site. While limiting access to the EOC through a single controlled point of entry enhances security of the site, consider an alternate point of access should the primary become compromised.[1]

- **Airfield and perimeter road access—**For sites within the airport secured perimeter (the AOA) consider locating the EOC in an area that avoids long routes around the airfield system from the point of entry. Locate the EOC far enough away from the airport perimeter fence to meet security stand-off guidelines. Larger open spaces between the EOC and the perimeter access road or fence will mitigate the effects of potential threats or attacks against the facility.[2]
- **Parking—**Provide ample, well-lit parking in a secured environment with enough vehicle capacity to serve all stakeholder requirements, including small vehicles, buses, mobile command units, and large emergency response vehicles used by fire departments.
- **Building access—**If the area is on the SIDA side of the terminal or within the existing AOA that require badged or permit access to the site, appropriate measures will need to be taken to allow personnel secure access to these areas. Additionally, assess the location for ADA compliance concerning vehicular parking and building access. Section 7 provides additional information on Building Accessibility.
- **Counter-surveillance—**Assess views of the building from the non-secure side of the site; e.g., does the building have unrestricted visual access from street corners, is its purpose obvious, can potential surveillance be detected via CCTV or security patrols?

Serviceability and Infrastructure

To determine the ability of the new site to support the required infrastructure needed for access, operation, and support functions of the EOC, consider the following.

- Adequate, convenient, and quick access to the site or building is critical during an emergency crisis.
- The site must support required infrastructure such as water, power, gas, HVAC, fiber cabling, and communications systems. While an ethernet provides the most reliable, stable, and fast source for LAN communication, Wi-Fi provides freedom of travel around the building and site. However, consider evaluating Wi-Fi for both inside and outside sources that may interfere with network stability, which is critical for data communication during emergency situations. Likewise, for voice communication, evaluate potential cellular interference from nearby structures or existing building materials. Ideally, the new site will be located away from radio interference and tall metal structures.[4]
- Assess the site for convenient and adequate access to fire hydrants in case of fire.[3]
- Review the site for critical redundant utility systems.
- Assess the new building location for potential vulnerabilities to wind-blown debris.[4]
- Ideally, the new EOC facility will be located on higher ground to avert potential flooding.
- If desired, assess the site for opportunities to implement sustainable design measures such as LEED. At a minimum, consider best practices associated with sustainable design.
- Consider conducting a noise analysis to determine if adjacent surroundings have the potential to affect the operation of the EOC and, if so, what steps will be taken to mitigate the noise issue.

Safety and Security

Safety and security are essential in ensuring the site will function and remain secure during emergent situations. Security of the site will be discussed in the TVA section. Consider the following principles[5] outlined below when evaluating the site's ability to protect the EOC from outside threats.

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- **Wayfinding and signage—**Wayfinding is important to the security of the site. Measures need be taken to reduce the number of persons with knowledge concerning sensitive areas of the site such as communications or utility areas. A means to provide stakeholders with the required information to access the EOC will be established and distributed on a need-to-know basis only.
- **Controlled access—**Consider a pull-off area at an entry point with adequate stand-off distance for vehicle checks, if parking is set within the controlled access area. Avoid perpendicular or straight-line of access to the new site while considering parallel access that mitigates the potential for vehicles to ram the building from the site entrance. Study the new site for service access and its location with respect to potentially affecting critical functions of the building, such as utilities, from a blast event. Avoid access drives into or under the building. Other site design techniques that help keep vehicles within the roadway include curbs, trees, berms, speed bumps, etc. These measures, however, will need to be studied to avoid interfering with emergency response times.
- **Parking—**If possible, prohibit parking in areas below the building. If the EOC is within a secured zone, limit interior parking to employees. Provide proper setbacks from the general parking areas to mitigate potential blast effects of VBIED.
- **Lighting—**Evaluate lighting plans for effectiveness to prevent or detect potential intruders from accessing the controlled area of the site or building. Consider backup lighting or emergency power as appropriate. Cover critical areas with lighting, including site access, building perimeter areas, any essential utility locations such as power, communications, and water distribution.
- **Weather protection—**Avoid new sites at risk for flooding or excessive ponding. Provide the new building with proper structural protection against earthquakes, and high wind events such as those from tornadoes and hurricanes.

Size

The size of the site plays a key role in the ability of the EOC to provide all the necessary functions and supporting infrastructure, including safety and security needed to operate the facility. Section 7 provides information on size and layout requirements for the building itself. Consider the following when evaluating the new EOC site:

- Identify all stakeholders to ascertain the site's ability to support the required building envelope and supporting parking spaces.
- The initial driver to design will be the intended ConOps of the facility—how it will work, who will be there, policies and procedures during an event response—then design to support that function.
- Assess the site for flexibility to expand the EOC if required beyond its planned envelope, including the associated infrastructure requirements needed to operate a larger facility as well as security enhancements required for proper protection.
- Assess the site for adequate clear zones and setbacks from the perimeter, which provide necessary security for the building.
- The site must be clear from RPZ penetrations.

Versatility and Adjacency

Flexibility is key to successful EOC site planning including the ability to accommodate emerging technologies and future functional requirements. See Section 7 for additional information on building flexibility and adaptability. Consider the following factors to ascertain the new site's ability to provide the following:

- If the EOC will be co-utilized with different non-emergency functions, conduct a study to determine if separate secured access and increased security such as wall hardening is required and any impacts to the other tenants.

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- If the facility will be co-utilized or shared with other stakeholders such as Operations, Emergency Management, Fire, Police, etc., a survey of the existing building's area will be required to ensure proper space is available to meet the needs of all the stakeholders' programmatic requirements.
- Assess locations within the non-secure portions of the terminal for any potential security vulnerabilities, especially various attack vectors using public space.
- Non-dedicated space will promote quick set up and start up for EOC functions during crisis management. See Section 7 for additional information on co-utilization.
- Study remote sites under consideration for an EOC from the main stakeholder locations for quick and convenient access during emergency situations. See Section 7 for additional information on adjacencies.

TVA

This process may be used when analyzing site alternatives from a facility TVA and risk perspective to help the airport narrow down the selection process to the optimum location from a TVA perspective. When considering risk factors of threat (likelihood of that threat and extent of damage or impact) associated with the facility vulnerability, this approach endeavors to reach an overall risk ranking for the use of airport managers.

The analysis should consider location and resilience of potential sites under consideration, the operational characteristics and a set of assumptions associated with security measures envisioned for each. The specific vulnerability analysis against the following attack scenarios is recommended:

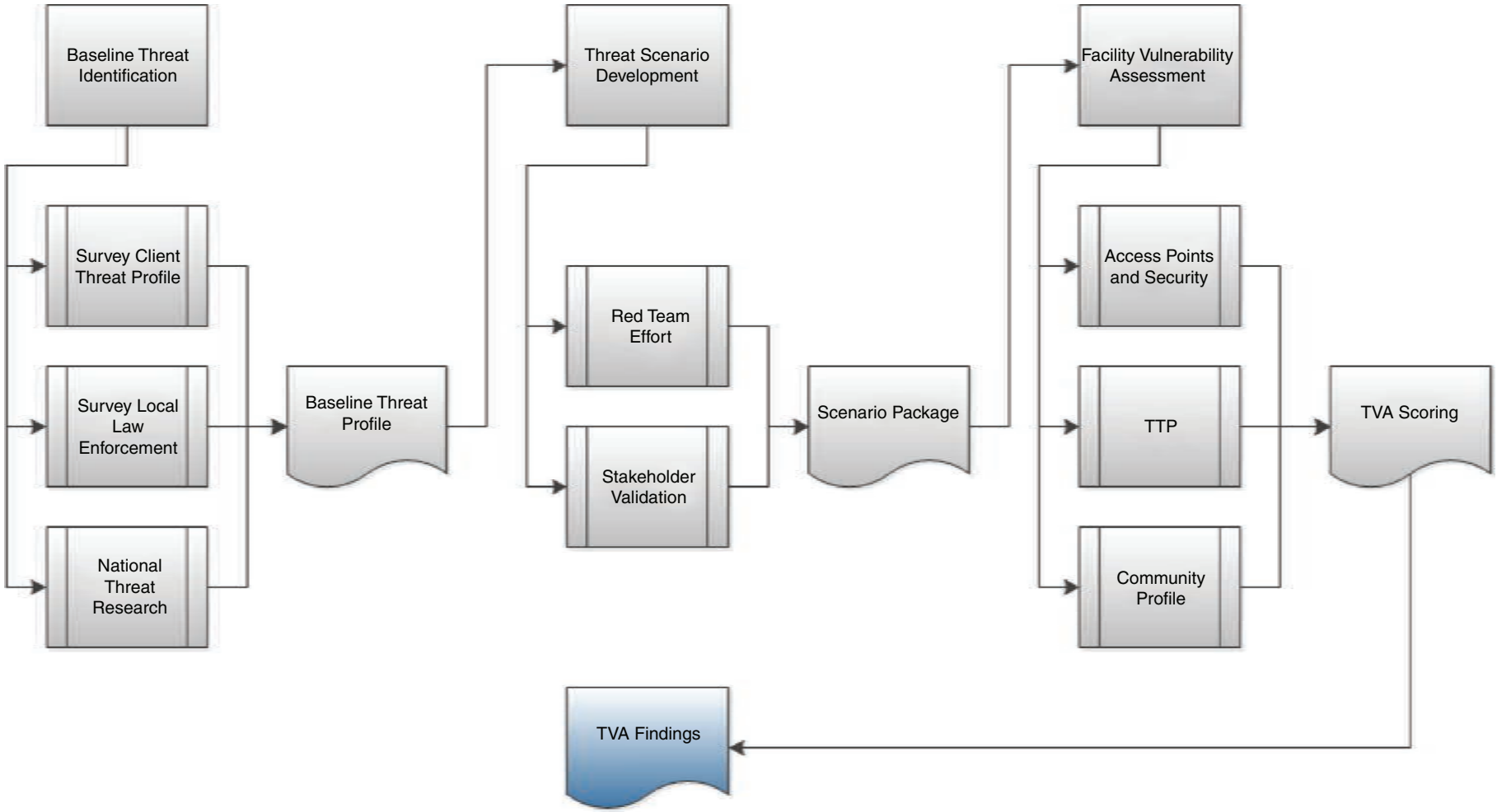
- Active Shooter (including coordinated armed assault, hostage takers, and lone wolf attacks);
- VBIED (including IEDs placed on the target and suicide bombers);
- Chemical, Biological, Nuclear (CBN) (including a radiological dispersing device); and
- Natural Causes (earthquake, tornado, etc.).

This approach applies a recognized methodology used by practitioners in DHS and other federal, state, and local agencies for human-caused threats. The approach necessarily involves airport law enforcement, adjacent law enforcement agencies (including state and local, FBI, TSA, and CBP, as appropriate), as well as emergency responders, ARFF, adjacent fire and EMS responders, and state emergency management.

This method attempts to identify potential threats at a high level, considering both national-level threat assessments and any reasonable, localized criminal activities that may be associated with or modeled with attack vectors on the facility. Once the threats can be bucketed into categories, scenarios will be developed, again at a high level, to assist in determining how they may be executed in the existing environs at the airport. This process is known as "Red Teaming," which assigns the task of formulating reasonable attack profiles against the facility. The final piece of the threat assessment will be to determine the risk of such an attack, weighing *impact* of the attack against *likelihood* to determine risk and considering known practices already in place called Tactics, Techniques, and Procedures (TTP). Figure 5-1 provides an overview of the TVA process, covering the steps necessary to successfully measure risk associated with each potential site.

Conduct site surveys to determine vulnerability issues resident with each consideration, using the scoresheet shown in Figure 5-2. Conduct these assessments with the background information derived from local SMEs, research of local criminal profiles, and national antiterrorism/counterterrorism documents, allowing the assessor to view the vulnerability assessment through the attacker's eyes. Finally, the process delivers a high-level rating that allows the user to apply results to decision making and displays scenario-specific vulnerabilities for each site.

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Source: Faith Group, LLC.

Figure 5-1. TVA process.

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VULNERABILITY ASSESSMENT SCORESHEET

SITE/TARGET NAME:	TOTAL SCORE RATING:	Value

1. Level of Visibility:
Assess the awareness of the existence and visibility of the target to the public.

- **0 - Invisible:** Existence secret/Classified location
- **1 - Very Low Visibility:** Existence not publicized
- **2 - Low Visibility:** Existence public but not well known
- **3 - Medium Visibility:** Existence known locally
- **4 - High Visibility:** Existence known regionally
- **5 - Very High Visibility:** Existence known nationally

2. Criticality of Target Site to Jurisdiction:
Assess usefulness of assets to local population, economy, government, etc. Potential targets deemed essential to the continuity of the jurisdiction.

- **0 -** No usefulness
- **1 -** Minor usefulness
- **2 -** Moderate usefulness
- **3 -** Significant usefulness
- **4 -** Highly useful
- **5 -** Critical

3. Impact Outside the Jurisdiction:
Assess the effect that loss would have outside of the jurisdiction.

- **0 -** None
- **1 -** Very Low
- **2 -** Low
- **3 -** Medium
- **4 -** High
- **5 -** Very High

4. Potential Threat Element (PTE) Access to Target:
Assess the availability of the target for ingress and egress by a PTE.

0 - Restricted: Security patrol 24/7, fenced, alarmed, Video Surveillance System (VSS), controlled access requiring prior clearance, designated parking, no unauthorized vehicle parking within 300 feet of facility, protected air/consumable entry.

1 - Controlled: Security patrol 24/7, fenced, alarmed, controlled access of vehicles and personnel, designated parking, no unauthorized vehicle parking within 300 feet of facility, protected air/consumable entry.

2 - Limited: Security guard at main entrance during business hours, fenced, alarmed, controlled access of visitors, designated parking, no unauthorized vehicle parking within 300 feet of facility, protected air/consumable entry.

3 - Moderate: Controlled access of visitors, alarmed after business hours, designated parking, no unauthorized vehicle parking within 50 feet of facility, protected air/consumable entry.

4 - Open: Open during business hours, locked during non-business hours, unprotected air/consumable entry.

5 - Unlimited: Open access, unprotected air/consumable entry.

5. Potential Target Threat of Hazard:
Assess the presence of legal Weapons of Mass Destruction (WMDs) material, Chemical, Biological, Radiological and Nuclear (CBRNE) in quantities that could be the target of a terrorist attack or would complicate the response to an incident at that facility.

0 - None: No WMD materials present.

1 - Minimal: WMD materials present in moderate quantities, under positive control, and in secured locations.

2 - Low: WMD materials present in moderate quantities and controlled.

3 - Moderate: Major concentrations of WMD materials that have established control features and are secured in the site.

4 - High: Major concentrations of WMD materials that have moderate control features.

5 - Very High: Major concentrations of WMD materials that are accessible to non-staff personnel.

6. Potential Target Site Population Capacity:
Assess the maximum number of individuals at a site at any given time.

- **0 -** 0
- **1 -** 1-250
- **2 -** 251-5,000
- **3 -** 5,001-15,000
- **4 -** 15,001-50,000
- **5 -** > 50,001

Figure 5-2. Vulnerability assessment scoresheet.

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VULNERABILITY ASSESSMENT SCORESHEET

SITE/TARGET NAME:	TOTAL SCORE RATING:	
		Value
7. Potential for Collateral Mass Casualties: Assess potential collateral mass casualties within a one-mile radius of the target site. **0 -** 0-100 **1 -** 101-250 **2 -** 251-5,000 **3 -** 5,001-15,000 **4 -** 15,001-50,000 **5 -** > 50,001		
RAW SCORE (Add Lines 1-7)		
Basic Target Vulnerability Assessment Rating: Convert total score to a rating number from 1-12 using the following key. Transfer the final rating to top right-hand box in this form. 0 - 2 pts = 1 3 - 5 pts = 2 6 - 8 pts = 3 9 - 11 pts = 4 12 - 14 pts = 5 15 - 17 pts = 6 18 - 20 pts = 7 21 - 23 pts = 8 24 - 26 pts = 9 27 - 29 pts = 10 30 - 32 pts = 11 33 - 35 pts = 12		

Source: Faith Group, LLC.

Figure 5-2. (Continued).

The more common attack vectors utilized at airports by terrorists in terms of attack trajectories tend to remain in the realm of low-complexity/high-impact events. IEDs, whether suicide, stand-alone, VBIED, or a combination of delivery methods, or small arms assaults/active shooter attacks, remain the vector of choice for terrorist/extremist operatives. The Guidebook will first provide an approach to dealing with the IED/VBIED threats.

From the IED/VBIED perspective, this TVA approach assumes that all sites require a standard package of blast protection and a 300 foot stand-off distance from any public parking area, secured parking for non-badged responders and visitors, a hardened perimeter utilizing the airport fencing and gate standards guidelines, such as K12 rated[7] gate drop arms (see Table 5-1); that each site should have the necessary CCTV and ACS in place to provide the requisite level of security; and that each facility should be outfitted with blast mitigation building systems, including window glazing, limiting windows along public roadways, blast deflection wall systems, and designs that allow for blast wave compression and management.

When discussing stand-off distances, several sources point to safe distances for parking structures, passenger terminals (PAX), large gathering spots, or other potential locations where a large

Table 5-1. Department of State barrier ratings K12 rated gate drop arms.

DoS Rating	Speed of Vehicle At Impact in mph (kph)	Kinetic Energy	Max. Allowable Penetration of Vehicle
K12	50 mph (81 kph)	1,250,000 ft-lbf (178,812 kgf-m)	
K8	40 mph (64 kph)	800,000 ft-lbf (110,600 kgf-m)	
K4	30 mph (48 kph)	450,000 ft-lbf (62,212 kgf-m)	
L3			3 ft (0.91 m)
L2			3 to 20 ft (0.91 to 6.1 m)
L1			20 to 50 ft (6.1 to 15.2 m)

* Based on 15,000-lb (6,818-kg) vehicle weight

Source: Department of State standard and found in TSA Airport Design Guidelines, 2011.

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Table 5-2. TSWG blast effects chart.

Type of Explosive	Explosive Capacity in TNT Equivalents	Lethal Air Blast Range
Pipe Bomb	5 lbs. (2.3 kg)	
Briefcase, Backpack, or Suitcase Bomb	50 lbs. (23 kg)	
Compact Sedan (in trunk)	500 lbs. (227 kg)	100 ft. (30 m)
Full Size Sedan (in trunk)	1,000 lbs. (454 kg)	125 ft. (38 m)
Passenger or Cargo Van	4,000 lbs. (1,814 kg)	200 ft. (61 m)
Small Box Van (14 ft. box)	10,000 lbs. (4,536 kg)	300 ft. (91 m)
Box Van or Water/Fuel Truck	30,000 lbs. (13,608 kg)	450 ft. (137 m)
Semi-trailer	60,000 lbs. (27,216 kg)	600 ft. (183 m)

Source: Transportation Security Working Group, Terrorist Bomb Threat Stand-off (Card), Government Printing Office.

IED could be used. Most prudent planners will point to a box van or cargo van for a baseline factor in terms of potential explosive power. As Table 5-2 from the TSA Transportation Security Working Group (TSWG) indicates, these kinds of vehicles can pack explosives with lethal air blast ranges of between 200 and 450 feet. These standards are in keeping with the Bureau of Alcohol, Tobacco, and Firearms (BATF) guidance, also adopted by the FBI's National Counterterrorism Center's (NCTC's) blast damage chart.

Positive factors in ranking vulnerability include access to the public safety staff (operations, ARFF, EMS, and police). For instance, if the necessary infrastructure is already in place or adjacent to the site (utilities, IT infrastructure, and others), the site will most likely gain some advantages in the selection process. Consider all the baseline data for each site in the process and roll that data into the overall analysis of the TVA.

Threat

As a starting point for threat assessment associated with airport operations, utilize the 15 National Planning Scenarios identified by the DHS (see list below) when conducting threat assessments for national critical infrastructure. These scenarios can serve as an initial list of potential threats, though not all apply to all airport site locations. For example, an improvised nuclear device, pandemic influenza, earthquake, hurricane, and foreign animal disease are not considered appropriate threats for consideration in this Guidebook. Either they will cause a regional disaster on a big enough scale to trigger an Incident of National Significance (INS), or they are probably not considered likely threats from an EOC facility perspective (earthquake or hurricane).

- Nuclear—Improvised Nuclear Device,
- Biological—Aerosol Anthrax,
- Biological—Pandemic Influenza,
- Biological—Plague,
- Chemical—Blister Agent,
- Chemical—Toxic Industrial Chemical,
- Chemical—Nerve Agent,
- Chemical—Chlorine Tank Explosions,
- Natural—Earthquake,

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Table 5-3. List of recent airport attacks.

Attack Site	Weapon Used	Date
Fort Lauderdale Hollywood International Airport	Lone Wolf/Active Shooter	2017
Will Rogers World Airport	Lone Wolf/Active Shooter	2016
Brussels Airport	VBIED	2016
Atatürk Airport	Armed Assault	2016
Sabiha Gökçen Airport	VBIED	2015
Los Angeles International Airport	Lone Wolf/Active Shooter	2013
Frankfurt International Airport	Lone Wolf/Active Shooter	2011
Glasgow International Airport	VBIED	2007
New Orleans International	Lone Wolf/Active Shooter	2002

Source: Faith Group, LLC.

- Natural—Hurricane,
- Radiological—Radiological Dispersal Device ("Dirty Bomb"),
- Explosives—Improvised Explosive Device,
- Biological—Food Contamination,
- Biological—Foreign Animal Disease, and
- Cyber.

In addition, airports are no strangers to active shooters, as outlined in Table 5-3.

The TSA *Recommended Security Guidelines for Airport Planning, Design and Construction* provides a sample chart of threats (see Table 5-4) from a historical perspective. These attacks can be traced to terrorist organizations and individual lone wolf attackers. While there may not be a specific terrorism target profile aimed at any specific airport, the fact is that air travel, and the FAA National Air Space System (NAS), are long-standing targets for terror attacks. These attacks can also take on the profile of a homegrown or domestic terrorist who may not necessarily be responding to orders from established leaders of terror organizations. Rather, they indoctrinate the philosophy of that organization and move on their own volition.

Table 5-4. Examples of terrorist attacks and weapon[8].

Type of Attack	Historical Examples	Type of Weapons
Explosive and Incendiary Devices	2011—Attack on Domodedovo Airport, Moscow, Russia	Suicide bomb; Incendiary device
	2010—Hidden explosives in cylinder of thermal liquid containers at screening checkpoint	Improvised explosive device
	2010—Discovery of explosive devices hidden in printer toner cartridges on all-cargo flights from Yemen	Improvised explosive device
	2010—Incendiary devices mailed to Maryland and Washington, D.C., area facilities	Incendiary device
	2009—Attempt to detonate device on-board Northwest Airlines Flight 253	Concealed body-worn plastic explosives
	2007—Attack on Glasgow International Airport	Incendiary device
	2001—Attempt to detonate device on-board American Airlines Flight 93	Concealed body-worn plastic explosives
	2001—World Trade Center 1995—Oklahoma City bombing	Proximity bombs, incendiary & secondary devices
Stand-Off Attack	2001—Tamil Tiger mortar attack and bombing of Sri Lanka's national Airport	Anti-tank rockets, mortars
Cyber Attack	2002—Code Red Worm	Worms, Viruses, Denial of Service Programs
Chem-bio, Radiological & Nuclear (CBRN)	1995—Aum Shinrikyo Sarin agent release in Tokyo Subway	Chemical, biological, or radiological or nuclear aerosolized

Source: TSA *Recommended Security Guidelines for Airport Planning, Design and Construction,* May 2011.

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Likewise, TSA has documented that Command Centers or Security Operations Centers can be specifically threatened by many potential issues. The potential attack profiles are in the range of those discussed in Table 5-4.

This study is focused strictly on the likelihood and impact of terrorist attacks like those outlined in Table 5-4. However, the site vulnerability analysis does consider adjacent environmental elements that do affect the severity of a given threat, including proximity and response time associated with law enforcement (a positive impact), and proximity to HAZMAT or other man-made disaster incidents (a negative impact).

Attack Scenarios

The Guidebook suggests that threat analysis be accomplished utilizing a specific vulnerability analysis against the following attack scenarios:

- **Active Shooter** (including coordinated armed assault, hostage takers, and lone wolf attacks);
- **VBIED** (including IEDs placed on the target and suicide bombers); and
- **CBN** (including a radiological dispersing device).

Utilize supporting scenario development to consider how an attack may be planned and executed. While these attack profiles do not have evidence to support an acute or active motivation for the attack, it will consider the likely attack, develop likely approaches, and use them to determine facility vulnerability to such scenarios. Under Homeland Security Presidential Directive 16 (HSPD-16), the DHS analysis of attack scenarios associated with airport infrastructure has narrowed the focus of attacks believed to be the most likely or most attractive attack profiles for airports.

Scenarios from the HSPD-16 include the list in Table 5-5. PAX attacks are included to assess potential risk associated with locating the EOC in or adjacent to passenger movement areas.

Based upon expert judgment, there are three violent threat areas to consider in this process, developed from the lists discussed above: (a) active shooter, (b) VBIED, and (c) CBN. These scenarios are by no means all inclusive, but they represent what the analysis considers the *most likely* attack profiles based upon the information and history available.

Table 5-5. HSPD-16 threat list for airports.

Target	Attack	Scenario
EOC	Suicide Bomber	Attacker enters facility with 20 lb. of explosives.
EOC	VBIED	Attacker drives truck of explosives into facility.
EOC	IED	Attacker brings roll-along bag containing explosives into facility, leaves it behind and remote detonates.
EOC	Active Shooter	Gunman enters facility and begins killing people.
EOC	Hostage Taker	Attacker enters facility and takes hostages.
EOC	CBN	Attacker enters facility with dispersing device containing chemical, radiological or biological agents.
Passenger Terminal (PAX)	Suicide Bomber	Attacker enters facility with 20 lb. of explosives.
PAX	VBIED	Attacker drives truck of explosives into facility.
PAX	IED	Attacker brings roll-along bag containing explosives into facility, leaves it behind and remote detonates.
PAX	Armed Assault	Handful of attackers enter facility during peak and begin killing people.
PAX	Active Shooter	Gunman enters facility and begins killing people.

Source: Homeland Security Presidential Directive.

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Suicide Bomber—This being a classic terrorist modality, it presumes an attacker could enter the facility (either posing as an "insider" delivery man or other innocuous visitor) with a nominal amount of explosives and shrapnel strapped to his or her body and initiate the explosive upon entering the center. The explosion will destroy the equipment and personnel exposed to the blast within the confines of the EOC, rendering it unusable, and leaving the EOC staff either dead or critically injured.

VBIED—This attack envisions up to 400 pounds of explosives in a vehicle (sedan), driven into the building or wall and igniting. The ensuing blast will most probably collapse the building wall, throwing chunks of the facility throughout the operating areas and rendering it unusable, with associated injury and death of its occupants. Not only will the adjacent wall be affected, but potentially the underground infrastructure around the car will also be compromised due to cratering, collapsing the facility and causing widespread casualties.

IED Insertion into Facility—This scenario uses similar entry techniques as the suicide bomber, except that the device is left behind and remotely detonated. The device size could range from 30–150 pounds depending on the package and cause extensive damage (in a confined space, 30 pounds of explosives will cause multiple blast and overpressure injuries to any humans exposed, most likely render the facility unusable).

CBN—A more complex scenario in some cases, this attack profile envisions the attacker gaining access to the air handling equipment, water supply, or simply entering the facility and releasing an agent. Depending on the agent, the result could range from immediate effects (VX/Sarin or Mustard Agent) to incapacitating agents with symptoms that have delayed onset (Botulism or Anthrax). Another potential route of attack could be an insider-spawned attack, where a refueling truck or other such HAZMAT gains relatively close access to facilities located on the AOA. While it has been some time since any attacker has used chemical or biological agents as a weapon[9], the threat still exists and, from an asymmetric attack perspective, remains a viable option.

Armed Assault, Active Shooter and Hostage—This scenario envisions an armed assault in the building interior to either inflict maximum casualties or take a hostage(s). The active shooter will begin discharging his weapon upon entry into the facility, or at some point when the largest target group is in range, inflicting maximum casualties. Unless there is a sworn officer on duty in the facility (armed), the shooter will be unopposed. The damage will be done to both system and operators with total damage depending on extent of the rampage.

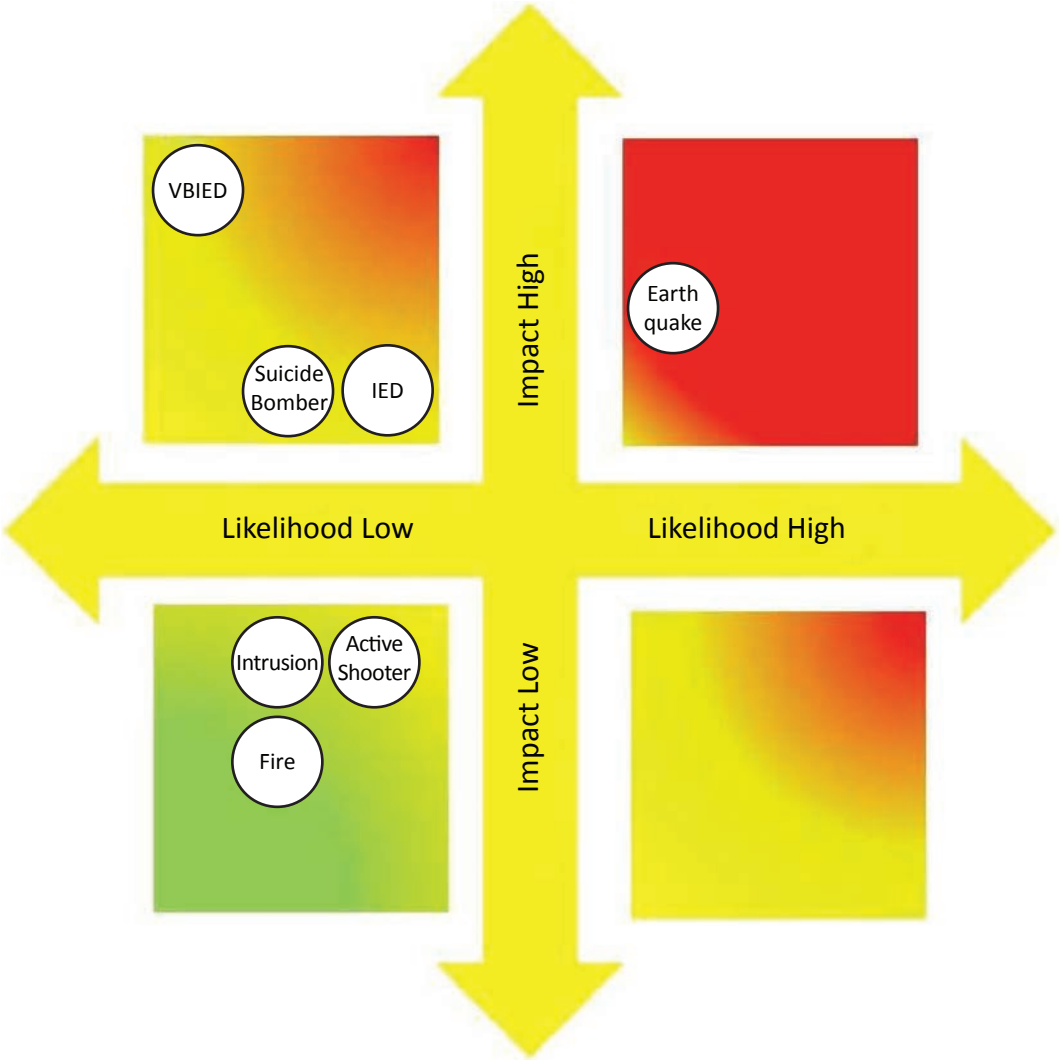
A mapping technique displayed in Figure 5-3 is an example of how the TVA stakeholders can come to a consensus regarding risk of any event. It is used to consider the scenario from the attacker's perspective, associated with the specific asset, and what the stakeholders and SMEs assess the likelihood and impact of each attack. Based upon expert judgment on how attack scenarios may be carried out, the potential difficulty encountered by the attacker in the execution, and the potential outcome, the vulnerability is ranked as High, Serious, or Low to assist in guiding resource allocation decisions. Add these findings to the individual assessment checklists outlined in the following pages to reach an overall perspective of the assessment.

Table 5-6 displays the range of assessment elements. On the vertical axis of the Criticality Matrix is the range of vulnerability, ranging from A–E: (A) "*Very Easy*," indicating virtually open and unfettered access to the target to (E) "*Too Difficult,*" which, in the opinion of the assessor, will be associated with a scenario virtually impossible from the attacker's perspective.

The horizontal axis of the Criticality Matrix considers the potential impacts of the attack, ranging from I–IV: (I), "Loss of Life," indicating strong possibility of casualties, to (IV), "No Injury, No Service Impact." Where these assessments come together is an overall assessment of risk

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Specific Threat Scenario Mapping



Source: Faith Group, LLC.

Figure 5-3. Scenario mapping against threat.

Table 5-6. Consequence Mapping Chart (TSA, 2011).

Vulnerability of Target	Impact		Criticality Matrix
A Very easy	I	Loss of Life	
B Somewhat Easy	II	Serious injury, major service impact	
C Difficult	III	Minor injury, minor service impact	
D Very Difficult	IV	No injury, no service impact	
E Too Difficult	--	---	

H=High S=Serious L-Low

	I	II	III	IV
A	H	H	S	S
B	H	H	S	L
C	H	S	L	L
D	S	L	L	L
E	S	L	L	L

Source: TSA *Recommended Security Guidelines for Airport Planning, Design and Construction,* May 2011, Appendix A.

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including (H) "High," (S) "Serious," and (L) "Low." This assessment used in conjunction with the site Vulnerability Assessment Scoresheet (Figure 5-2) provides a measured sense of risk for each site.

While none of these tools is perfect, used together they can help the evaluator consider the problem from several perspectives and prevent overlooking a potential threat. This is termed as "mirror imaging" of the analysis. Mirror imaging is an intelligence analysis term used to describe how an analyst will consider the potential threat based upon their personal, moral, and ethical compass, or on TTP. To reduce this, it is recommended that the threat be analyzed from many different angles and perspectives, which these tools enable, polling many stakeholders in order to fully vet the findings.

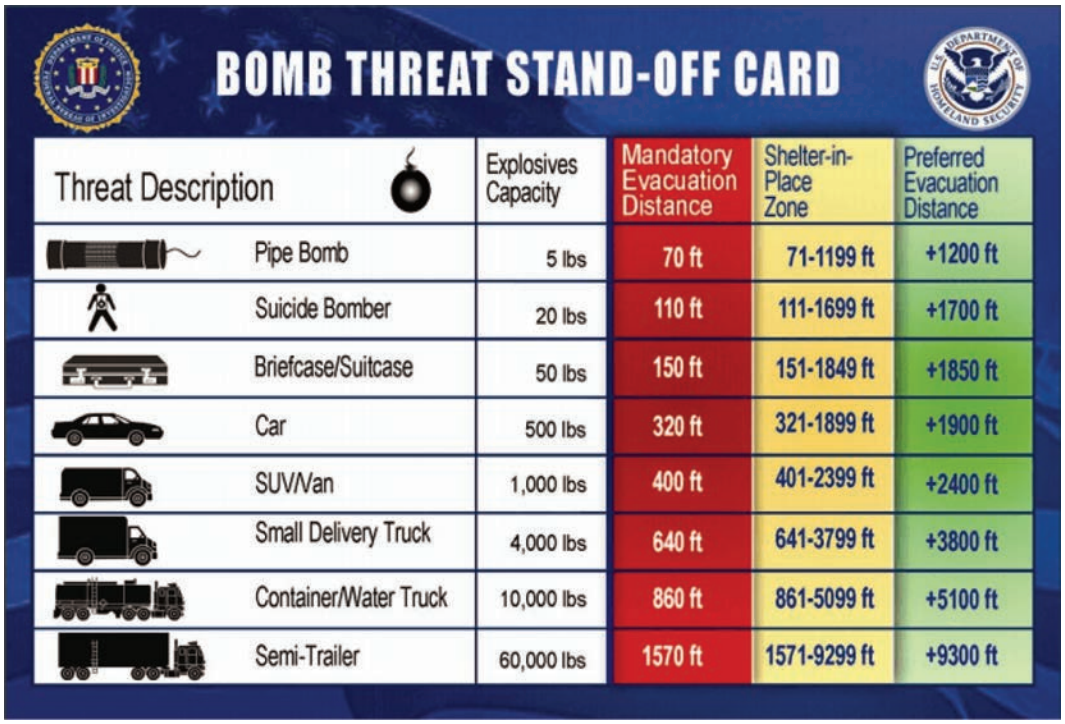
Site-Specific Assessment

To determine overall facility threat and risk, utilize the Vulnerability Assessment Scoresheet (Figure 5-2) when making site-specific assessments. The form considers the seven factors that help better understand the risk associated with each facility, or in this case, each facility location. This assessment considers the overall resilience of the facility to withstand the proposed attack scenarios and will help direct the owner to a more balanced decision regarding site selection.

- **Level of Visibility—**This factor considers the overall SA of the existence of the facility and its functions in the community and the public. If, for instance, the facility has large signs indicating its location and purpose, and is adjacent to easy access from major roadways, it offers a higher risk of being targeted.
- **Criticality of Target Site to Jurisdiction—**For this study, it is assumed that all sites carry the same ranking for this assessment element. As the central point for command and control for airport public safety, it is deemed to be at the "High" level of criticality to the operation. While the airport will not necessarily stop operating without it, law enforcement, operations, and other emergency response events will become challenging.
- **Impact Outside the Jurisdiction—**This assesses the potential impact to other jurisdictions, including national ramifications. For instance, if an IED and/or CBN attack was staged at a major airport or its EOC (especially if coincident with a second attack on similar target), it could potentially trigger a ripple effect across the nation. Again, the assessment in this element will remain constant for each site.
- **PTE Access to Target—**This element considers the obvious: access control, fencing, barriers, vehicle parking lot adjacencies, ease of access and egress to and from major roadways, adjacencies to public roads, and CCTV and other limiting measures to physical and visual access to the facility. Additionally, stand-off distances will be assessed based upon the FBI's NCTC Bomb Stand-off Distances Charts (see Figure 5-4 and Figure 5-5).
- **Potential Target Threat of Hazard—**There may be adjacencies such as fuel storage areas or other threats to consider, such as mobile HAZMAT threats (fuel tankers and freight rail).
- **Potential Target Site Population Capacity—**This assessment is based on estimated census of the site and adjacencies. Each site will be assessed to determine the ranking.
- **Potential for Collateral Mass Casualties—**This will assess potential collateral damage within a one-mile radius of the site. This potential for collateral mass casualties assessment considers WMDs (large explosives and biological/chemical weapons) and the value of locating this site away from large populations.

Once completed, each site should be assigned a Basic Vulnerability Assessment Rating from 1 to 12, 12 being the highest risk from a threat and vulnerability perspective. While still depending on many subjective assessment tools, utilizing the Vulnerability Assessment Scoresheet (Figure 5-2) will help decision makers ascertain, from a risk perspective, which site is most advantageous. See Figure 5-6 for a site evaluation plan checklist.

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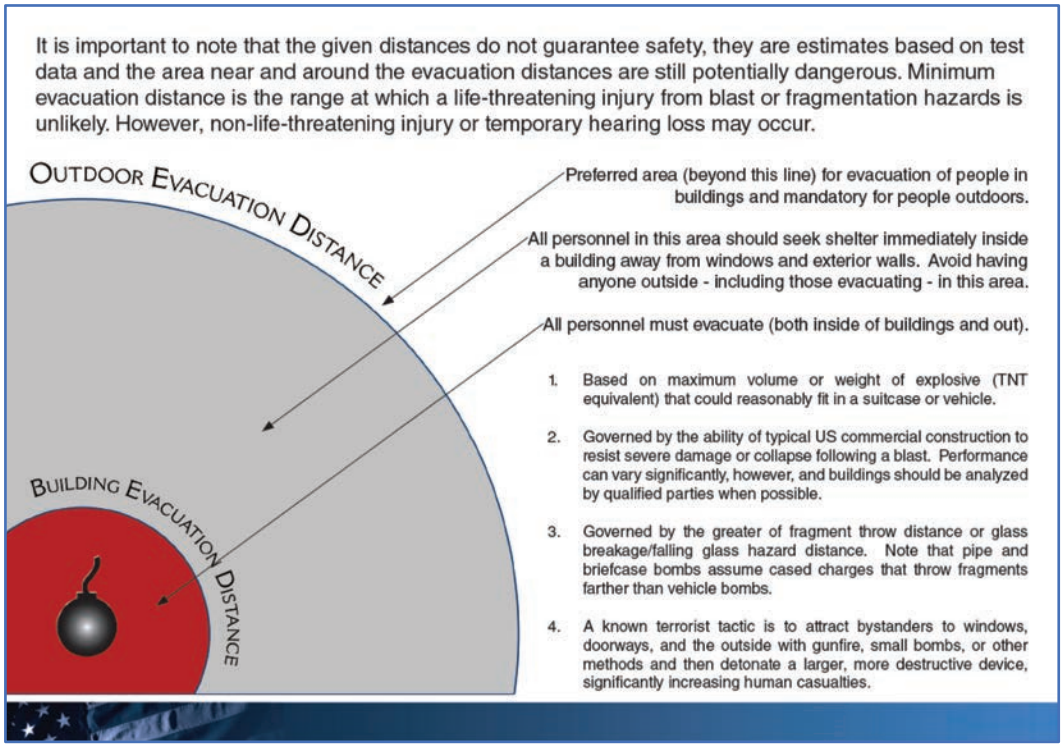


BOMB THREAT STAND-OFF CARD

Threat Description	Explosives Capacity	Mandatory Evacuation Distance	Shelter-in-Place Zone	Preferred Evacuation Distance
Pipe Bomb	5 lbs	70 ft	71-1199 ft	+1200 ft
Suicide Bomber	20 lbs	110 ft	111-1699 ft	+1700 ft
Briefcase/Suitcase	50 lbs	150 ft	151-1849 ft	+1850 ft
Car	500 lbs	320 ft	321-1899 ft	+1900 ft
SUV/Van	1,000 lbs	400 ft	401-2399 ft	+2400 ft
Small Delivery Truck	4,000 lbs	640 ft	641-3799 ft	+3800 ft
Container/Water Truck	10,000 lbs	860 ft	861-5099 ft	+5100 ft
Semi-Trailer	60,000 lbs	1570 ft	1571-9299 ft	+9300 ft

Source: FBI.

Figure 5-4. Bomb stand-off chart.



Source: FBI.

Figure 5-5. Stand-off distance chart.

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Checklist: Site Evaluation Plan

The evaluation checklist provided below identifies key criteria and issues to consider when evaluating sites for a potential EOC. The checklist is organized using the same categories as previously described in this chapter.

Existing and New Facilities

Accessibility

- ❑ ADA compliance
- ❑ Driving and parking
- ❑ Public roadway access
- ❑ Perimeter road access
- ❑ Airfield access
- ❑ Quick access from terminal and other stakeholder locations
- ❑ Secure

Serviceability and Infrastructure

- ❑ Site must support required infrastructure needed to run the EOC
  - ❑ Utilities
  - ❑ Primary and emergency power
  - ❑ Water
  - ❑ Communications (landline, cellular, LAN, Wi-Fi)
  - ❑ Redundant infrastructure
- ❑ Fire hydrant access
- ❑ Locate away from radio interference and tall metal structures
- ❑ Avoid vulnerabilities to wind-blown debris
- ❑ Locate on higher ground
- ❑ Noise mitigation

Safety and Security

- ❑ Wayfinding and directions to the EOC should be distributed on a need-to-know basis
- ❑ Controlled access with secondary access considerations
- ❑ Sterile vs. non-sterile area
- ❑ Vulnerability to HAZMAT (e.g., freight train, truck traffic)
- ❑ Protection from potential blast effects
- ❑ Adequate site area for stand-off distances
- ❑ Adequate site lighting
- ❑ Weather protection
- ❑ Conduct TVA

Size

- ❑ Adequate to support required building envelope and supporting parking spaces
- ❑ Flexibility to expand facility if required beyond current envelope
- ❑ Adequate clear zones and setbacks
- ❑ Clear from RPZ penetration

Versatility and Adjacency

- ❑ Will facility be co-utilized with a different non-emergency function?
- ❑ Will it be shared with other agencies (i.e. emergency management, fire, and police)?
  - ❑ If yes, determine needs of all stakeholders
- ❑ Can site support quick access/start up of facility, i.e., location to terminal and other important stakeholders?

ConOps

- ❑ Consult ConOps plan and update as required
- ❑ Once site selection(s) is finalized, update ROM costs
- ❑ Once funding is approved, update ConOps

Source: Faith Group, LLC.

Figure 5-6. Site evaluation plan checklist.

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Other Considerations

Consider a backup site location should the primary EOC become compromised and unable to perform the functions of a fully operational EOC. Ideally, the backup site should be capable of immediate activation and located to minimize the factors that disabled the primary site. Consider alternate access routes and a separate power grid from the primary site. Should the backup site be co-utilized with other non-emergency agencies, consider studying impacts to the resident agency to understand if their functions can still be performed or if temporary suspension of their activities is warranted. The minimum infrastructure required to run and support the alternate facility includes all the considerations of the primary EOC site[10].

Another consideration for an alternative site involves a ready "To Go Bag" with resources necessary to establish a temporary EOC anywhere. "Bag" provisions will ideally include phones, radios and chargers, pencil, paper, all applicable response binders, checklists, forms, as well as a flashlight, emergency-kit stabilized food, and water for at least 48 hours. Do not keep the provisions at the airport. They can be kept at an alternative, off-site facility or in the personal care of pre-assigned positions such as the EOC manager and command staff.

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SECTION 6

Ergonomic and Human Factors

Introduction

EOCs and associated spaces are complex environments supported by many integrated technologies. As such, a detailed operational review is critical prior to the development of a layout to accommodate the necessary human machine interface (HMI) relationships appropriately. Complete task analysis and operational planning should be carefully and completely developed prior to any facility planning, to guide the design, construction, and eventual operation of the facility successfully. See Figure 6-1 for an overview of ergonomic EOC planning.

The following sections will look at all three major aspects of ergonomics (physical, cognitive, and organizational) as well as human factors and how they should be considered within the airport EOC environment. General layout and design recommendations should be made based on theoretical research as well as current best practices and ergonomic standards. Lessons learned and case studies should include not only airport EOC application but also other mission-critical EOC applications.

Survey Responses

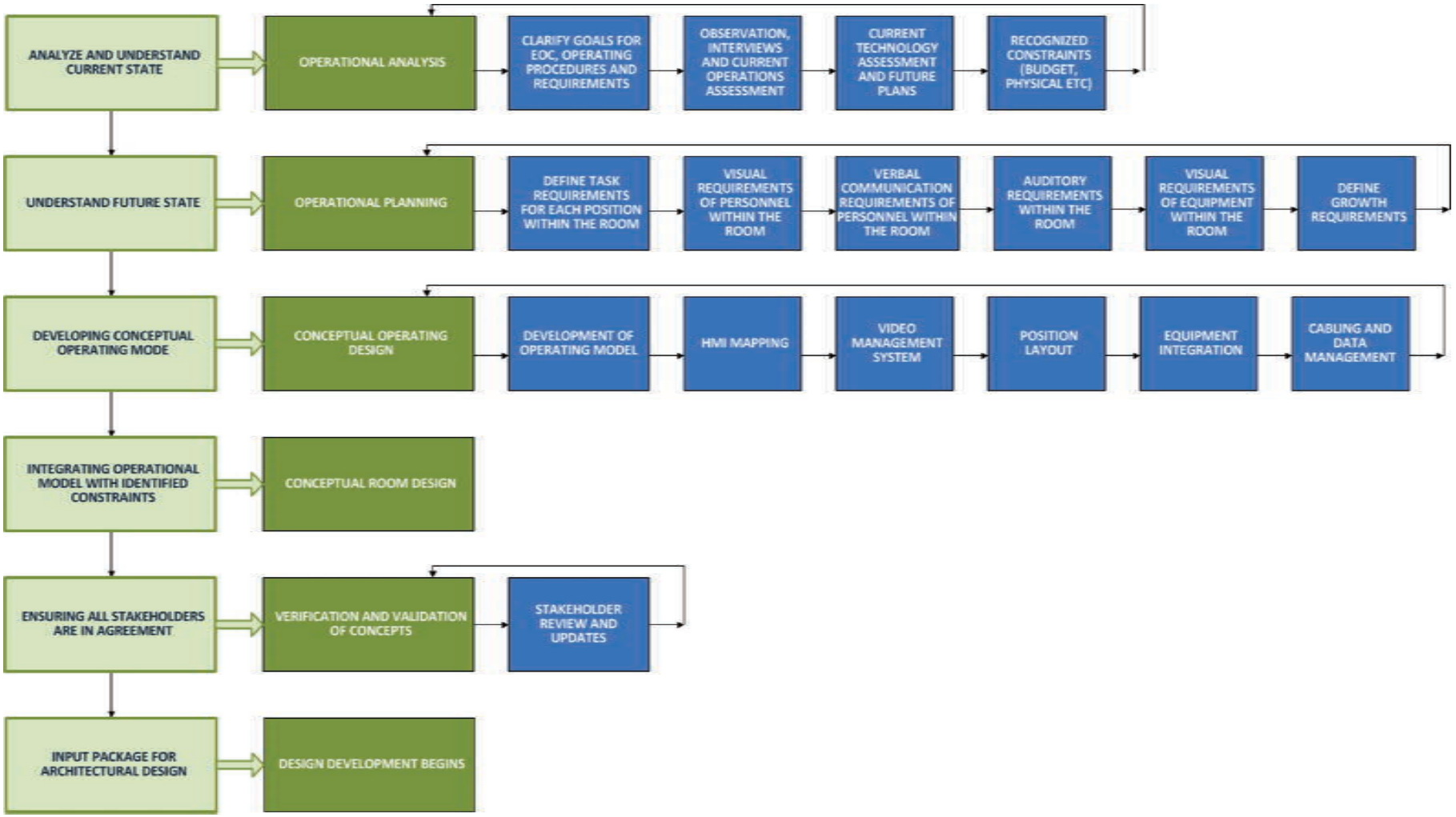
Through surveying of 27 airports of various sizes, much feedback was received regarding the ergonomics and human factors of current EOCs. Although not always a 24/7 control room, when activated, an EOC becomes a mission-critical facility with the potential for extended use. Therefore, the planning and design needs to meet or exceed the standards of a control room—and the human factor element is paramount.

Survey responses showed several trends across airports of all sizes, the most prominent trends indicating that ergonomics and human factors are below standard, and in many cases, have been neglected. Not one airport surveyed had an ergonomic mandate in place. This is largely due to the lack of awareness as to the impact that these factors have on the overall effectiveness of the EOC environment, especially in long periods of operation.

The following were the most prominent items in the related feedback:

- **Traffic flow—**Nearly every survey indicated that flow was inefficient, or restricted when the room was populated with the required personnel.
- **Overall size—**Many EOCs have been placed in a certain space "because it was available" and are often not adequately sized for the number of personnel required during activation.
- **Noise—**Noise from multiple conversations, phones, and equipment was noted as an interruption to decision making and communication.
- **Work surface area—**Given the number of people, variety of operations, and amount of equipment, the amount of work surface area was often reported as inadequate.

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Source: Evans Consoles.

Figure 6-1. Ergonomic approach to EOC designs.

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- **Poor configuration (no breakout rooms or break areas)—**Inadequate space for non-task activities, such as space for food and drink, bags, and clothing as well as the lack of any quiet space for rest during lulls in activity was noted.

Human Factors and Ergonomics

Human factors and ergonomics, as related to the design of an EOC, consider a human-centered design approach to optimal system design that incorporates the interactions among three interrelated categories: organizational, cognitive, and physical considerations. The term "ergonomics" is often associated with the physical aspects of the system (equipment, workstation, environmental conditions, etc.), and sometimes this is termed "micro-ergonomics," whereas the term "human factors" is often used to refer to the "cognitive ergonomics" aspects of a system. Organizational ergonomic considerations are also often termed "macro-ergonomics." A brief description of each of these considerations is given below.

Organizational Ergonomics

Organizational ergonomics is a branch of ergonomics that is concerned with the interaction among humans, technology, and other elements in a system. This category of ergonomics looks at issues such as social interactions and patterns of communication, leadership, and teamwork. This is a critical element of control center design to ensure adequate task loading and effective communication flow in both normal and upset scenarios.

Cognitive Ergonomics

Cognitive ergonomics is concerned with the mental processes around a combination of tasks. It is a study of how people observe information, the way in which people understand its implications, and how that determines the decisions people make. It is important to recognize the constraints of human cognitive abilities and accommodate those within a functional design. This is particularly important in a mission-critical facility where multiple processes and alarms are monitored by one person. Understanding responding staff's memory, perception, reasoning, and motor response in relation to operational needs is critical. Such considerations are especially important given that EOC operating conditions are likely to present people with multiple stressors, especially time stress and multitask loading. Cognitive ergonomics also considers issues of SA, which are critical to the effective performance of the EOC.

Physical Ergonomics

Physical ergonomics focuses on the body's response to physical work demands and is concerned with the impact of the physical work environment (repetition, vibrations, workstation arrangement, etc.) on the responding staff. Designs for physical ergonomics considerations include the consideration of biomechanics, anthropometric data, and employee habits. When designing with physical ergonomics in mind, it is important to identify primary, secondary, and tertiary tasks at the workstation. These categories should be reviewed in normal and critical operational modes and the workstations set up to accommodate the most ergonomic positioning in both operational scenarios.

Theoretical Research

The goal of human-centered ergonomic design is to create a system in which human participants perform at an optimal level with minimum risk of error, regardless of how stressful the

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situation might be. To do this requires considering characteristics of the work to be performed (i.e., the tasks, operations, and organizational factors), the technology to be used to perform that work (both the hardware and the software required), and the physical setting in which work is being performed, which includes consideration of the environmental conditions (temperature, light, noise, vibration, ventilation, air quality), the physical workstations (chair, desk, keyboard, mouse, screen, etc.), and the architectural design (spatial layout, windows, doors, color). This ergonomic approach to systems is an iterative process that proceeds through a series of phases as the system design evolves from concept to operational.

Standards

The International Organization for Standardization (ISO) has promulgated a series of standards concerned with the ergonomic principles for designing various arrangements of rooms and spaces in a control suite. These standards provide guidance on the physical layout of an EOC, including the design of workstations and ambient environmental considerations: thermal environment, air quality, lighting environment, acoustic environment, vibration, aesthetics, and interior layout design. The following standards are available:

- ISO 11064-1:2000 Ergonomic design of control centers—Part 1: Principles for the design of control centers:

  This standard specifies ergonomic principles, recommendations and requirements to be applied in the design of primarily non-mobile control centers. It covers all types of control centers and addresses the expansion, refurbishment, and technological upgrades of control centers.
- ISO 11064-2:2000 Ergonomic design of control centers—Part 2: Principles for the arrangement of control suites:

  This standard specifically deals with the ergonomic design principles for the arrangements of rooms and spaces in a control suite. It includes identification of functional areas, space provisions for each functional area, operational links between functional areas and optimal layouts. The ergonomic design principles are based on an analysis of functions and tasks that must be performed in the EOC.
- ISO 11064-3:1999 Ergonomic design of control centers—Part 3: Control room layout:

  This standard includes ergonomic design requirements, recommendations and guidelines for primarily non-mobile control room layouts, workstation arrangements, the use of off-workstation visual displays, and control room maintenance.
- ISO 11064-4:2013 Ergonomic design of control centers—Part 4: Layout and dimensions of workstations:

  This standard specifies ergonomic design principles, recommendations, and requirements for the design of seated, visual-display-based workstations with emphasis on layout and dimensions, along with some consideration of standing workstations. It assumes that workstations will incorporate flat-display screens.
- ISO 11064-5:2008 Ergonomic design of control centers—Part 5: Displays and controls:

  This standard presents ergonomic design principles, requirements, and recommendations for displays, controls, and their interaction in the design of control-center hardware and software.
- ISO 11064-6:2005 Ergonomic design of control centers—Part 6: Environmental requirements for control centers:

  This standard presents environmental requirements for the thermal environment (in temperate regions), air quality, lighting environment, acoustic environment, vibration, aesthetics

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and interior design, along with recommendations for the ergonomic design, upgrading, or refurbishment of control rooms and other functional areas within the EOC.

- ISO 11064-7:2006 Ergonomic design of control centers—Part 7: Principles for the evaluation of control centers:

  This standard establishes ergonomic principles for the evaluation of control centers, including the control suite, control room, workstations, displays and controls, and work environment.
- ANSI/HFES 100:2007 Human Factors Engineering of Computer Workstations:

  This standard provides specific guidance for the design and installation of computer workstations, including displays, input devices, and furniture. The standard can be applied to a range of environments and accommodate a wide variety of users.
- Canadian Standards Association (CSA)-Z412-00 (R2016)—Guideline on Office Ergonomics:

  This guideline provides step-by-step instructions of the ergonomics process for the optimal design of office systems where computers are being used, including the design of jobs and work organization, layout of the office, environmental conditions, and workstation design. It is intended predominately for office workers and employers who are responsible for health, safety, and/or ergonomics programs in the workplace.

Ergonomic Operational Design Guidelines and Considerations

When in operation, the EOC can be a stressful setting in which difficult human decision making will need to happen as quickly and accurately as possible. The wide range of alternative scenarios that could occur also requires that the EOC be somewhat flexible in its capabilities. In such challenging situations, it is extremely important that attention be paid to optimizing the ergonomic design of all relevant aspects of the system.

Programmatic Requirements for the EOC

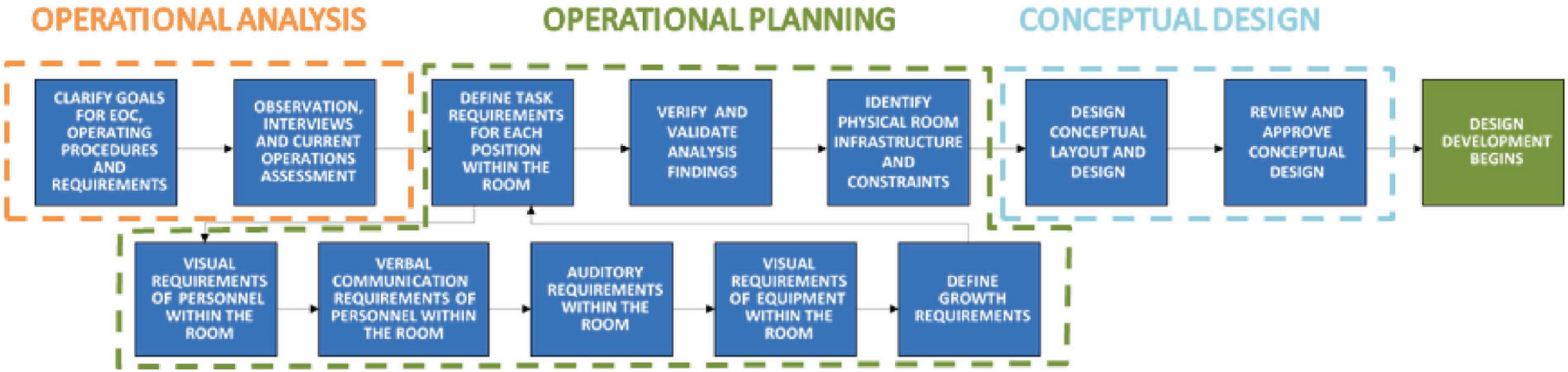
The programming and design process should be aimed at identifying and achieving institutional-driven goals in addition to accommodating all requirements identified during the operational analysis and operational layout planning phases of design. Programmatic requirements for mission-critical operations typically include mission goals, performance, operational requirements, space needs, functional adjacencies, and technical requirements. Additionally, programmatic requirements may include provisions for long-term future use, expansion, or facility flexibility. A Programmatic Process chart can be found in Figure 6-2.

Typically, it is suggested that EOC programming be executed as an inclusive process that engages all stakeholders who will share in the management, use, construction, and operation of the facility. In some circumstances, security, or other considerations may drive the extent to which some programmatic requirements of the facility may be shared among stakeholders. The intent of early engagement is to gather and document the needs and concerns of the participating stakeholders. Encouraging frequent communication and feedback among the stakeholder group throughout the programming and design process is advisable. Figure 6-11, at the end of this section, provides a general checklist to be followed during an ergonomic and operational assessment.

Operational Analysis

Information gathered from the various stakeholders will vary. All information is useful to developing a control center that is responsive to organizational objectives and user

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Source: Evans Consoles.

Figure 6-2. Programmatic process for EOC design.

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needs. In general, stakeholders may be expected to provide useful programmatic information as follows:

Executive and Operational Management

- What are the functional objectives?
- Are there technical or aesthetic institutional standards to incorporate?
- What are the overall tasks to be performed?
- What is the planned staffing for activation?
- What ancillary spaces are required to fully support EOC operations?
- What are the current operational challenges to be addressed?
- How do the EOC room parameters change between short- and long-term activations?
- How does management see the individual staff requirements in terms of tasks to be completed and required interactions with other operators, supervisors, and managers?

EOC Operators

- What are the positions within the control center?
- What are the individual task loads for each position?
- What equipment are they currently using?
- How is their time during an average hour divided among each piece of equipment?
- How does that change during a long-term activation?
- What types of errors typically occur?
- How frequently do they occur and when do the operators themselves experience highs and lows in energy and awareness?

IT and Systems Integrators

- What equipment is used or proposed for use in the control center?
- Where is the equipment best located?
- What challenges are anticipated in setting up, operating, and maintaining the equipment?
- How will the control center systems be integrated with remote operations, facilities, and equipment?

Space Requirements

EOC facilities can vary greatly in terms of purpose, size, population, and programmatic complexity. Most are unique in their requirements and physical configuration. These characteristics can depend on several factors such as airport size, location, city/state procedures, etc. In specific and limited circumstances, modular and pre-fabricated control centers are available; however, these products would typically not meet the needs of large airports.

Operational Planning

When designing the EOC itself, it is important to first review the current operating model to highlight which processes are working well and which could be better optimized. This review includes current technology used and planned upgrades, operational goals, shift length, and changes in the typical length of EOC activation based on the emergency operating plans. Once the current operating state is fully defined, it is important to outline the future operating plans. This includes identifying the visual requirements needed for each task or position, future position planning, and the communication needs of each group within the EOC. How will they be communicating and who are they speaking with? Facilitating this communication through adjacencies in the layout can cut down on noise and distraction to others within the EOC.

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Before a conceptual plan can be developed, the individual positions within the room and their equipment requirements need to be identified. The amount of equipment and identification of primary, secondary, and tertiary sources of information will directly affect the furniture requirements within the room. Once layout of the equipment for each position has been identified, the optimal furniture can be selected and an appropriate layout designed.

Conceptual Design Considerations

The following utilizes the standards and recommendations discussed in the preceding topics of this section to outline general control center design guidelines as they relate to the general mission requirements of an Airport EOC. Further detail on these topics can be found in Section 7 of this Guidebook.

The layout of an EOC can be critical in assisting smooth operational processes. There are many factors to consider when laying out an EOC and adjacent areas. The physical layout of the EOC should reflect the organizational processes to be supported within the EOC. It is critical to complete full operational planning, including adjacency study, of critical facilities as well as staff positions and equipment prior to determining any layout selections. This will ensure that the desired process and goals are supported by the facility. Please refer to Figure 6-6 for a sample adjacency analysis and further adjacency discussion. Although this image focuses on adjacencies between rooms in a facility, the same chart can be created for adjacencies among positions within the operations center.

EOC activations are often long-term events, covering natural break times in a work day. As well, it is not uncommon for there to be participants within the EOC facility who do not have badge access within the area and are escorted on site. It is important to consider and accommodate this as completely as possible when planning an emergency operations suite. For employee comfort, safety, and responsiveness, the following areas within the common secure perimeter should be considered: adjacent kitchen/kitchenette facilities, restroom facilities, smoking areas, and prayer/quiet rooms.

Co-Location of Operation Facilities

What is Co-Location? In addition to EOCs, most airports typically include operational space for many other functions, including safety and security, direct facility operations, maintenance and utility operations, fire, emergency response, and police and security operations.

Historically, most organizations have developed separate system monitoring functions running in parallel and independent of each other. Contemporary control suites/EOCs are designed to support integrated operations that bring multiple functions together under one roof. This has proven to be very beneficial as organizations are able to leverage efficiencies at multiple levels to improve their operation. These control suites, designed as multi-functional, collaborative environments, are widely seen in many of the following markets:

- Public safety,
- Process control,
- Utilities, and
- Transportation.

The emerging trend in co-location of facilities has been facilitated by the rapid growth in technology in the fields of process monitoring and control, communication, audio–visual (AV), and IT. Most control applications are now available electronically with remote reporting and interface capabilities. These developments, combined with new flexibilities in modern video management systems, allow advanced content sharing and collaboration, which has increased

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the modern control facility's ability to manage multiple operations and share information within a single facility.

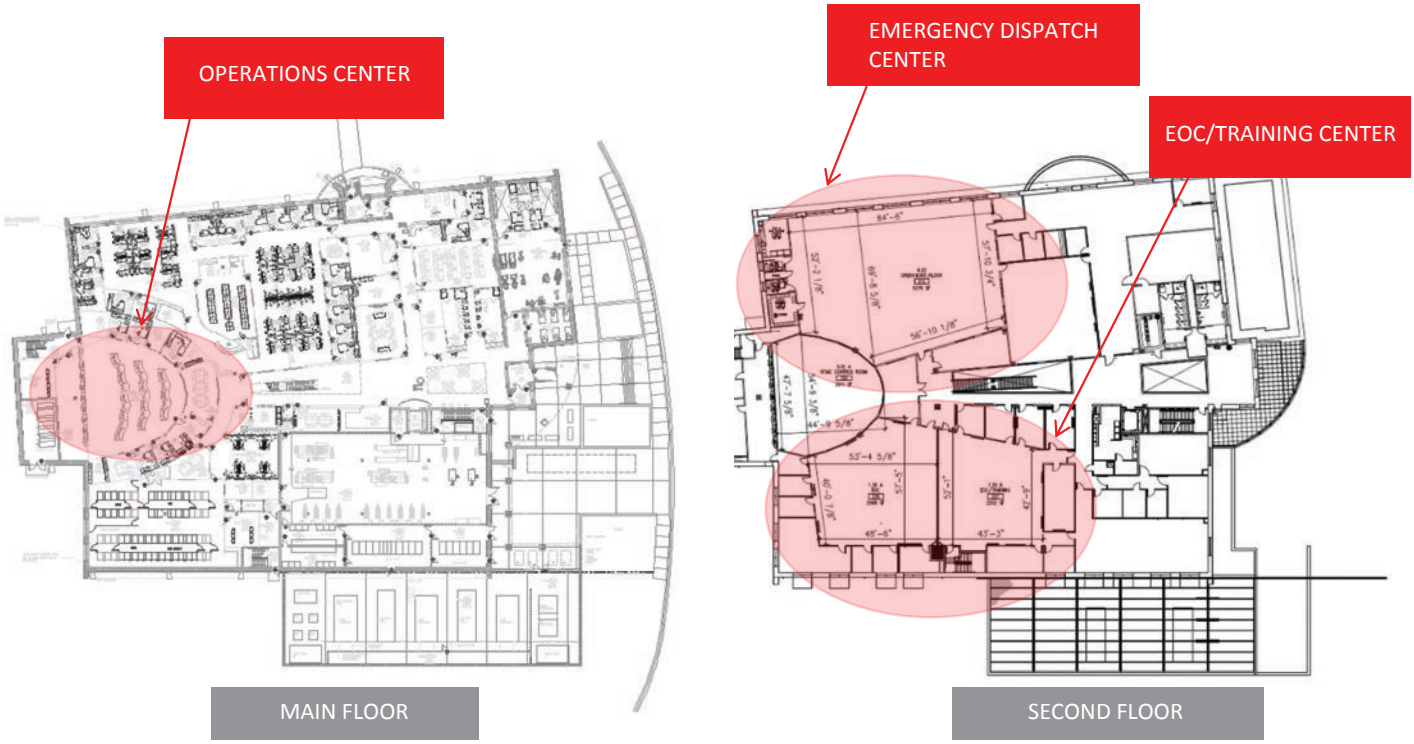
Advantages of Co-Location? Often, there is a benefit to co-locating emergency response and operations facilities to optimize SA and communication. Co-location of facilities can offer many operational advantages, unique design characteristics, and reduce overall cost for construction, maintenance, and technology. Some of those efficiencies include:

- Collaborative discovery of operational efficiencies,
- Elimination of duplicate facilities,
- Enhanced security,
- Technology consolidation and reduced maintenance,
- Improved communications, and
- Functional cross training.

An example of floor layouts for co-located facilities can be found in Figure 6-3 and Figure 6-4.

In the examples presented in Figures 6-3 and 6-4, the main floor houses the operations center, with a two-story high AV wall and ceiling. On the second floor is the multi-function EOC/training center, as well as the security center, both with sightlines to the operations center and the AV wall.

Multi-use Areas. Often, an EOC is undermined by the fact that the facility is not used often. This can make it difficult to obtain the necessary budget to maintain the EOC facility to the level required in any given emergency. As shown below, there are many possible set-ups for an EOC, most that leave the EOC empty during non-emergency events. Many facilities are adopting a multi-use design that allows the EOC to act as a training or simulation room when not in use



Source: Evans Consoles.

Figure 6-3. A sample floor plan of a co-located facility.

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Source: Evans Consoles.

Figure 6-4. A photographic sample of a co-located facility.

during an emergency. Focusing on design flexibility can allow a single room to support multiple functions, as illustrated in Figure 6-5.

When to Consider Co-Location or Multi-use Areas?

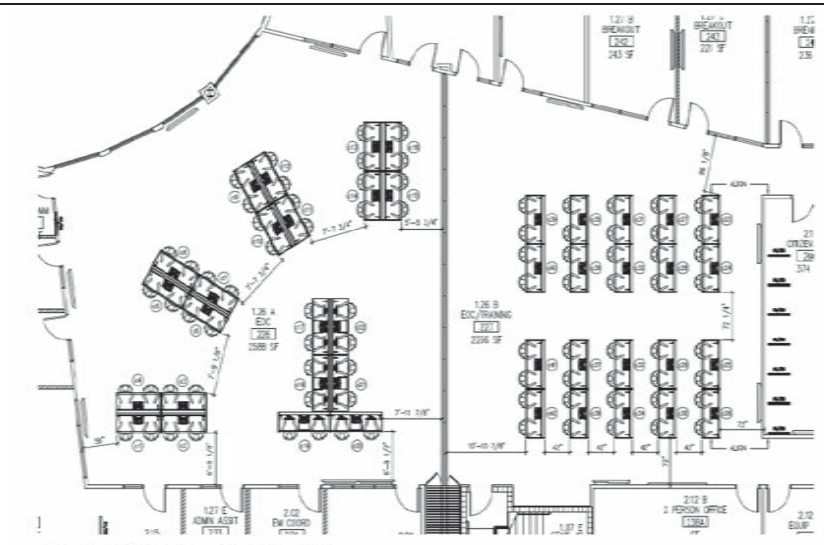
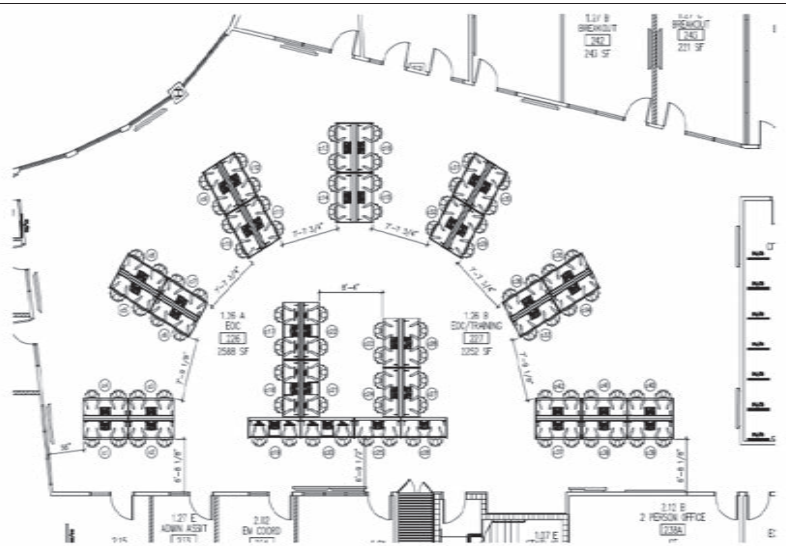
Consideration of a consolidated operations center should always include an assessment of existing conditions. The scope of this assessment will vary from airport to airport and should address all the considered operations, not just emergency. Assessments to include in an emergency operations master plan can include:

Location Assessment. Existing operations centers, the associated space they occupy, and the systems monitored should be mapped on an airport-wide basis. An assessment should be made regarding how efficiently the space is being used and whether the available space affects the performance of the existing operations in normal and emergency situations. Consideration may also be given to the relationship between the EOC and associated field staff.

Use and Interaction Assessment. The current use of existing EOCs should be understood and documented. Current and potential interactions among existing EOCs, control rooms, systems, operators, and field staff should be evaluated to identify needed linkages and potential consolidation benefits. It is important also to understand operational contingency planning and the technology and communication systems that support the backup operations. Figure 6-6 demonstrates a sample adjacency and interaction analysis.

Technology Assessment. The AV and IT technology infrastructure should also be evaluated in terms of type, capacity, condition, and suitability for supporting operational objectives. The technology assessment may also include an evaluation of how existing space can support future technology upgrades.

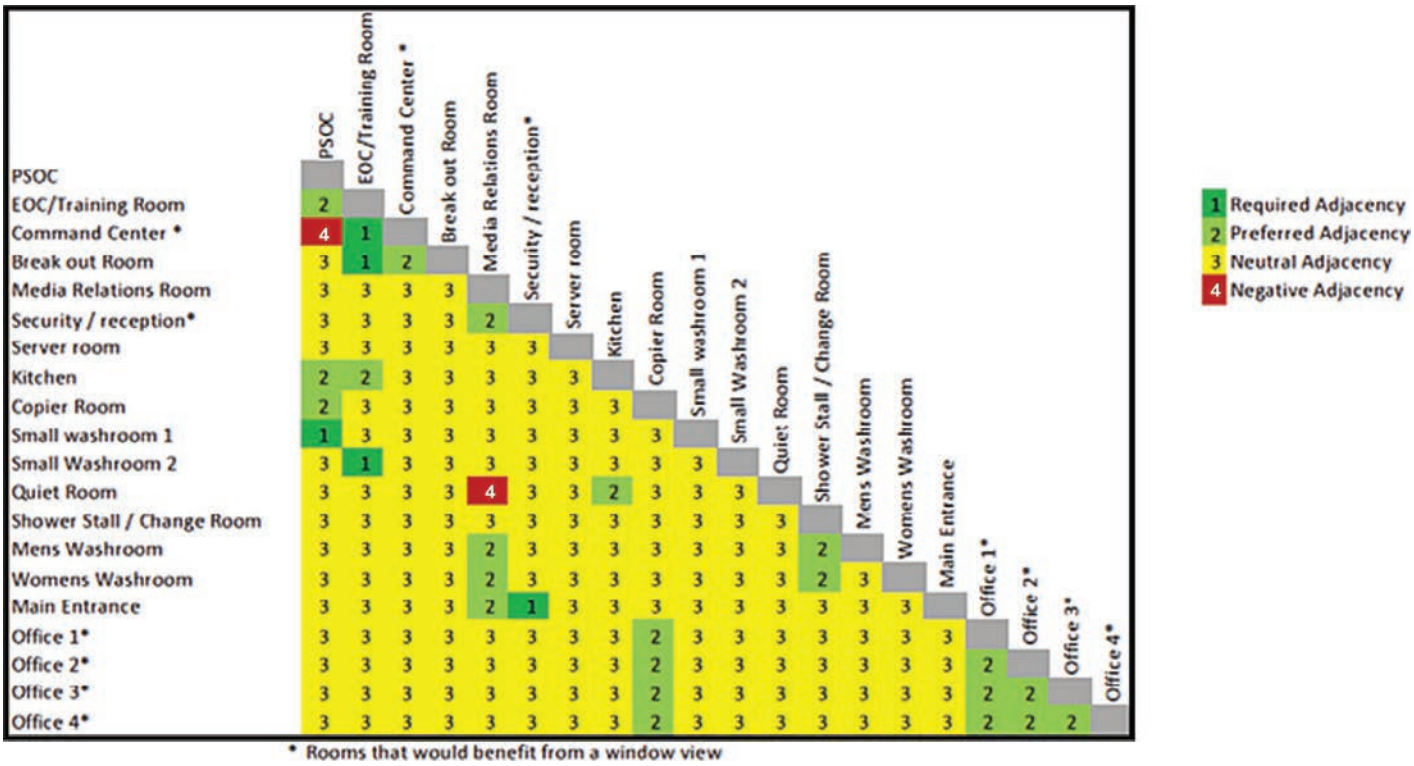
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Training Scenario	Activated Scenario
	

Source: Evans Consoles.

Figure 6-5. An illustration of a multi-use warm (semi-activated EOC in a constant ready state but not fully activated) EOC. One side of the facility is an active EOC with limited personnel, and the other is used for training purposes. During an emergency event, the moveable wall system is removed, and the tables easily converted to accommodate the operations of a full EOC activation.

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Source: Evans Consoles.

Figure 6-6. Sample adjacency analysis for room adjacencies (PSOC = port security operations center).

Threat Assessment. As a matter of routine operations, most airports should have a threat assessment and associated contingency planning for each control facility, as was discussed in Section 5 of the Guidebook. This information should be compiled and factored into the evaluation of existing control room operations and potential consolidation strategies.

New facilities allow for optimum site planning, access control, security, hardening, and other requirements typically associated with mission-critical functions. However, while these factors are relatively easy to incorporate into greenfield facilities, it is somewhat more challenging for existing facilities. Often built decades ago, the infrastructure, security, vulnerability to threat, and technology of existing operations centers may not readily allow for consolidation. This often leads facilities to continuing to operate on a dispersed model. In the end, this can often be a significantly more expensive operating model.

EOC Style

- **Stand-alone**—Only set up in the case of an emergency event,
- **Stand-alone warm EOCs**—Continuously staffed by a limited number of personnel who would be participants in the EOC when activated, or
- **Dual use**—A multi-functional room that is used for other purposes during non-emergency events, but that can be quickly and easily set up as an EOC when necessary.

In any of these scenarios it is important to design the facility assuming a full EOC activation.

Inside the EOC there should be sufficient space as well as easily accessible power and data to accommodate the number of personnel, including future growth assumptions (usually 25%), support personnel and necessary AV, printers, and excess computer requirements. It is important

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to keep in mind that overhead power and data drops create disrupted sightlines to common AV, as well as limit the flexibility of the layout design. A raised access floor with power and data cabling accommodated within the plenum (space under raised flooring which is used for running cables and utilities and also often used for air circulation) is recommended.

Shared Visual Displays. The audio–visual system in an EOC is an essential element that can greatly contribute to the overall quality, safety, and efficiency of the activated EOC. AV system designs can vary greatly depending on the site, equipment, as well as operational model and communication/collaboration requirements. The placement of the AV system(s) should be determined once the desired layout is complete to maintain ergonomic sightlines for operational efficiency.

Video Management Systems. Modern video management systems can be a key component of advanced SA for those inside and outside the EOC. The ability to share and view information between positions via wall-mounted systems, conference or breakout rooms, and even mobile devices increases the efficiency of decision making and assists in limiting unnecessary personnel inside the EOC. Many video management systems will include collaboration tools for interactive information sharing between mobile sites and the EOC, allowing for quicker updates and command disbursements.

Suggested "Low-tech," Traditional Displays. Although the advances in modern technology are drastic and have increased SA exponentially, there is still a recognized need for traditional displays and equipment within an EOC. Standard whiteboards, laminated setup instructions, printed procedure manuals, and physical incident command forms (with appropriate writing implements) are still invaluable for some incidents. Although traditional displays and forms can lead to longer reaction times and reduced information capture/sharing, they are essential for incidents where there are power issues within the EOC.

Specific Human Factors and Ergonomics Considerations for Furnishings and Equipment

This section will describe the furniture requirements for successful interactions between operators and their physical environment to maximize SA and performance. This section assumes that operators will be able to move and change posture during EOC operation. The recommendations also assume that operators will vary in size and that the adjustability will accommodate a range of staff sizes (5th percentile female to 95th percentile male).

In general, when designing a workstation, it is recommended to focus on the personnel and equipment independently of the physical workstation itself. Ideally, the console should be designed according to the ergonomic placement of the equipment, rather than the other way around (see Figure 6-7). In an EOC there is potential for several different operators to work at different times at the same workstation. Flexibility of all aspects described in the sections below will allow for adjustability of equipment for users of varying sizes.

Computer Equipment. Placing the computer equipment to be used by each staff person in the optimal ergonomic reach and visual zones for each person will improve performance and will help determine the work surface area that is required along with the EOC layout options.

Computer Equipment—Work Surface Heights. The optimal location of computer equipment depends in part on the anthropometric dimensions of the individual staff person. For the United States, the average woman is 5′4.2″ (163 cm) tall, and the average man is 5′9.6″ (176.9 cm). However, that range of human sizes varies considerably beyond this, such that for a U.S. woman, the height varies between a 5th percentile (4′11.8″; 152 cm) and a 95th percentile

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Source: Evans Consoles.

Figure 6-7. Prioritizing the workstation design.

(5′8.5″; 173.9 cm) and for a U.S. man this varies between a 5th percentile (5′4.8″; 164.5 cm) and a 95th percentile (6′2.5″; 189.3 cm) (PeopleSize, 2008).

Seated Working. Seated work is less tiring than standing work, but it requires that the staff person also has a chair, which increases the amount of space needed for each workstation. For seated working the work surfaces can either be:

- **Seated fixed height**—Operators are not all a standard size so when a work surface for seated work has a fixed height then it will result in variations in personnel posture depending on their seated height. In general, for optimal keyboard and mouse position on a fixed desk or a fixed table, the surface height should be between 23.2″–28.3″ (59–72 cm) high.
- **Seated adjustable height**—For optimal keyboard and mouse position on an adjustable height desk, the surface height should be adjustable between at least 21.6″–28.7″ (55–73 cm) high.

Standing Working. The advantage of standing work is it requires less space because of the lack of a chair. The disadvantage is that over time standing is more tiring than sitting. For standing work, the work surfaces can either be:

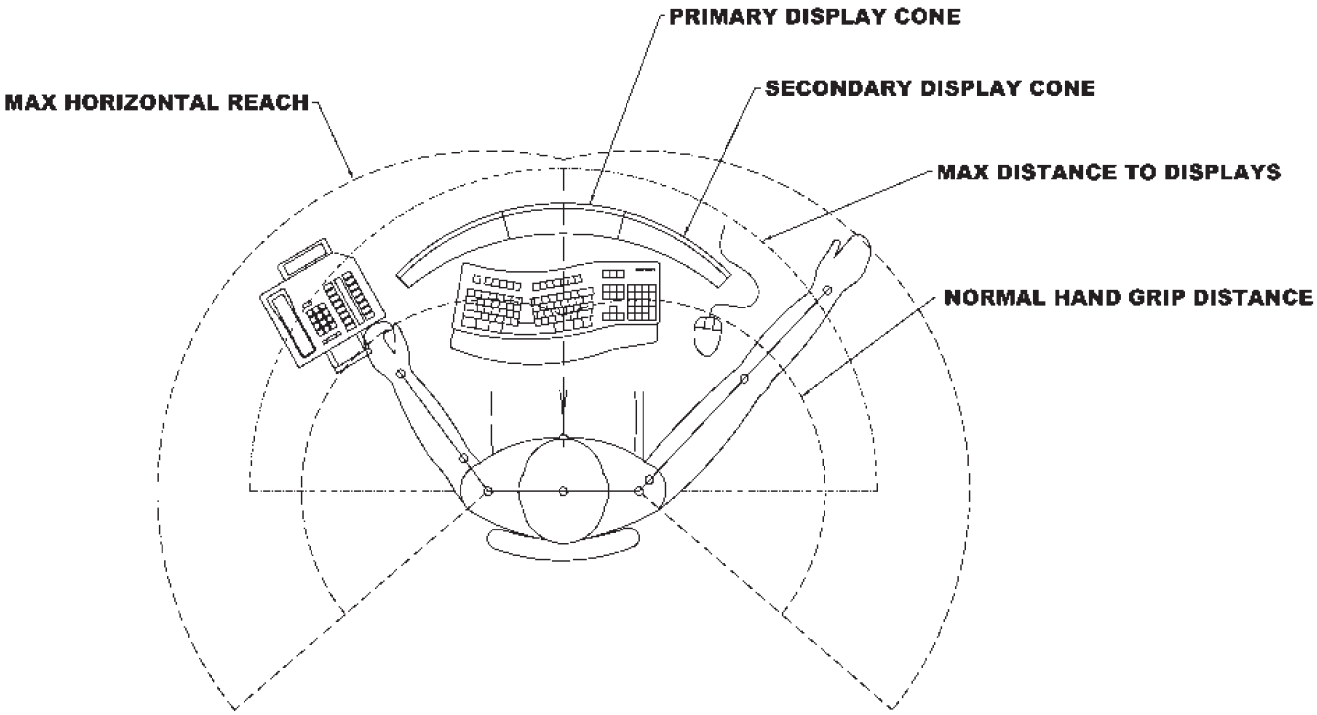
- **Standing fixed height**—Operators are not all a standard size so when a work surface for standing work has a fixed height then it will result in variations in personnel posture depending on their height. In general, for optimal keyboard and mouse position on a fixed desk or a fixed table, the surface height should be between 39.4″–43.3″ (100–110 cm) high to accommodate either female or male operators.
- **Standing adjustable height**—For optimal keyboard and mouse position on an adjustable height desk, the surface height should be adjustable between at least 36.2″–46.8″ (92–119 cm) high to accommodate either female or male operators.

It is possible to retrofit a sit–stand height-adjustable work surface to a regular table or desk surface so that staff can alternate between working sitting and working standing to better maintain their performance and comfort over time.

Equipment and cabling requirements also need to be accommodated by the workstation design. Separation of power and data cabling, as well as accessibility for efficient maintenance, all need to be considered during the design process.

Computer Equipment—Horizontal Work Area. The horizontal work area is typically organized for reach and for vision (Figure 6-8). For reach, the horizontal work area is organized into a primary zone of convenient reach where usual work tasks are performed, e.g., for computer

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Source: Evans Consoles.

Figure 6-8. Horizontal ergonomic work areas.

work, this is where the keyboard, mouse, or other input device should be located; a secondary zone of convenient reach is one where occasional work tasks are performed, e.g., for computer work with separate displays, the secondary zone is where displays should be positioned for comfortable viewing. The remaining horizontal space is non-work area, which often is used for storing infrequently used items.

If a person is using more than one screen, then a curved arrangement of monitors can be used, and this can accommodate two, three, four, or more computer monitors. The horizontal area required will depend on the number of screens being used. The size of the work surface area will be affected by whether a desktop, laptop, or tablet is being used to perform the task.

- For a desktop computer, the design of the keyboard and other input devices, such as a mouse or a trackball, will affect the required work surface area, as well as the size and number of monitors. In addition, the amount of hard copy paperwork required and any other required technologies, e.g., landline phone, printer, etc., will affect the work surface size, design, and layout.
- If the workstation is a console, then it is important to determine the console base by considering the equipment to be housed within the console. If system units are to be used at the console, how many computers will need to be accommodated? Are there any specific storage requirements at the console position? Will the console be capable of a sit/stand adjustment?

The horizontal work area should allow for a comfortable viewing distance to the computer screen(s). For those with normal vision, or normally corrected vision, a single computer screen should be centered on the head position of the staff person so that they are looking straight ahead. The screen should be positioned at a comfortable distance between 20″ (50 cm) to 30″ (76 cm), depending on the size of the critical information to be viewed. Ideally, if a separate screen is being used, as with a desktop computer, then this should also be on an articulating arm to allow easy position adjustment. If the screen is that of a laptop computer then it is probably too low for a

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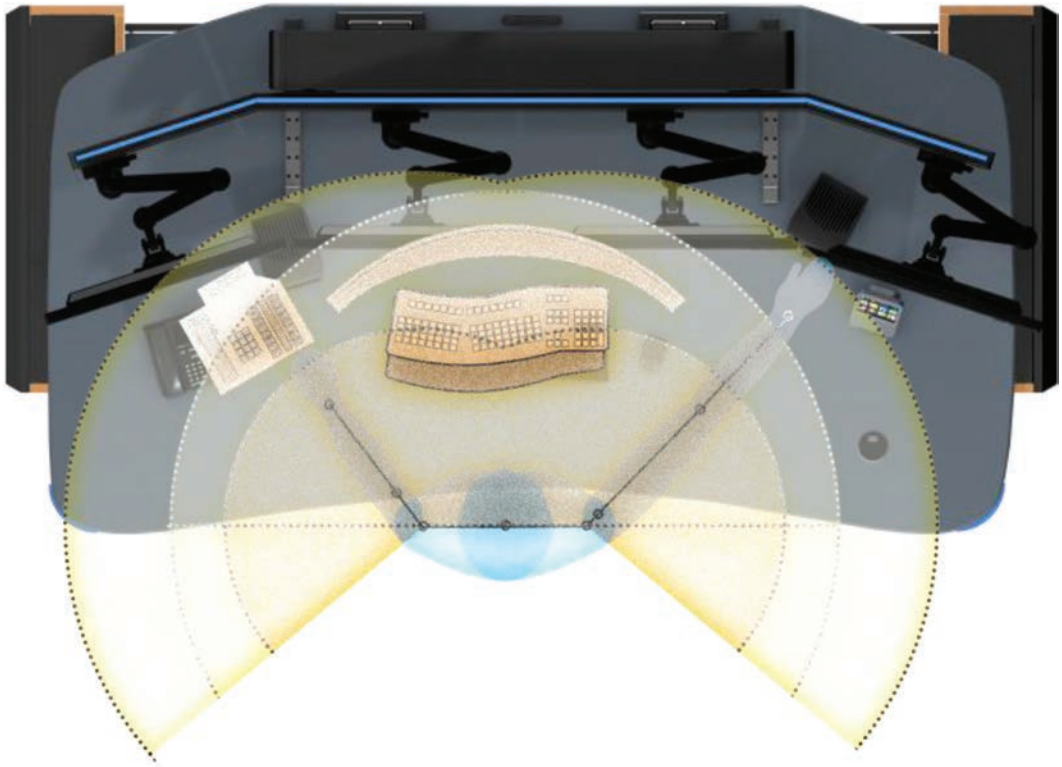
fixed height work surface, causing uncomfortable neck flexion. Ideally, a laptop screen should be connected to an external monitor for easier viewing or the laptop should be placed on a riser to elevate the screen. An external keyboard and mouse may be required for the laptop in this case.

If two screens are being used with one as a primary screen and the second one as an advisory screen, the primary screen should be positioned as described above with the secondary screen placed either to the right or left of the primary screen and at an angle so that a small horizontal head turn will bring the advisory screen into clear view. If two screens are being used and they are equally important, then the two screens should be angled on either side of the midline of vision so that they form a gentle V-shape. If this is the configuration, then choosing computer monitors that have a very small bezel is recommended.

If more than two screens are used, they should be arrayed in a curve to follow the line of vision. The easiest guideline for this is to sit upright, fully extend an arm horizontally with middle finger out pointing, and they should almost touch the center of each of the screens as the arm is moved from left to right. See Figure 6-9.

Computer Equipment—Vertical Work Area. The vertical work area is typically organized into a primary zone of convenient reach where usual work tasks are performed, typically at elbow level. For computer work this is the height of the work surface where the keyboard, mouse, or other input device should be located.

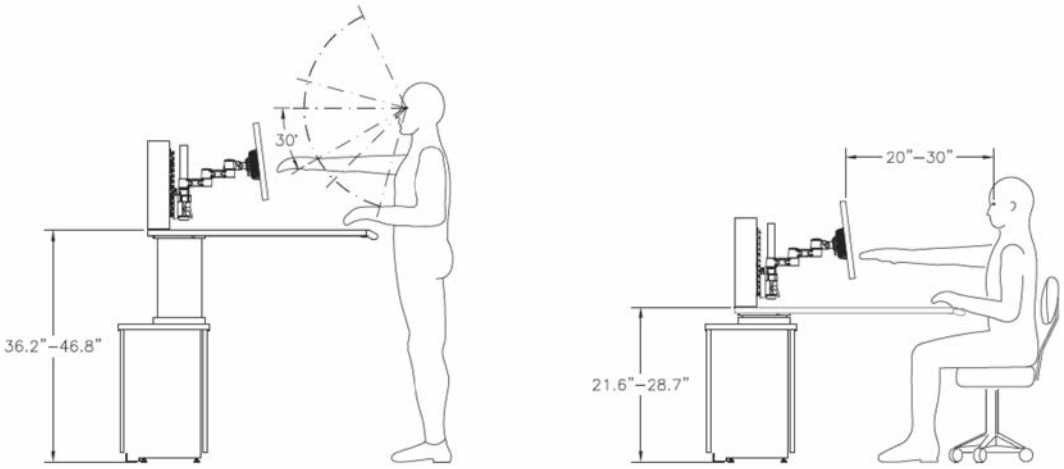
For optimal visibility, the computer monitor(s) should be positioned so that the center of the monitor is below the horizontal ear and eye-line (an angle of around 20° below the horizontal). The screen(s) should be adjustable so that it can be tilted for clear vision without any reflected glare from overhead lighting or windows. See Figure 6-10.



Source: Evans Consoles.

Figure 6-9. Workstation with two or more screens.

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Source: Evans Consoles.

Figure 6-10. Vertical ergonomic work areas.

Seating

Ergonomic Office Chair. If a person will be sitting to undertake much of their work in the EOC, then it is important that they are sitting in an ergonomic chair that can be adjusted to fit their body. The criteria for an ergonomic chair are specified in national standards (e.g., USA: American National Standards Institute (ANSI)/Human Factors and Ergonomics Society (HFES) 100-2007 Human Factors Engineering of Computer Workstations; Canada: CSA-Z412-00 (R2016), Guideline on Office Ergonomics). The following are the basic criteria for a suitable ergonomic chair:

- Easy to adjust from a seated position, comfortable, and appropriate to the task being undertaken;
- Seat height-adjustable, preferably utilizing a gas lift for ease of adjustment;
- Seat pan with a curved, "waterfall" front edge, to minimize pressure on the underside of the thighs;
- Backrest that is height, angle, and depth adjustable and provides good lumbar support;
- Covered by a breathable material;
- A five-star base for stability and safety; and
- Chair casters appropriate for floor covering.

Arm rests are optional. They help decrease the forces on the shoulders and back but can limit access to the desk. Some workers prefer to rest their forearms on the work surface. If provided, it is preferable the armrests are height- and position-adjustable. If the chair has arms, then it is important that these can be lowered so that they do not interfere with access to the work surface.

If a person must look at information displays for long periods of time, (especially for regular and social media monitoring) and these are mounted above head level, then the chair should provide a high back with head and neck support so that the displays can be comfortably viewed from a reclined seated posture.

Alternative Seating. The EOC may have alternative seating designs. If a person is standing to do work, then a stool can be provided for occasional sitting. If a person needs to move locations, then a stool with casters can be provided. If operators need to hold meetings fewer than 15 minutes long, then these can be undertaken while standing. For meetings lasting longer than 15 minutes, some type of chair is recommended. This can be a fixed height conference chair or a stool if the meeting is less than 60 minutes, but for longer meetings an ergonomic chair should be available.

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Other Furniture Considerations. If a person must refer to paper documents, especially any large format paper documents or heavy manuals, then adequate work surface should be provided. This can be achieved using tables or other horizontal work surfaces.

If there is a need for the storage of any materials in the EOC, then adequate storage should be provided (e.g., file drawer, mobile file pedestal).

If any food or beverages are to be prepared or consumed in the EOC then adequate equipment should be provided (e.g., refrigerator, coffee maker). If food is to be prepared or consumed in the EOC then adequate surfaces for both should be provided.

The furniture in the EOC should be configured so that operators have adequate clearance to adjust their furniture and to move around the EOC as required. The EOC spatial layout should be optimized to facilitate this ease of movement as well as ease of ingress/egress for the EOC.

Environmental Considerations

Lighting, acoustics, technology, and furniture design will be discussed in detail, and recommendations will be made which highlight the major elements of an effective airport EOC design.

Windows. The addition of windows to an EOC is recommended for psychological and physiological reasons, however, while providing a view to the outside of the EOC, windows also can create a significant difference in luminance within the room, resulting in glare and a possible drop in performance. If windows are not a primary source of information, the layout should be designed so that they are neither directly behind nor in front of the EOC operators. User-controlled blinds are also recommended to give personnel the ability to eliminate any potential sources of glare. Wherever possible, workstations should be located a minimum of 9'10" from any windows.

Lighting Design. Consideration should be given to the different visual demands expected of the EOC operators and their ability to perceive information from electronic screen sources and written text. The ability to perceive and react to both stimuli in short- and long-term activations is critical in the EOC facility. Lighting should be designed to allow flexible levels through dimming controls and zoning in different task areas. It is recommended that illuminance levels on works surfaces be maintained at recommended levels. This can be best achieved using a combination of ambient and task lighting. Please see Section 7 for more details.

Acoustic Design. Understanding how operators work within a given area will determine how acoustics need to be addressed. Currently, dissatisfaction with noise levels is believed to be one of the biggest disruptors of performance, yet it is often one of the last issues addressed in the design of buildings. The goal of acoustic treatment is to maximize the operators' SA of alarms and processes being monitored while also minimizing noisy distractions, and this includes distracting conversations. It is important that auditory alarms are discriminable from one another and against the background noise level in the EOC. It is crucial to position operators to optimize operational adjacencies and collaboration tasks, which requires an in-depth understanding of their workflow and interactions.

The ideal ambient noise level in an emergency operations environment is 30–35dB (ISO). The potential for building components (walls, ceilings, doors, windows, etc.) to contribute to minimizing sound transmission from space to space is significant. Please see Section 7 for more details.

See Figure 6-11 for a checklist on operational and ergonomic assessment.

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Checklist: Operational and Ergonomic Assessment

This checklist outlines the steps to be taken to complete an operational and ergonomic assessment if an outside advisor has not been consulted.

Operational Analysis

- ❑ Clarify goals for EOC, including operating procedures and requirements
- ❑ Complete observation, interviews, and current operations assessment activities
- ❑ Complete a current technology assessment and consider future plans
- ❑ Identify recognized constraints (budget, physical, etc.)
- ❑ Define the typical length of EOC engagement
- ❑ Define the typical shift length for EOC position

Operational Planning

- ❑ Define task requirements for each position within the room
- ❑ Identify visual requirements for personnel within the room
- ❑ Identify verbal communication requirements among personnel within the room
- ❑ Identify auditory requirements within the room (alarms, radios, etc.)
- ❑ Identify visual requirements to equipment within the room
- ❑ Define growth requirements for the facility
- ❑ Develop EOC model (hot, warm, cold, or multi-use facility)

Identify Position Requirements

- ❑ Define number of console positions
- ❑ Define the equipment that needs to rest on the work surface for each position
- ❑ Define the equipment needed in each workstation for each position
- ❑ Identify the need for storage furniture within the room
- ❑ Identify any requirements for mobility of positions and equipment
- ❑ Decide if task assignment and shift length require a seated or standing position

Identify Sightline Requirements

- ❑ Identify all shared visual displays within the room
- ❑ Identify facility constraints for AV screens (including any ocular acuity distances from AV manufacturer)
- ❑ Complete a video management system needs assessment
- ❑ Define redundancy plans

Conceptual Design

- ❑ Select optimal layout for EOC based on recognized operational plan above
- ❑ Select optimal furniture and equipment based on recognized operational plan above
- ❑ Determine placement of shared devices
- ❑ Identify HMI mapping needs
- ❑ Select video management system
- ❑ Define position layouts for EOC

Verification of Concept

- ❑ Conduct ergonomic assessment of proposed concept
- ❑ Conduct traffic flow analysis
- ❑ Conduct communication flow analysis

Source: Faith Group, LLC.

Figure 6-11. Operational and ergonomic assessment checklist.

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SECTION 7

Architectural Considerations

Introduction

This section addresses how to bring into physical form the goals established in the planning process outlined in Section 4, at a site established with the guidance of Section 5, and incorporating the human-centered design principles of Section 6. Architectural design considerations encompass initial analysis of known goals and data, leading to programming of functional and space needs, and finally integration into design of physical spaces and/or buildings.

Architectural development goes together with site selection (see Section 5). Major architectural considerations should inform site selection, and the building design must respond to site forces, including building orientation, access, and so forth. In addition to site issues, initial design begins with programming space needs and adjacencies. Involve the various stakeholders early in design, including EOC participants as well as facilities and maintenance staffs, and so on. Involve multiple design team disciplines, such as architects, electrical engineers, mechanical engineers, security consultants, IT consultants, and sustainability consultants early to achieve an integrated building design. EOCs have many unique design considerations.

Adjacencies

Smooth operation of an EOC involves cooperation among diverse participants. Plan to facilitate cooperation by organizing functional spaces for convenient interaction. This can be realized at multiple levels. Some agencies, such as police, fire, Transportation Security Administration (TSA), and local emergency management may determine that a JOC is the best approach to sharing information and support. Adjacent centers will have to be carefully considered for threats and vulnerabilities. While adjacencies can be helpful, they may also represent a higher valued target.

When possible, consider locating the EOC near or adjacent to functions that can aid in its mission. An adjacent airport communications center could simplify efforts during an emergency to stay connected to critical information, and allow individuals involved with both operations to devote attention to both.

There can be advantages to locating the EOC near the daily work locations of personnel involved in EOC operations. This will allow convenient, almost immediate access to the EOC in the event of sudden emergencies, such as crashes. The official in charge of maintaining and managing the EOC would find advantages from proximity to the regular work area. This project's survey revealed instances of emergency management directors using EOC spaces or adjacent breakout spaces for daily office space.

Considerable attention is needed for adjacency of rooms that make up an EOC suite, including breakout and support spaces (discussed below). Some, such as a flexible-use conference room,

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might benefit from immediate adjacency, and possibly even direct visual connection by means of a glass wall or window, equipped with blinds offering separation when appropriate. Other spaces are more appropriately positioned somewhat further from the action. Certain agencies may prefer spaces that offer a degree of separation while still being near enough for interaction. Rooms set aside to offer a break from potential tense and prolonged operations, for resting or personal communication and the like, will benefit from at least acoustical separation.

Adjacency to facilitate coordination among personnel extends to the specific layout of the principal EOC facility itself. Characteristics of various types of layouts are outlined further below. One type of layout may emphasize hierarchical arrangements, and another will lend itself well to connecting teams of individuals into subgroups, for example.

Co-utilization

It is to be expected that many airports will leverage EOC facilities for alternative day-to-day uses when not needed for relatively rare emergency operations, if for no other reason than to reduce capital and operational costs. Despite this strong temptation, it is worthwhile to carefully consider the pros and cons of this major decision. When there is a determination to co-utilize spaces, it is important to consider how best to minimize the potential disadvantages of this choice.

Among the airports surveyed for this Guidebook, a minority of airports have dedicated EOCs. The officials responsible for such dedicated facilities were quick to point out that even for facilities that had significant disadvantages, such as inadequate size and inconvenient or vulnerable locations, given a choice they would much rather accept the inadequacies of their facilities than work with idealized facilities that, due to co-utilization, would require 30 minutes or more to activate.

Dedicated EOCs can start up for emergencies almost immediately. Wall displays, computers, communications, furniture, and equipment are ready for use. If devoting the entire EOC facility to emergency use is not possible, consider a hybrid solution, dedicating at least the principal, large room for EOC use, with adjacent breakout spaces utilized for conference rooms or similar uses. This can allow immediate activation of the EOC, while secondary uses that tend to have fewer specific technological requirements can be put into use shortly thereafter.

When co-utilization is necessary, make sure to prioritize the needs of the EOC. The most common shared use of the large, principal EOC facility is for a meeting or presentation space. Display monitors integral to emergency operations can easily adapt to daily presentation use. Acquire furniture that will meet the EOC requirements and can be quickly and easily configured for EOC use from the potentially different daily arrangements. Create adjacent storage rooms for needed furniture and equipment.

Where individuals will occupy offices intended for EOC use, perhaps for breakout rooms and the like, it can be difficult to adapt these spaces for different uses, especially quickly. When this is the case, it may be necessary to design rooms that are larger than would otherwise be necessary. This could allow a segment of the room to be outfitted with desk and files and other personal office accoutrements, while leaving adequate space for whatever is needed for the EOC operations, such as a small meeting table.

Emergency situations can occasionally persist for prolonged periods of time. Plan for alternative accommodations for people or functions displaced by emergency operations. The EOC can easily dominate its assigned area. If it is in an area that is ordinarily not within a secured area, security measures will cut off access. The facilities within the EOC area, including restrooms and

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other potentially shared spaces, will become unavailable, meaning alternative facilities must be provided for normal operations in the remainder of the building.

Room Organization

Airports of different sizes, in different parts of the country, with different organizational structures and different stakeholders, will be best served by a variety of approaches to floor plans.

Size

The size of an airport's operations will influence the size and complexity required for the EOC. First, identify emergency operations procedures to anticipate the needs. Determine the agencies and stakeholder organizations that may play a role and calculate the number of persons from each that may participate. Reach out to these organizations to confirm a mutual understanding of expectations. List the airport-related departments and personnel that may be involved. Confirm or establish SOPs as part of the incident action plan. The resulting combined list should provide the number of personnel that may need to be accommodated in the EOC broken down by group, role, or affiliation. Consider that, over time, there may be additional stakeholders such as increased airport staff, new airlines, changes in government agencies and so forth. Build in room for flexibility and adaptability. Although it may be true that many emergencies will not require the full range of participants, it is also true that regular training will likely include individuals from all the groups, plus some backups. Construction budgets are always a consideration—be aware that many, if not most, airports surveyed routinely remarked about inadequate space.

Once procedures are defined, the number of participants required to occupy the primary EOC room can be determined. The actual amount of space required will depend upon the number of participants, the way the room will be organized, and the geometry of the space available. Smaller rooms will be somewhat less efficient in terms of square feet per person due to requirements for circulation, etc. For all these reasons, the space requirements cannot be fully understood until design options are studied. However, for initial planning purposes, a rough rule of thumb can be helpful.

- ISO 11064 suggests 9 m^2 per working position for a control center (ISO 11064-4 Ergonomic Design of Control Centers Part 4). This is approximately 96 square feet per position. Since, for the EOC, the working positions are not permanently staffed and may have less equipment, for planning purposes, this area would likely include the EOC main room and some basic support spaces.

Consider life safety when planning an EOC location. Coordinate with local code requirements. Be aware that the main room of an EOC will typically be large enough to require two exit pathways. As an example, the 2015 International Building Code requires that for an assembly occupancy such as a conference room greater than 750 square feet in area there be at least two out-swinging exit doors, separated by a distance not less than 1/3 the diagonal measurement of the room (for a building with a full sprinkler system).

Layout

EOC primary spaces have tended to have furniture, displays, and equipment arranged as variations of several layout categories, each with its own advantages and disadvantages, and each being suited to an airport's needs.

The following illustrations are simplified. In addition to the furniture shown below, main EOC rooms may include workstations for communications specialists or others, and space for seating at the perimeter of the room.

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Boardroom

Among the most common arrangements, the boardroom or conference room layout, as its name suggests, has a linear central table with seats on either side, or in larger EOCs, a rectangular grouping of tables with seats at the perimeter facing inward.

This layout has the advantage of maximizing face-to-face contact and communication. If the table or groups of tables bow out at the center of the long sides, this can improve communication for those sharing the same side of the table. There can be a focus (as in a boardroom) on the head of the table, where the person or persons in charge of operations can be positioned (either at the head of the table, or at a separate, perpendicular head table), or on where the principal display screens can be located.

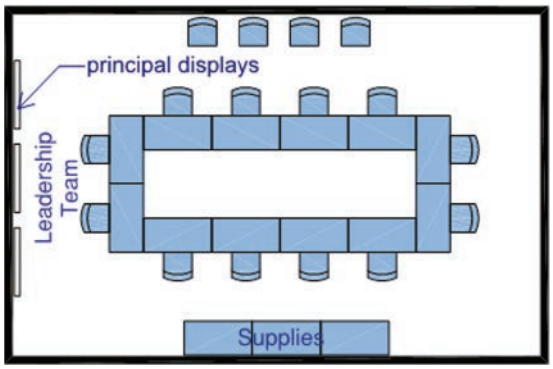
While cross-table, face-to-face communication is maximized in the boardroom layout, it can be somewhat difficult to see operations leaders and displays at one end, both because participants must turn sideways to do so, and because the arrangement can leave only a narrow view aperture to the end of the table since the view is blocked at the sides by the other participants. It can also be difficult to communicate with others at the other end of the same side of the table. Small group or team discussions involving more than two adjacent participants can be challenging and distracting to others nearby. Graphic displays along the side walls can be blocked from view by those at the opposite side of the table or may require turning around to see.

The boardroom layout, Figure 7-1, might lend itself best to small EOC facilities with fewer anticipated attendees.

Mission Control

The National Aeronautics and Space Administration (NASA) Mission Control Center, familiar to many from the Apollo moon missions of the 1960s and 70s, is a fitting model for this layout. Desks are laid out in rows, straight or slightly curved, facing one wall where principal displays and operational leaders are arrayed. This might also be called a classroom layout, except that this term conjures for most an image of rows of desks facing the narrow end of a room, whereas the more common mission control arrangement arrays desks along the long axis, facing the wider wall of a room.

This layout maximizes common attention to the central issues at hand, as displayed at the front of the room or as presented by the officials positioned at a head table. Visibility of the main presentations is enhanced for all, except for those at the head table who would have to turn



Source: Alliance.

Figure 7-1. Boardroom or conference style EOC layout.

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around, unless there are supplemental displays on other monitors. Communication between those at the head table and those at the rows of desks is straightforward.

Prioritization of central focus is at the cost of ease of communication among various parties. Some participants will have their backs to others, and side by side with other team members, or possibly separated by some distance, rather than face to face. Arranging rows along the long axis of the room (as opposed to classroom style) and curving the rows can reduce some of the problems, but a certain degree of communication obstruction is inherent.

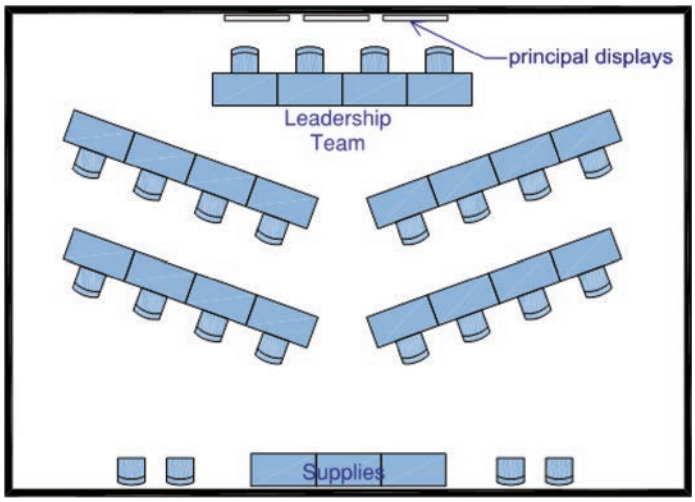
There is also some inherent inefficiency in the use of space due to the tables having seating on only one side, requiring more circulation and access space overall. This may be exacerbated if the tables are arranged at angles or in a curve. An example of a mission control layout is seen in Figure 7-2.

Teams

A "teams" layout places separate clusters of tables surrounded by chairs, grouping together participants with common roles or interests. There is likely still a head table for operational leaders and a focus for critical displays.

Smaller tables surrounded by participants with common duties or interests maximize their interface and can help facilitate common work efforts. During an incident, it will be natural for subgroups to develop, no matter what the layout, so the teams approach accepts that reality and optimizes for it. Focusing work efforts and related discussions at separate subgroup tables can reduce distractions for other groups.

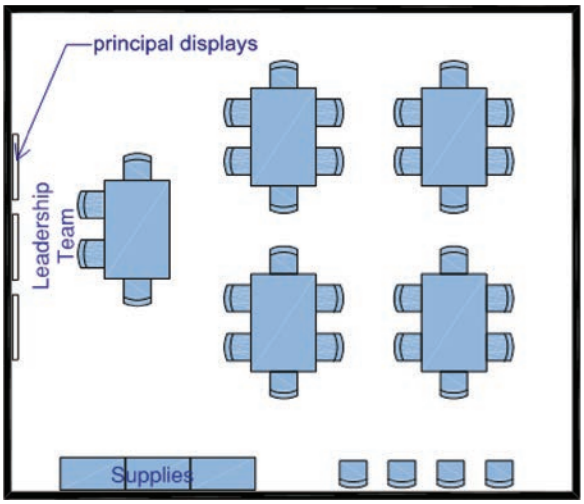
This arrangement clearly focuses on small group efforts at the expense of a centralized focus. It may be more difficult for leaders to get the attention of the groups, and some of those seated at the small tables are bound to be in less ideal positions for viewing the head table or main displays. With separate tables comes the challenge of participants being separate from one another; some participants will be facing the backs of others. In addition, despite efforts to anticipate the appropriate grouping of individuals, it may be difficult to organize groups that may not all be the same ideal size; circumstances may change the appropriate team dynamic, leaving participants "stranded" in the wrong team. Circulation space required around the table clusters can demand a relatively large square footage per person. An example of a teams layout is seen in Figure 7-3.



Source: Alliiance.

Figure 7-2. Mission control style EOC layout.

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Source: Alliiance.

Figure 7-3. Teams style EOC layout.

Radial Teams

In a "radial" teams layout, there is a head table and primary display wall, often along the shorter wall of a rectangular room. Linear or V-shaped tables are positioned along view lines radiating from the head table. The ends of the tables are left open for sightlines and are most effective if arranged as an elongated V-shape to open views for all participants. This is in some ways a hybrid of other layouts, and a variation on the sometimes-cited bullseye layout, which due to its problematic centralized head table is not suggested.

The individual linear tables offer the advantage of enhanced face-to-face communication, as well as allowing participants to be grouped for enhanced interaction. The linear tables may accommodate more seats than a traditional teams layout, allowing greater flexibility in group sizes, with possibly more than one team sharing a table. The radial orientation maintains some focus on the head table and primary displays.

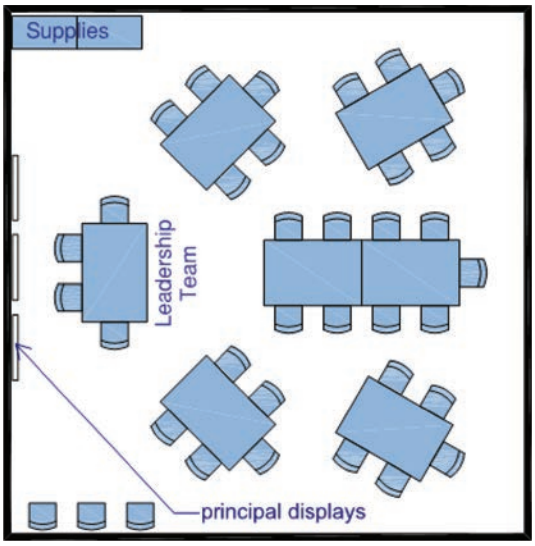
Radial layouts may be difficult to execute in other than large EOCs with many participants. Separate tables still have the challenge of being separate from other participants. A radial layout, as well as V-shaped table groupings, have inherent inefficiencies of space utilization, requiring a greater amount of space de voted to circulation due to the geometry fitting less neatly in what is usually a rectangular room. A trapezoidal room might be designed to work with a radial layout but might in turn create geometric challenges within the larger building layout. An example of a radial layout is seen in Figure 7-4.

Breakout Spaces

Regardless of airport size, take care not to assume that an EOC is composed solely of a single, large room. Although there is typically such a room at the heart of an EOC, it is always advantageous, and to some extent necessary, to have associated breakout and support spaces.

Every incident brings unique challenges and complexities, and the many participants are present because each has a role to play. Because all are working for a common cause, it is sensible to bring them together in a common space, but because there inevitably will be numerous separate but simultaneous efforts, there can be a great deal of noise, commotion, and distraction. It is therefore extremely helpful to have smaller, alternative breakout rooms for concentrated tasks. This can decrease the commotion in the main room, and at the same time offer the breakout

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Source: Alliiance.

Figure 7-4. Radial teams style EOC layout.

group a space with fewer distractions. These are typically set up as conference rooms of the size needed to seat the anticipated group.

A breakout room that is set aside but not assigned to any group can adapt to various needs. Other rooms might be assigned for specific agencies or groups, such as FBI, TSA, or airline liaisons. There may or may not be assigned seating as well as in the main EOC room for some or all the parties, depending on the anticipated interaction. A room immediately adjacent to the main EOC room might be set aside for conferencing by principal participants and leaders.

During a prolonged and hectic operation, it can be beneficial to have space available to briefly get away from the action, to have a side conversation, make a private phone call, eat a meal, or simply take a few moments to relax. Although one would not want to be too far from the action, this area should have good acoustic separation.

Joint Information Center (JIC)

A JIC adjacent to the EOC allows organizations to gather and get immediate updates from the EOC manager and craft responses to be released to the media in a quiet and private setting. The square footage will depend on the complexity of events at any specific airport and the number of stakeholders who would need to meet in a JIC. Ideally, this is practiced by airports during the required triennial drill exercise for a worst-case scenario event or a real-world event has taken place and those historical numbers for JIC responders should be documented by the airport.

Press Briefing and Work Area (Remote)

Contrary to a JIC, press briefing spaces should not be located near the EOC. Ideally, they are remotely located either outside the terminal or, if need be, in a structure such as an aircraft hangar or FBO. Press briefings often can be chaotic, loud, and disruptive. Several pre-established areas should be designated for the media who should be informed of each location. Have the locations named and shown on a map to be distributed to news agencies so that one of the first messages released is where they should respond. Ideally, equip some of these locations with external power for media personnel to be able to plug in equipment. This keeps many of the media responders sequestered in one area, except for any organized tours. Feed the media information regularly, even if it is to say there are no updates.

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Family Assistance Center (Remote)

Airports need to have a pre-established first-response and subsequent long-term area for friends and families that is not near the EOC. The family assistance center is not the sole responsibility of an airline, and, in fact, if the event is not airline related but perhaps a natural or terrorist driven disaster, it is fully incumbent on the airport to care for families and other loved ones. A family assistance center, particularly early on in a catastrophic event, is often very loud and emotionally charged. Several sites for family assistance response should be coordinated in advance to allow for flexibility depending upon the type and location of the event. Facilities to consider may include airline club lounges that are secure and usually in the sterile areas, making it difficult for unwanted media to access. They could also be in a hangar or FBO facility where airfield access is also difficult for media to penetrate.

There should be a secondary long-term briefing and quiet area for the families, perhaps at a nearby hotel. Arrangements should be made in advance, and space immediately reserved upon the catastrophic event taking place to secure adequate facilities in what will be a space-competitive environment with outside responders (NTSB, FBI, media etc.). Some hotels will make agreements with agencies to ensure that space for friends and families is available. Once national and international media are made aware of a catastrophic event, they may immediately book available nearby hotel space for days or even weeks.

Considerations for short- and long-term family assistance facilities should include:

- No visual sightline of the scene from the facility,
- Highly secured, with either an ACS or with law enforcement,
- Large space with private rooms for counseling and grieving,
- Adequate restroom facilities,
- Access to food and beverages, and
- Able to support communications needs.

For further information on family assistance centers, this Guidebook references *ACRP Research Report 171: Establishing a Coordinated Local Family Assistance Program for Airports*, which provides guidance to airport personnel when assisting victims and families affected by an aviation disaster. This report incorporates practices for planning an effective response while coordinating with different partners. The guidance is adaptable to both general aviation and commercial service airports of any size. The report includes a description of key terminology, federal regulatory and statutory requirements, history and background of the Aviation Disaster Family Assistance Act, and development of a strategic plan for creating and implementing a local airport victim and family assistance program.

View the toolkit that includes customizable checklists and forms airports can use to support their family assistance program, training courses that provide an overview of the guidebook, and Homeland Security Exercise and Evaluation Program-compliant materials at http://www.trb.org/ACRP/ACRPReport171.aspx.

Support Spaces for EOC

Sanitation and Hygiene. Access to restrooms is an obvious need. Make sure that facilities are adequate, accommodate all gender needs, and are accessible for disabled persons. Keep in mind that the number of occupants in an emergency operation may be a good deal higher than there might otherwise be in the building or area, and the fixture count should be appropriate for the higher number. The operations can be long-term and having shower facilities can be a plus. Restrooms should be maintained at a lower air pressure than other areas to control odors. Make sure access to restrooms does not cross security barriers, requiring escort.

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Food Service and Eating. Anticipate how food service will be provided. Will meals be prepared off-site and delivered to the EOC, or does there need to be some capacity for preparation? Even if prepared food is to be delivered, it is still advantageous to have common break room facilities, with a sink, coffee machine, and microwave. Will participants be expected to eat at their stations, or will there be a break room with adequate tables and seating to allow for this function in shifts? If there is a flexible, unassigned breakout space, it could serve as a dining area, but beware that circumstances may be such that the room is occupied when needed for eating. Kitchen and break rooms should be acoustically separated from the general EOC areas.

Resting and Sleeping. With long-term emergency operations, places to sleep, or at least rest, are invaluable. Whereas a dedicated location with a bed, couch, or reclining chair is ideal, having such furniture in a break room, flex space, or quiet room can still be helpful. Having foldable foam mattresses in storage can serve additional people if unusual circumstances demand.

Private, Quiet Room. In the stress and commotion of emergency operations, participants will appreciate an opportunity to briefly get away from the action, for a private side conversation, a personal phone call, or even some quiet time. Such a room need not be large, including perhaps a couple of chairs and small table. The room should be acoustically isolated from the general EOC areas.

Smoking Area. If a smoking area is provided, it should be separate from other occupied areas, including break rooms and private, quiet rooms. Adequate ventilation must be provided. Exterior areas might be designated, keeping in mind the amount of time required to get to and from the EOC. Exterior smoking areas should not be immediately adjacent to building entrances. Check institutional policies and local laws for smoking restrictions.

Storage. Storage space is needed for equipment and supplies for any EOC installation. For a dedicated EOC, it may be a simple matter of storing general supplies, backup equipment, hard copies of manuals, etc., and a small room, closet, or even storage cabinets might be adequate. It is preferable that EOC storage is in an area separate from other general or maintenance use.

When the EOC is going to be used for another purpose daily, there can be a great deal more storage space needed, depending upon what the daily use will be. Will a daily use conference room, for example, have all the same furniture that will be needed for the emergency function, so all that might be required would be possible rearrangement? If not, then there would need to be a room, preferably immediately adjacent, that can hold the needed EOC furniture, as well as accept whatever furniture might not be needed during EOC operations.

Special equipment, such as laptop computers and radios, need to be at the ready, including a storage arrangement that will allow them to be kept charged and periodically checked for readiness and system updates. Carts equipped with power and charging equipment can provide a dedicated storage location and simplify deployment. Ideally, wall-mounted display monitors and communications equipment can be permanently installed to reduce the time and complexity of setting up technology. Other wall displays, such as large aerial maps, will need convenient storage if they cannot be permanently in place.

Consider short-term storage that might be needed during emergency operations. Provide locations for participants to safely store their coats and personal belongings.

Printing and Copying. Technology and space should be provided for hard copy printing and copying. Include storage for printing supplies, and make sure computers are networked as needed to allow printing.

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Design Considerations for Airport EOCs

Furniture and Equipment

Select furniture in part based on ergonomic criteria as described in Section 6. Consider ways furniture and equipment selections enhance or inhibit the emergency mission.

Accessibility

All facilities need to accommodate the needs of a wide range of users. The Americans with Disabilities Act Accessibility Guidelines (ADAAG) set standards that should be considered the minimum baseline. New construction and remodeling in the United States is required by law to follow these standards. When existing facilities are used or are retrofit for EOC use, it is important to make sure all facilities are accessible. There are instances where existing buildings or spaces within have been "grandfathered in" if there have not been any changes in many years. There are other instances where certain spaces have not been made fully accessible due to the nature of the users, or because they were merely made "ADA ready," set up to allow later modifications to conform. Perhaps an upper level or mezzanine of an older firefighting facility was accessible only by stairs, since the users were all assumed to be able-bodied firefighters. Since those needing access to the EOC may have accessibility challenges, it is necessary that all facilities, including restrooms, break rooms, and so forth can meet their needs.

Protective Design

Once security and threat assessment has been conducted in accordance with Section 5, influencing the site selection, the potential perceived threats must be addressed through security measures and in the building design. The cause of the emergency that necessitated putting the EOC into service may also threaten the EOC itself.

Where the site has exposure to public ways (thus allowing a human threat), or exposure to the outside (thus allowing potential natural threats), determine what level of protection will be required. It is virtually impossible to prepare for every degree of every type of threat. There are specialty design consultants who can assist both with the evaluation of appropriate protection levels as well as with the associated construction requirements.

Human Threats. Some human threats (see Section 5) are addressed by limiting access with basic security systems, like access-controlled gates and doors and surveillance cameras. Beyond this are concerns about attacks against EOC facilities or against the buildings in which they are contained. Mitigation of threats by blast or by vehicle incursions is first handled by site measures around the building perimeter. Use physical obstructions such as bollards and earthworks (ditches and berms) to keep vehicles at a distance. That distance, plus the determined threat level, will determine the protective design strategies that will be needed. Walls, columns, even glass curtain wall systems can be designed to protect occupants from various levels of percussive blasts, though the systems themselves may be damaged.

Natural Events. Building design can also mediate against natural threats (see Section 5). Again, site measures come first, such as avoiding flood plains.

Design for the effects of earthquakes varies regionally depending on earthquake zone and local building codes. As with human threats, evaluate the degree of protection desired—is it wanted to provide protection beyond the code minimums? What risks do earthquakes pose to infrastructure, such as water and sewer, or natural gas?

Weather threats also vary regionally. Where flooding is a risk, whether from overflowing rivers or surface flooding, the EOC should be in a building safely located or protected from the threat, including assuring access to and from the building. In regions susceptible to hurricanes, physical protection can be provided, especially to glass windows and doors. This protection is

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best achieved with impact-resistant glazing and frames, or done in a way that is easily deployable, such as with exterior coiling shutters. In areas vulnerable to tornadoes, it is worthwhile considering measures to reduce the risk. The likelihood of a direct hit is almost certainly low, but the potential damage extraordinarily high. Accordingly, providing complete protection is difficult and costly. Glass windows and doors can be protected in a similar manner as with hurricane protection, though impact threats in a tornado are even higher. Exterior walls such as tilt-up concrete panel construction can offer a high level of protection. As with human threats, special design analysis, beyond what is normally required for conventional construction will be required to provide significant protection from weather-related threats, especially tornadoes.

Protective design measures can be costly and can create unfavorable design limitations (such as limiting windows and associated daylight and views). It may be determined that the level of protection for the EOC complex itself is required to be higher than that of the remaining building, and so more aggressive measures might be taken for that area. Consider co-location of other critical facilities, such as communications centers, in the highly protected area.

Basic building services are critical to maintain. Take care that emergency power is available for all critical functions from a generation source that is itself protected from threats and that has the capacity to serve for whatever extended period can be envisioned. Also, ensure that ventilation and conditioned air systems are served by emergency power and can be maintained through an emergency. Likewise, protect communications systems.

Indoor Environment

Lighting. Effective lighting is integral to the use of any indoor space. This is not as simple as providing high light levels, which, in fact, can be a source of visual discomfort, as well as a waste of energy. Different functions have different appropriate average light levels. Five foot-candles (fc) is adequate for corridors, 30–50 fc at desk level is appropriate for private office lighting, and 30 fc is recommended in conference rooms. Specific detailed work can benefit from up to 100 fc, best achieved where necessary by localized task lighting. How lighting is designed is even more important than the light levels. Light fixtures should be selected to avoid glare at viewing angles that can be visually distracting or reflect off computer or video monitors. This is accomplished by careful selection of fixtures and lenses. However, maximum cutoff of light angles is not necessarily the ideal. Visual perception is most comfortable and effective in response to appropriate contrast levels. It is best to provide general illumination of wall and ceiling surfaces so that the human eye does not have to constantly adjust between the well-lit work surface and a relatively dark background.

Light emitting diode (LED) light sources can result in energy savings and less frequent replacement. The technology also can offer a range of color choices. As incandescent lights are phased out, personal preferences are evolving toward acceptance of "cooler" (less yellow) light, closer to daylight, but light that is too cool generally seems unnatural indoors. LED applications are evolving that can change color, with warmer light in the early morning and evening (mimicking light at sunrise and sunset), and cooler in the daytime. Such systems could aid visual comfort during long-term events.

Lighting consumes a great deal of energy. Automated sensors can ensure that lights are used when they are needed. As opposed to occupancy sensors, which activate lights when someone enters a room, it is usually preferable to use vacancy sensors. A person must manually activate the lights, but the sensors will turn them off when the room has been vacant for a set period.

Daylight and Views. In most workplaces, access to daylight and exterior views are important features of a quality indoor environment. Studies show improvements to productivity with such conditions. With shorter-term use, views and daylight might seem less important in an EOC.

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Protective design measures might eliminate, or at least reduce, the availability of windows. When possible, these amenities can be beneficial, but daylight must be well controlled. Direct solar penetration must be prevented with physical barriers such as overhangs, sunshades, and window blinds to prevent glare.

Acoustic Control. EOC operations can be hectic. One of the greatest challenges to effective operation is minimizing distractions and maintaining good verbal communication. Poor room acoustics will exacerbate these challenges. With multiple simultaneous conversations, the most important measure is to create a high level of absorption. Typical acoustical panel ceilings are effective, as well as carpeted floors. Beyond these routine approaches, sound absorbing panels can be added to walls. Parallel walls can create an annoying condition called "flutter echo." Absorptive materials with a noise reduction coefficient (NRC) of 50–80 applied to the upper portion of each parallel wall can address this concern.

Sound masking, a low-level, constant ambient noise can help focus listening and dampen distracting outside conversations and noises. While this is sometimes achieved, often inadvertently, by the hum of mechanical equipment, systems are available utilizing, usually, ceiling-mounted speakers with sound that is designed and can be controlled for this purpose. HVAC sound levels for meeting spaces should not exceed a noise criteria/room criteria (NC/RC) of 30.[11]

Make sure between-room partitions are designed to contain sound in individual rooms and avoid noise from exterior sources. Extending partitions to the underside of floor or roof decks above is much more effective than relying on ceiling materials, as often done in offices with acoustic blankets near the edges. Install sound gaskets on doors. Sound transmission class (STC) is a measurement procedure for sound transmission. For normal privacy levels, STC 52–55 is recommended, with STC greater than 55 for confidential conditions.[12] A typical steel stud partition with gypsum board facing has a rating of STC 40. Performance can be increased with acoustical blankets in the stud space, with resilient channels or offset studs, and with special membranes. Other partition construction, such as concrete masonry, can be effective in reducing sound penetration, but might need treatments such as gypsum board over resilient channels to address the full range of sound frequencies.

Technology can also help reduce noise levels. Use a microphone and headsets to avoid broadcasting audio beyond those who need to hear it, and utilize breakout spaces to relocate subgroups that might be getting loud enough to be a distraction or hindrance to others.

Technology and Communications Integration

Technology and communications considerations vary from airport to airport depending on size, complexity, and vulnerabilities. At a minimum, an airport EOC should have at least two monitors and the ability to display a computer screen. The two monitors should be able to broadcast both news and weather. It is no longer so imperative to stage desktop computers today. Some airports maintain a bank of laptops at the ready, keeping them charged and with up-to-date software. For co-utilized space, laptops can be kept in a cart, ready to be rolled out for distribution. Some participants will bring their own laptops. Care should be taken about allowing outside laptops to connect directly to a network. Wi-Fi access will commonly be made available, allowing access to the Internet while protecting the local network. Consider a router specifically for the EOC so there is not competition for bandwidth.

When integrating technological equipment into furniture or buildings, be wary of solutions that are highly customized to specific equipment, such as displays that are built into walls. As technological change becomes ever swifter, the lifespan of technological hardware grows ever shorter. In a surprisingly short period of time, what seemed a cutting-edge merging of function

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and technology can become an anachronistic curiosity. Be mindful as well as to how intuitive a new technology or application will be to use, and how smoothly it is likely able to be integrated to work with other systems. This is not a suggestion to avoid new and useful technology. Simply look beyond the novelty and evaluate the best, most usable and adaptable way to integrate such technology and equipment.

A communication plan needs to be established and practiced for EOC operations. Communications includes activation of the EOC. There is software available for mass notifications that work well to notify individuals and groups of an event and what is expected concerning a response. Pre-defined response groups may help with early organization to accommodate an event that will be longer than a 24-hour period. A "Shift A" and "Shift B" approach works well with the use of 12-hour shifts. If the event occurs during the hours of Shift A, Shift B personnel know either not to come to the airport until their shift or to leave and get some rest to be prepared to come in and relieve Shift A. Ensure that only those required and requested personnel respond to the EOC.

Other communication considerations are within the EOC. It is imperative there be a single, designated person having contact with the field IC. More than one contact can cause inaccurate information and unnecessary workload. Consider a designated radio channel or phone line. EOCs can become very loud very quickly with phones, radios and multiple discussions taking place at the same time. Noise levels can be minimized using headsets for both radios and phones. A custom headset should be available for everyone who would regularly respond to the EOC. Phones with visible displays instead of a ringtone are also effective in reducing noise levels. Varied color lights can indicate exactly which position is being called, such as green for operations section, red for logistics section and yellow for planning section. The lights may be on the phone as well as elevated on a pole at the position to be easily seen around the room.

Resilience, Flexibility, Adaptability

Unanticipated change is not merely a concern with emerging technology. No matter the degree of planning, there are bound to be changes and unexpected developments or evolving situations. Regardless of technology, new or old personnel need to stay current, and the EOC needs to be used repeatedly so that persons are comfortable with the technology and properly trained. Staffing an IT person in the EOC helps with adaptability, especially with changes in technology.

The most obvious potential change is in capacity. Over time, emerging demands, addition or diversification of government agencies, added airport departments, additional airlines, airport growth, and so forth can add space and equipment needs. There also can be evolution of preferred procedures and room organization; having more than the minimum space will allow adjustments and experimentation. When space and budget allow, it is best to design somewhat larger than the minimum demanded by initial projections. Further, design can anticipate potential future expansion, especially with new construction. Siting, building orientation, exits, structure, and exterior wall materials are all considerations when designing with expansion in mind. It is important that this be a stated goal in the early design phases.

The concept planning section of the Guidebook (see Section 4) should be used periodically as a tool to help the airport evaluate current and future needs, and, when appropriate, to initiate the process of migrating to a different space or building a new one. The more the EOC is used, the easier it will be to assess evolving needs. Use of the EOC will develop a working culture for personnel and stakeholders to better understand when to respond, in what capacity, and how to work with each other during what is usually a stressful situation.

Figure 7-5 is a checklist for Architectural Considerations for Building an EOC.

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Checklist: Architectural Considerations for Building an EOC

Following is a simplified list of many of the important architectural considerations for developing a successful EOC.

Analysis

- ❑ Complete data collection, initial planning, and site selection processes (see earlier sections)
- ❑ Identify threat conditions related to selected site or site options
- ❑ Identify protective design measures responding to human and natural threats of selected site(s)
- ❑ Consider when it is feasible to take advantage of professional design consultation

Programming

- ❑ Identify desired adjacencies with other or related functions, such as communications facilities
- ❑ Consider whether a dedicated EOC is feasible
- ❑ If co-utilization is necessary, determine what the regular use will be, understanding that EOC function will have to take precedence when needed
- ❑ Identify backup EOC location(s) and take measures to provide portability of essential materials and systems
- ❑ Identify the number of potential EOC participants
  - ❑ Determine participating agencies and stakeholders, and number of participants for each
  - ❑ List participating airport departments and personnel
- ❑ Establish minimum area of basic facilities for initial planning purposes
- ❑ Identify desired breakout spaces, including both assigned and flexible spaces
- ❑ Identify important support spaces
  - ❑ Restrooms
  - ❑ Food service and eating
  - ❑ Resting and sleeping
  - ❑ Private, quiet room
  - ❑ Storage (more space needed if a co-utilized EOC facility)
  - ❑ Printing and copying

Design

- ❑ Consider advantages and disadvantages of the various EOC layout options
- ❑ Investigate furniture and equipment needs with attention to ergonomics (see Section 6)
- ❑ Coordinate with appropriate agencies and departments to ensure security needs are accomplished
- ❑ Coordinate with local codes and code officials to maintain egress and other requirements
- ❑ Ensure facilities are fully accessible, meeting ADAAG requirements at minimum
- ❑ Carefully consider the acoustic environment and take measures to reduce communication problems and stress
- ❑ Employ effective lighting strategies, also considering controlled daylight and views
- ❑ Carefully integrate technology, emphasizing ease of use and flexibility
- ❑ Establish and practice a communication plan
- ❑ Plan for resilience
- ❑ Plan for adaptability

Source: Faith Group, LLC.

Figure 7-5. Architectural considerations for building an EOC checklist.

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Sources Related to Key Technical Sections

Section 3 Roles and Responsibilities

ACRP Report 103: A Guidebook for Integrating NIMS for Personnel and Resources at Airports, H. Stambaugh, M. Argabright, H. Benaman, and M. Cheston, Transportation Research Board of the National Academies, Washington, D.C., 2014.

Section 4 Planning Processes

"Building and Exercising Your Command Center," Steven C. Davis, All Hands Consulting, www.davislogic.com/BuildandEXyourEOC.pps.

"Emergency Operations Center (EOC)—Design, Operation and Management," Dave Roop, TIFAC-IDRiM Conference, Oct. 2015, Delhi, India. http://www.academia.edu/18067421/EMERGENCY_OPERATIONS_CENTER_EOC_-_DESIGN_OPERATION_and_MANAGEMENT.

"Emergency Operations Center Assessment Checklist," FEMA (website) https://www.fema.gov/emergency-operations-center-assessment-checklist.

2000 Emergency Response: A Guidebook for First Responders During the Initial Phase of a Dangerous Goods/Hazardous Materials Incident, USDOT, 2000.

Airport Command Center, Phoenix Sky Harbor International Airport, Final Draft Report, Counter Technology, Inc., January 30, 2004.

Aviation's Continuing Critical Vulnerabilities to Vehicle-Borne Improvised Explosive Devices, Individual Suicide Bombers, Suitcase Explosive Devices Attack at Airports, Billie H. Vincent, October 2012, www.asi-transec.com.

Common Issues in Emergency Transportation Operations Preparedness and Response: Results of the FHWA Workshop Series, Houston, Nancy. Booz Allen Hamilton, Federal Highway Administration, 2007, 16p, http://trid.trb.org/view/807151.

Abstract: Between May 2002 and June 2005, the Federal Highway Administration (FHWA) and Booz Allen Hamilton conducted workshops on Transportation Operations Preparedness and Response in 30 regions across the United States. This report consolidates the common issues identified during the 30 workshops. Issues are not presented in priority order, but rather have been grouped in common categories. The categories are as follows: Interagency Coordination and Communication; Emergency Operations; Equipment; Intelligent Transportation Systems (ITS); Mutual Aid; Threat Notification, Awareness, and Information Sharing; Policy; and Threat and Vulnerability.

Continuity of Operations Plans for Transportation Agencies, Edwards, Frances L., and Goodrich, Daniel C. Mineta Transportation Institute; San Jose State University; California Department of Transportation; Department of Homeland Security; Research and Innovative Technology Administration, 2011, 684p, http://trid.trb.org/view/1124548.

Abstract: This report covers the research conducted to determine how to integrate Continuity of Operations/Continuity of Government Plan (COOP/COG) into the overall NIMS approach to emergency management for a state-level transportation agency. It provides an approach to align the COOP/COG element of the state DOT with the state DOT's emergency operations center (EOC). It provides checklists that show the

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Design Considerations for Airport EOCs

primary functions of each state-level transportation agency EOC position. It provides training materials on the system and the relationships among EOC positions.

Emergency Operations Centers Handbook, Federal Emergency Management Agency, CPG 1–20, May 1984.

Generic Continuity of Operations/Continuity of Government Plan for State-Level Transportation Agencies. Edwards, Frances, L., Goodrich, D. C., and Medigovich, W. M. San Jose State University; California Department of Transportation; Department of Homeland Security; Research and Innovative Technology Administration, 2011, 101p, http://trid.trb.org/view/1118138.

Abstract: The Homeland Security Presidential Directive 20 (HSPD-20) requires all local, state, tribal, and territorial government agencies, as well as private-sector owners of critical infrastructure and key resources (CI/KR) to create a Continuity of Operations/Continuity of Government Plan (COOP/COG). There is planning and training guidance for generic transportation agency COOP/COG work. This report covers the research conducted to determine how to integrate COOP/COG into the overall NIMS approach to emergency management, including a connection between the EOC and the COOP/COG activity.

Handbook of Emergency Management for State-Level Transportation Agencies, Edwards, Frances L., and Goodrich, Daniel C. San Jose State University, California Department of Transportation, Department of Homeland Security; Research and Special Programs Administration, 2010, 114p, http://trid.trb.org/view/918699.

Abstract: This report describes a research project intended to lay the groundwork for establishing priorities that would lead to a mature management capability for public agencies for emergencies, disasters, and catastrophes. State transportation agencies are required to have plans for the continuity of their government functions during any catastrophic disaster, as well as for the continuation of the essential services that they provide. This report outlines the hierarchy of emergency plans. Appendices include lists and templates.

Ideal EOC, http://www.emergencymgmt.com/disaster/Ideal-EOC.html.

Recommended Security Guidelines for Airport Planning, Design and Construction, TSA, May 2011.

TCRP Research Results Digest 87: Emergency Preparedness, Response, and Recovery in the Transit Industry, Transportation Research Board of the National Academies, Washington, D.C., 2008, 38p, http://trid.trb.org/view/859150.

Abstract: This digest summarizes the mission performed March 16–31, 2007, under TCRP Project J-03, "International Transit Studies Program." It includes an overview of the mission that investigated emergency preparedness programs in the public transportation systems in several Asian cities (Hong Kong, Beijing, Tangshan, Seoul, Kobe, and Tokyo). The report begins in Section I with general observations by the mission team about the state of transit emergency preparedness in the cities visited. In Section II, information gathered about the policies and programs in place in each of the transit environments is provided. Section III shows how the different transit systems manage key emergency functions, including the following: control centers, IT and telecommunications, interagency coordination, public communications, and evacuation planning.

Utility Systems Terrorism Countermeasures for Command, Control, Communications, Computer, Intelligence, Surveillance and Reconnaissance (C4ISR) Facilities, Department of the Army, 21 February 2006.

Section 5 Site Selection

Building and Exercising Your Command Center, Steven C. Davis, All Hands Consulting. www.davislogic.com/BuildandEXyourEOC.pps

Emergency Operations Center (EOC)—Design, Operation and Management, Dave Roop, TIFAC-IDRiM Conference, Oct. 2015, Delhi, India. http://www.academia.edu/18067421/EMERGENCY_OPERATIONS_CENTER_EOC_-_DESIGN_OPERATION_and_MANAGEMENT.

"Emergency Operations Center Assessment Checklist," FEMA (website). https://www.fema.gov/emergency-operations-center-assessment-checklist.

2000 Emergency Response Guidebook, USDOT, 2000

Airport Command Center, Phoenix Sky Harbor International Airport, Final Draft Report, Counter Technology, Inc., January 30, 2004.

Aviation's Continuing Critical Vulnerabilities to Vehicle-Borne Improvised Explosive Devices, Individual Suicide Bombers, Suitcase Explosive Devices Attack at Airports, Billie H. Vincent, October 2012, www.asi-transec.com.

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Bomb Threat Stand-Off Distances, FBI/NCTC, http://www.nctc.gov/witsbanner/docs/2009_report_on_terrorism.pdf

Building and Infrastructure Protection Series, Reference Manual to Mitigate Potential Terrorist Attacks Against Buildings, Department of Homeland Security, FEMA-/426BIPS-06, October 2011

Common Issues in Emergency Transportation Operations Preparedness and Response: Results of the FHWA Workshop Series. Houston, Nancy. Booz Allen Hamilton; Federal Highway Administration, 2007, 16p, http://trid.trb.org/view/807151.

Abstract: Between May 2002 and June 2005, the Federal Highway Administration (FHWA) and Booz Allen Hamilton conducted workshops on Transportation Operations Preparedness and Response in 30 regions across the United States. This report consolidates the common issues identified during the 30 workshops. Issues are not presented in priority order, but rather have been grouped in common categories. The categories are as follows: Interagency Coordination and Communication; Emergency Operations; Equipment; ITS; Mutual Aid; Threat Notification, Awareness, and Information Sharing; Policy; and Threat and Vulnerability.

Continuity of Operations Plans for Transportation Agencies. Edwards, Frances L.; Goodrich, Daniel C. Mineta Transportation Institute; San Jose State University; California Department of Transportation; Department of Homeland Security; Research and Innovative Technology Administration, 2011, 684p, http://trid.trb.org/view/1124548.

Abstract: This report covers the research conducted to determine how to integrate Continuity of Operations/Continuity of Government Plan (COOP/COG) into the overall NIMS approach to emergency management for a state-level transportation agency. It provides an approach to align the COOP/COG element of the state DOT with the state DOT's emergency operations center (EOC). It provides checklists that show the primary functions of each state-level transportation agency EOC position. It provides training materials on the system and the relationships among EOC positions.

Emergency Operations Centers Handbook, Federal Emergency Management Agency, CPG 1–20, May 1984.

Exploring Terrorist Targeting Preferences, Martin C. Libicke, Peter Chalk, and Melanie Sissons, The Rand Corporation, 2007

Generic Continuity of Operations/Continuity of Government Plan for State-Level Transportation Agencies. Edwards, Frances L., Goodrich, D. C., and Medigovich, W. M. San Jose State University; California Department of Transportation; Department of Homeland Security; Research and Innovative Technology Administration, 2011, 101p, http://trid.trb.org/view/1118138.

Abstract: The Homeland Security Presidential Directive 20 (HSPD-20) requires all local, state, tribal and territorial government agencies, and private-sector owners of critical infrastructure and key resources (CI/KR) to create a Continuity of Operations/Continuity of Government Plan (COOP/COG). There is planning and training guidance for generic transportation agency COOP/COG work. This report covers the research conducted to determine how to integrate COOP/COG into the overall NIMS approach to emergency management, including a connection between the emergency operations center (EOC) and the COOP/COG activity.

Handbook of Emergency Management for State-Level Transportation Agencies. Edwards, Frances L.; Goodrich, Daniel C. San Jose State University; California Department of Transportation; Department of Homeland Security; Research and Special Programs Administration, 2010, 114p. http://trid.trb.org/view/918699.

Abstract: This report describes a research project intended to lay the groundwork for establishing priorities that would lead to a mature management capability for public agencies for emergencies, disasters and catastrophes. State transportation agencies are required to have plans for the continuity of their government functions during any catastrophic disaster, as well as for the continuation of the essential services that they provide. This report outlines the hierarchy of emergency plans. Appendices include lists and templates.

Ideal EOC, http://www.emergencymgmt.com/disaster/Ideal-EOC.html.

Recommended Security Guidelines for Airport Planning, Design and Construction, TSA, May 2011

TCRP Research Results Digest 87: Emergency Preparedness, Response, and Recovery in the Transit Industry, Transportation Research Board of the National Academies, Washington, D.C., 2008, 38p, http://trid.trb.org/view/859150.

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preparedness programs in the public transportation systems in several Asian cities (Hong Kong, Beijing, Tangshan, Seoul, Kobe, and Tokyo). The report begins in Section I with general observations by the mission team about the state of transit emergency preparedness in the cities visited. In Section II, information gathered about the policies and programs in place in each of the transit environments is provided. Section III shows how the different transit systems manage key emergency functions, including the following: control centers, IT and telecommunications, interagency coordination, public communications, and evacuation planning.

Utility Systems Terrorism Countermeasures for Command, Control, Communications, Computer, Intelligence, Surveillance and Reconnaissance (C4ISR) Facilities, Department of the Army, 21 February 2006

Section 6 Ergonomic and Human Factors

2000 Emergency Response Guidebook, USDOT, 2000

A Design Language for EOC Facilities: "Some thoughts on basic design approaches by Art Botterell," (http://flghc.org/ppt/09-10/TS%20EM7.pdf – Governor's Hurricane Conference):

- Seating: Boardroom, Mission Control, Marketplace, Bullseye
  "Emergency Operations Center Project Development and Capabilities Assessment," State of Florida Division of Emergency Management.
- Architects Design Group, Inc., http://www.floridadisaster.org/Response/engineers/eoc/eocguide.pdf.

Airport Command Center, Phoenix Sky Harbor International Airport, Final Draft Report, Counter Technology, Inc., January 30, 2004.

Architectural Lighting Design, 3rd ed., Gary Steffy, New York: John Wiley & Sons, Inc., 2008.

*Building and Exercising Your Command Cente*r, Steven C. Davis, All Hands Consulting, www.davislogic.com/BuildandEXyourEOC.pps.

"Emergency Operations Center Assessment Checklist," FEMA (website), https://www.fema.gov/emergency-operations-center-assessment-checklist.

*Emergency Operations Center (EOC)—Design, Operation and Managemen*t, Dave Roop, TIFAC-IDRiM Conference, Oct. 2015, Delhi, India, http://www.academia.edu/18067421/EMERGENCY_OPERATIONS_CENTER_EOC_-_DESIGN_OPERATION_and_MANAGEMENT.

Emergency Operations Centers Handbook, Federal Emergency Management Agency, CPG 1–20, May, 1984.

Getting the Big Picture. *ITS International*, Volume 20, Issue 3, 2014, pp. NA5-NA6, http://trid.trb.org/view/1322448.

Abstract: A new emergency management facility in Morris County, New Jersey, began operations last fall, using advanced visualization and communications technology. Planning for the facility began in 2007, and it serves as an addition to a public safety complex that serves most of Morris County. A variety of sources relay information to the facility, allowing situations to be monitored in real time. Specialists aggregate data feeds to provide useful visual information. The emergency center must be able to function under any condition, from earthquakes to electrical failure. The building has redundant systems, so that staff can continue working through catastrophic events. There are plans for additional emergency centers to be built in the region, which will be able to share networks with the Morris County facility, allowing emergency centers across the state to help each other in emergencies.

Ideal EOC, Emergency Management, http://www.emergencymgmt.com/disaster/Ideal-EOC.html.

Information Sharing Guidebook for Transportation Management Centers, Emergency Operations Centers, and Fusion Centers. Booz Allen Hamilton, HNTB Corporation, Federal Highway Administration, Houston, N., Wiegmann, J., Marshall, R., Kandarpa, R., Korsak, J., Baldwin, C., Sangillo, J., Knisely, S., Graham, K., and Easton, A. V., 2010, 144p, http://trid.trb.org/view/980782.

Abstract: This guidebook provides an overview of the mission and functions of transportation management centers, emergency operations centers, and fusion centers. The guidebook focuses on the types of information these centers produce and manage and how the sharing of such information among the centers can be beneficial to both the day-to-day and emergency operations of all the centers. Challenges exist to the ability to share information, and the guidebook addresses these challenges and options for handling them. The guidebook also provides some lessons learned and best practices identified from a literature search and interviews/site visits with center operators.

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Meeting the Expanding Operational and Security Related Demands on Operations Control Centers. Lengyel, Dennis. 2008 American Public Transportation Association (APTA) Rail Conference, American Public Transportation Association, 2008, 6p, http://trid.trb.org/view/874578.

Abstract: This paper describes how transit agencies have always relied on their operational control centers (OCCs) in order to improve the efficiency of their operations. Today the OCC and the operational staff are being tasked to play a role in security and incident management on both the railroad and in the region. These incidents can range from service disruptions to major weather or security-related events. To meet these new demands, the OCC must be designed to collect, process, and display information from more than the traditional train control and security control and data acquisition–/(SCADA-) related equipment on the property. The OCC must also function as a source of real time information for customer service personnel, transit police, and local and regional EOCs. In addition, the threats against transit systems–both physical and cyber—have dictated that OCC designs also include increased emphasis on security considerations. Although the OCC is not safety critical from a signaling perspective, these facilities have become mission critical with a growing number of properties implementing backup centers. These new demands on the functionality and performance of the OCC have led to new architectures and technologies used in recent OCC designs and implementations.

People Size 2008 (anthropometric database). Open Ergonomics. http://www.openerg.com/psz/anthropometry_data.html.

Room-to-Room Privacy and Acoustical Design Criteria, J. T. Weissenburger, SV Sound and Vibration, 2004.

S*tandard Guide for Emergency Operations Center (EOC) Development*, ASTM, http://www.astm.org/DHS/E2668.pdf.

The Lighting Handbook: Reference & Application, 10th Ed., Illuminating Engineering Society of North America, 2011. www.lightingdesignlab.com/sites/default/files/pdf/Footcandle_Lighting%20Guide_ Rev.072013.pdf

UFC 4-141-04 Emergency Operations Center Planning and Design, with Change 1, DOD, 9/1/2008, http://www.wbdg.org/FFC/DOD/UFC/ufc_4_141_04_2008_c1.pdf.

Unified Facilities Criteria (UFC) DOD Minimum Antiterrorism Standards for Buildings, DoD, 9 February 2012, UFC 4-010-01.

Unified Facilities Criteria (UFC) DOD Security Engineering Facilities Planning Manual, UFC 4-020-01, DoD, 11 September 2008. www.acoustics.com/conference_room.asp.

Section 7 Architectural Considerations

2000 Emergency Response Guidebook, USDOT, 2000.

Airport Command Center: Phoenix Sky Harbor International Airport, Final Draft Report, Counter Technology, Inc., January 30, 2004.

Architectural Lighting Design, 3rd ed., Gary Steffy, New York: John Wiley & Sons, Inc., 2008. www.acoustics.com/conference_room.asp.

Aviation's Continuing Critical Vulnerabilities to Vehicle-Borne Improvised Explosive Devices, Individual Suicide Bombers, Suitcase Explosive Devices Attack at Airports, B. H. Vincent, October 2012, www.asi-transec.com.

*Building and Exercising Your Command Cente*r, S. C. Davis, All Hands Consulting, www.davislogic.com/BuildandEXyourEOC.pps.

Building and Infrastructure Protection Series, Reference Manual to Mitigate Potential Terrorist Attacks Against Buildings, Department of Homeland Security, FEMA-/426BIPS-06, October 2011.

Bomb Threat Stand-Off Distances, FBI/NCTC, http://www.nctc.gov/witsbanner/docs/2009_report_on_terrorism.pdf.

A Design Language for EOC Facilities: "Some thoughts on basic design approaches by Art Botterell," (http://flghc.org/ppt/09-10/TS%20EM7.pdf – Governor's Hurricane Conference):

- Seating: Boardroom, Mission Control, Marketplace, Bullseye
  "Emergency Operations Center Project Development and Capabilities Assessment," State of Florida Division of Emergency Management.
- Architects Design Group, Inc., http://www.floridadisaster.org/Response/engineers/eoc/eocguide.pdf.

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"Emergency Operations Center Assessment Checklist," FEMA (website), https://www.fema.gov/emergency-operations-center-assessment-checklist.

Emergency Operations Centers Handbook, Federal Emergency Management Agency, CPG 1–20, May, 1984.

Exploring Terrorist Targeting Preferences, M. C. Libicke, Chalk P., and Sissons, Melanie, The Rand Corporation, 2007.

Getting the Big Picture. *ITS International*, Volume 20, Issue 3, 2014, pp. NA5-NA6, http://trid.trb.org/view/1322448.

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(FOUO) *Homeland Security Threat Assessment, Executive Summary,* Department of Homeland Security, August 2007.

Ideal EOC, Emergency Management, http://www.emergencymgmt.com/disaster/Ideal-EOC.html.

Implementing Security Improvement Options at Los Angeles International Airport, Donald Stevens, et al., The Rand Corporation, 2006.

Information Sharing Guidebook for Transportation Management Centers, Emergency Operations Centers, and Fusion Centers. Booz Allen Hamilton, HNTB Corporation, Federal Highway Administration, Houston, N., Wiegmann, J., Marshall, R., Kandarpa, R., Korsak, J., Baldwin, C., Sangillo, J., Knisely, S., Graham, K., and Easton, A. V., 2010, 144p, http://trid.trb.org/view/980782.

(FOUO) *Improvised Explosives Devices,* US Army TRADOC, July 2008

(FOUO) *Incidents at US Airports May Suggest Possible Pre-Attack Probing,* TSA, Transportation Intelligence Gazette, 20 July 2077

The Lighting Handbook: Reference & Application, 10th Ed., Illuminating Engineering Society of North America, 2011. www.lightingdesignlab.com/sites/default/files/pdf/Footcandle_Lighting%20Guide_Rev.072013.pdf.

Meeting the Expanding Operational and Security Related Demands on Operations Control Centers. Lengyel, Dennis. 2008 American Public Transportation Association (APTA) Rail Conference, American Public Transportation Association, 2008, 6 p, http://trid.trb.org/view/874578.

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Recommended Security Guidelines for Airport Planning, Design and Construction, TSA, May 2011.

Room-to-Room Privacy and Acoustical Design Criteria, J. T. Weissenburger, SV Sound and Vibration, 2004.

S*tandard Guide for Emergency Operations Center (EOC) Development*, ASTM, http://www.astm.org/DHS/E2668.pdf.

UFC 4-141-04 Emergency Operations Center Planning and Design, with Change 1, DOD, 9/1/2008, http://www.wbdg.org/FFC/DOD/UFC/ufc_4_141_04_2008_c1.pdf.

Copyright National Academy of Sciences. All rights reserved.

TCRP Research Results Digest 87: Emergency Preparedness, Response, and Recovery in the Transit Industry, Transportation Research Board of the National Academies, Washington, D.C., 2008, 38p, http://trid.trb.org/view/859150.

Abstract: This digest summarizes the mission performed March 16–31, 2007, under TCRP Project J-03, "International Transit Studies Program." It includes an overview of the mission that investigated emergency preparedness programs in the public transportation systems in several Asian cities (Hong Kong, Beijing, Tangshan, Seoul, Kobe, and Tokyo). The report begins in Section I with general observations by the mission team about the state of transit emergency preparedness in the cities visited. In Section II, information gathered about the policies and programs in place in each of the transit environments is provided. Section III shows how the different transit systems manage key emergency functions, including the following: control centers, IT and telecommunications, interagency coordination, public communications, and evacuation planning.

Unified Facilities Criteria (UFC) DoD Minimum Antiterrorism Standards for Buildings, DoD, 9 February 2012, UFC 4-010-01.

Unified Facilities Criteria (UFC) DoD Security Engineering Facilities Planning Manual, UFC 4-020-01, DoD, 11 September 2008.

Utility Systems Terrorism Countermeasures for Command, Control, Communications, Computer, Intelligence, Surveillance and Reconnaissance (C4ISR) Facilities, Department of the Army, 21 February 2006

(FOUO) *Vehicle Bomb Mitigation Guide,* US Air Force Handbook, 1 April 2006

Copyright National Academy of Sciences. All rights reserved.

Guidebook End Notes

1. *Unified Facilities Criteria (UFC) DoD Emergency Operations Center Planning and Design*. UFC 4-141-04, DoD, 11 September 2008, pp. 16, 19.
2. *Reference Manual to Mitigate Potential Terrorist Attacks Against Buildings*. FEMA 426, December 2003, pp. 126.
3. *Unified Facilities Criteria (UFC) DoD Emergency Operations Center Planning and Design*. UFC 4-141-04, DoD, 11 September 2008, pp. 15.
4. M. Carroll, H. Stambaugh, J. Kolesar, S. Berger, H. Benaman, and Z. Varwig. *ACRP Report 127: A Guidebook for Mitigating Disruptive Wi-Fi Interference at Airports*. Transportation Research Board of the National Academies, Washington, D.C., 2015.
5. *Reference Manual to Mitigate Potential Terrorist Attacks Against Buildings*. FEMA 426, December 2003, pp. 74.
6. A portion of the security guidelines outlined in this section is referenced from the *FEMA 426 Reference Manual to Mitigate Potential Terrorist Attacks Against Buildings*, December 2003, Chapter 2.
7. US Department of State standard associated with a gate system capable of keeping the truck bed of a 15,000 lb. truck to 36 inches penetration or less at speeds up to 50 MPH. *Recommended Security Guidelines for Airport Planning, Design and Construction*. Transportation Security Administration, May 2011, pp. 30.
8. *Recommended Security Guidelines for Airport Planning, Design and Construction*. Transportation Security Administration, May 2011, Appendix A, pp. 154.
9. Shinrikyo, A. 1995. Attack on subway in Tokyo using crudely constructed Sarin; Rajneeshees attacked restaurants in Oregon with Salmonella Typhimurium; and other similar attack profiles interrupted by FBI in pre-attack and planning stages.
10. *Unified Facilities Criteria (UFC) DoD Emergency Operations Center Planning and Design*. UFC 4-141-04, DoD, 11 September 2008, pp. 16.
11. *GSA Special Design Considerations*. www.gsa.gov/node/84139 (as of November 29, 2017).
12. Weissenburger, J. T. 2004. "Room-to-Room Privacy and Acoustical Design Criteria." *SV Sound and Vibration, February 2004*.

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Glossary of Terms

AAAE	American Association of Airport Executives
AC	Advisory Circular
ACRP	Airport Cooperative Research Program
ACS	Access Control System
ADA	Americans with Disabilities Act
ADAAG	Americans with Disabilities Act Accessibility Guidelines
ADO	Airport District Office
ADS	Addison Airport
AEOC	Airport Emergency Operations Center
AEP	Airport Emergency Plan
AIP	Airport Improvement Program
ALP	Airport Layout Plan
ANSI	American National Standards Institute
AOA	Airport Operations Area
AOC	Airport Operations Center
ARC	American Red Cross
ARFF	Aircraft Rescue and Fire Fighting
ATF	Bureau of Alcohol, Tobacco and Firearms
AV	Audio–visual
BGR	Bangor International Airport
CBN	Chemical, Biological, Nuclear
CBP	Customs and Border Patrol
CCTV	Closed-Circuit Television
CIP	Capital Improvement Program
CMAR	Construction Manager at Risk
ConOps	Concept of Operations Planning
COS	Colorado Springs Airport
CP	Command Post
CSA	Canadian Standards Association
CVG	Cincinnati/Northern Kentucky International Airport
DCA	Washington Reagan Airport
DEN	Denver International Airport
DFW	Dallas/Ft. Worth International Airport
DHS	Department of Homeland Security
DOC	Department Operations Center
DSM	Des Moines International Airport
EMS	Emergency Medical Services
EOC	Emergency Operations Center

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ESF	Emergency Support Functions
FAA	Federal Aviation Administration
FAR	Federal Aviation Regulation
FBI	Federal Bureau of Investigation
FBO	Fixed Base Operator
FC	Field Command
fc	foot-candles
FEMA	Federal Emergency Management Agency
FLL	Ft. Lauderdale Hollywood International Airport
GA	General Aviation
HAZMAT	Hazardous Material
HFES	Human Factors and Ergonomics Society
HMI	Human Machine Interface
HPSD	Homeland Security Presidential Directive
HVAC	Heating, Ventilation and Air Conditioning
IAD	Washington Dulles Airport
IAH	George Bush Intercontinental Airport
IC	Incident Commander
ICS	Incident Command System
ISM	Incident Support Model
IED	Improvised Explosive Devices
IND	Indianapolis International Airport
INS	Incident of National Significance
IRIS	Incident Resource Inventory System
ISO	International Organization for Standardization
IT	Information Technology
JAN	Jackson Evers International Airport
JIC	Joint Information Center
JOC	Joint Operations Center
LAN	Local Area Network
LED	Light Emitting Diode
LEED	Leadership in Energy and Environmental Design
LEO	Law Enforcement Officers
LNK	Lincoln Airport
MBS	MBS International Airport
MDT	Harrisburg International Airport
MEM	Memphis International Airport
MSP	Minneapolis St. Paul International Airport
NAS	US National Air Space System
NASA	National Aeronautics and Space Administration
NC/RC	Noise Criteria/Room Criteria
NCTC	National Counter Terrorism Center
NIMS	National Incident Management System
NRC	Noise Reduction Coefficient
NTSB	National Transportation Safety Board
PAX	Passenger Terminal
PDX	Portland International Airport
PHX	Phoenix Sky Harbor Airport
PTE	Potential Threat Element
ROM	Rough Order of Magnitude
RPZ	Runway Protection Zone

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RQE	Prosser Airport
SA	Situational Awareness
SEMS	Standardized Emergency Management System
SIDA	Security Identification Display Area
SJC	Mineta San Jose International Airport
SME	Subject Matter Expertise
SMF	Sacramento International Airport
SOP	Standard Operating Procedure
STC	Sound Transmission Class
SUS	Spirit of St. Louis Airport
TLH	Tallahassee Regional Airport
TRB	Transportation Research Board
TSA	Transportation Security Administration
TSWG	Transportation Security Working Group
TTP	Tactics, Techniques and Procedures
TUL	Tulsa International Airport
TV	Television
TVA	Threat and Vulnerability Assessment
UCC	Unified Command Control Center
VBIED	Vehicle Borne Improvised Explosive Devices
VIP	Very Important Person
WMD	Weapon of Mass Destruction

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Abbreviations and acronyms used without definitions in TRB publications:

A4A	Airlines for America
AAAE	American Association of Airport Executives
AASHO	American Association of State Highway Officials
AASHTO	American Association of State Highway and Transportation Officials
ACI–NA	Airports Council International–North America
ACRP	Airport Cooperative Research Program
ADA	Americans with Disabilities Act
APTA	American Public Transportation Association
ASCE	American Society of Civil Engineers
ASME	American Society of Mechanical Engineers
ASTM	American Society for Testing and Materials
ATA	American Trucking Associations
CTAA	Community Transportation Association of America
CTBSSP	Commercial Truck and Bus Safety Synthesis Program
DHS	Department of Homeland Security
DOE	Department of Energy
EPA	Environmental Protection Agency
FAA	Federal Aviation Administration
FAST	Fixing America's Surface Transportation Act (2015)
FHWA	Federal Highway Administration
FMCSA	Federal Motor Carrier Safety Administration
FRA	Federal Railroad Administration
FTA	Federal Transit Administration
HMCRP	Hazardous Materials Cooperative Research Program
IEEE	Institute of Electrical and Electronics Engineers
ISTEA	Intermodal Surface Transportation Efficiency Act of 1991
ITE	Institute of Transportation Engineers
MAP-21	Moving Ahead for Progress in the 21st Century Act (2012)
NASA	National Aeronautics and Space Administration
NASAO	National Association of State Aviation Officials
NCFRP	National Cooperative Freight Research Program
NCHRP	National Cooperative Highway Research Program
NHTSA	National Highway Traffic Safety Administration
NTSB	National Transportation Safety Board
PHMSA	Pipeline and Hazardous Materials Safety Administration
RITA	Research and Innovative Technology Administration
SAE	Society of Automotive Engineers
SAFETEA-LU	Safe, Accountable, Flexible, Efficient Transportation Equity Act: A Legacy for Users (2005)
TCRP	Transit Cooperative Research Program
TDC	Transit Development Corporation
TEA-21	Transportation Equity Act for the 21st Century (1998)
TRB	Transportation Research Board
TSA	Transportation Security Administration
U.S.DOT	United States Department of Transportation

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